

Compaction Permeameter 101

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Permeameter Components

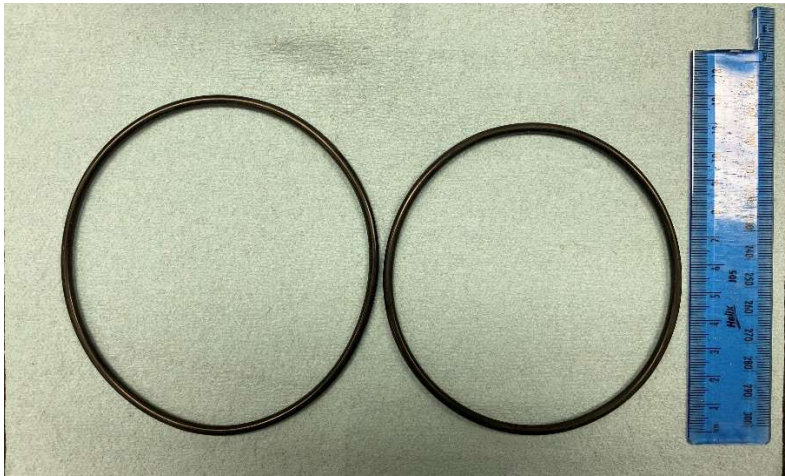
The following components are labelled as I refer to them in the rest of the instructions. These names are not necessarily the technical ones, but what works for me. For a more thorough description of each part, refer to the ASTM standards attached to the end of the document.



- | | |
|---------------------------------------|--|
| a. Porous Stones | g. Three Rubber O-rings. (One is slightly larger than the other two) |
| b. Upper plate and water Input Valves | h. Reservoir Stand (for manometer pipe) |
| c. Lid screws | i. Upper plate with swelling-reduction spring |
| d. Collar | j. Filter Papers |
| e. Sample Mold | |
| f. Mold Base / Lower plate | |

Below are the two different sized O-rings. The rightmost O-ring is the largest and is used to seal the space between the mold and the collar. The smaller O-rings are used on the upper and lower plates.

Also pictured are the two types of compaction hammers. The “modified” proctor hammer is on the left and the “standard” proctor hammer is on the right. When used to ASTM standards, each hammer imparts a specified compactive effort to the soil. (2,700 kN-m/m³ and 600 kN-m/m³, respectively).



For reference, the dimensions of the mold and the collar are below.



Soil Sample Preparation

Ideally, the soil sample used for this experiment fulfills the following criteria, but are not required:

- Soil is dry (for accurate water content determination)
- Full sample is roughly two kilograms in mass (so there is sufficient soil to fill the mold)
- Soil is “loose” and free of clumps/clods (unless clods are desired)
-

The amount of water added to the dry soil is related to the goals of the experiment and the soil type.

Typically, sandier soils require just enough water to hold the sample together and evenly. Sandy soils will generally develop flow and saturate fast enough that full saturation is not necessary. Clay rich soils will require more water and mixing time to evenly distribute the moisture. These clayey soils also tend to exhibit lower hydraulic conductivities, and I would generally recommend adding enough water to nearly saturate the sample. This may reduce the amount of time required to develop flow and fully saturate the sample during the course of the experiment.

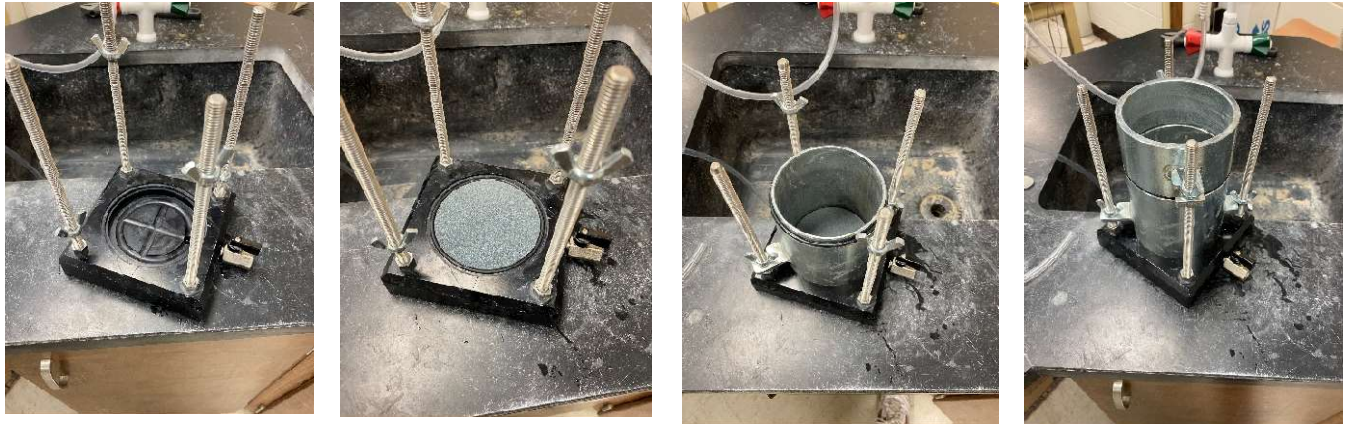


In some cases, the sample may need to be fully saturated before the experiment can begin. This can be accomplished using methods like those in ASTM D5084-24.

Sample Chamber Preparation

The sample chamber is prepped the same way for both falling head and constant head tests.

I typically start by rinsing out the base, one of the porous stones, and one of the smaller diameter O-rings. Place the wet porous stone, filter paper, and an O-ring into the base. Secure the mold and the collar by placing and twisting them until their “hooks” meet the threaded rods. Then gently tighten the wing nuts.



Record the mass of the empty chamber (the lower plate, porous stone, O-Rings, mold, and collar).

Similarly, record the mass of an oven tray and the mass of roughly 100 to 200 grams of soil on the tray. Place this tray of soil in the oven set to 110°C. The masses of wet and dry soil are used to determine the pre-test water content.



Begin filling the mold with soil. Generally, it should be filled in “lifts” and compacted with a proctor hammer between each lift. However, intensely compacted sediments may take a significant amount of time to develop flow. I have typically shaken and tapped the soil down with the hammer to gently compact it, instead of performing the standard number of blows for each lift. If tapping down the soil, tap in a circular motion with the hammer. Try to keep the top lift flat and even. Record a note of how the sample is compacted. Otherwise, ASTM standards for proper use of the compaction hammer are found in D698-12 and D1557-12.

Below is a picture of a lift being tapped down with the standard hammer.



If enough soil is available, the best course of action is to perform 3 lifts with compaction between each lift, filling the chamber above the top of the mold. Then, remove the collar and scrape the soil sample to be flush with the top of the mold (this allows for use of the swelling-reduction spring and easy volume calculations). When compacting above the mold, remove the O ring for

Next, the compacted sample must be weighed and measured.

Record the mass of the chamber with the compacted sample. If the sample is not flush with the top of the mold, record the depth of the sample from the top of the mold. This is required to calculate the volume of the soil sample. I like to take several depth measurements with the calipers around the circumference of the sample and average the depth. If it varies by more than a few millimeters, I will try to level out the surface for a consistent volume.



Now that the soil sample is prepared, the sample can be sealed.

Rinse the second porous stone, apply the filter paper, and place it on top of the sample. Insert a small O-ring into the upper plate. If there is a concern that the soil may swell, insert the swelling-reduction spring into the upper plate. This helps the sample retain a constant volume over the course of the test. The upper plate can be oriented in any direction.



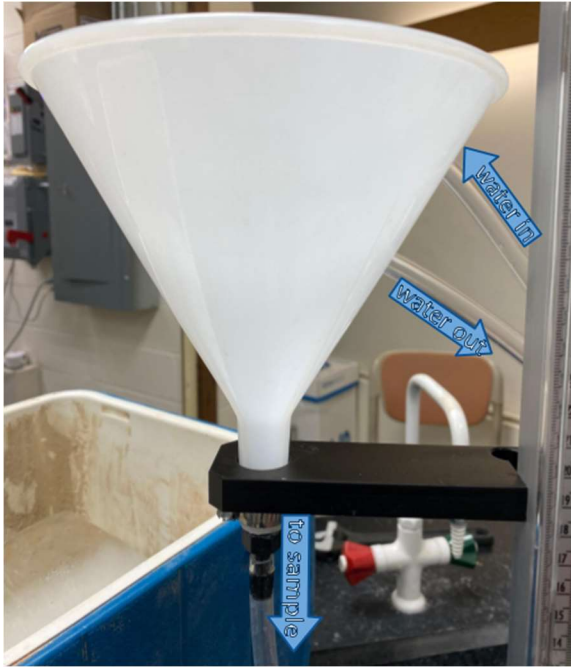
The wing nuts and screws do not need to be very tight. Generally, I tighten the upper plate until it just starts to give resistance. Then I would tighten the bottom most wing nuts and the collar wing nuts until they give resistance. If the screws are too tight, the O-rings may begin to deform, causing leaks.

Another note: the surfaces that seal the chamber should be kept clean to minimize any leaks. This is especially true when working with very fine soils or any glass beads.

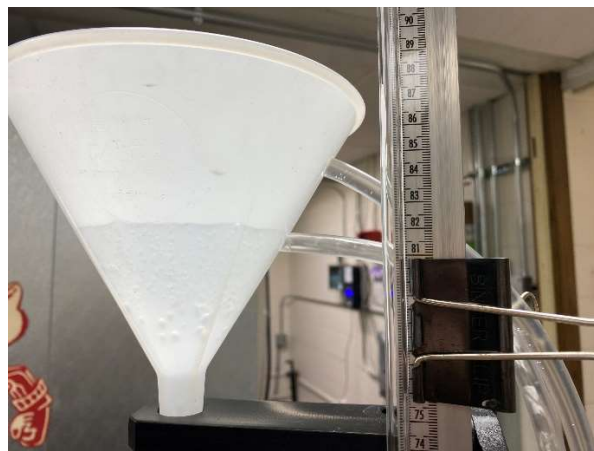
If any leaks are present, it *might* be fixed by tightening the upper plate and the wing nuts, otherwise surfaces that contact each other may need to be cleaned, requiring the chamber to be disassembled.

Constant Head Test

The constant head test does exactly as it implies. It keeps the water pressure on the sample constant throughout the course of the experiment. This is accomplished using a modified funnel as a reservoir. The reservoir maintains the desired head through two tubes; one connected to a nearby faucet and the other diverting excess water to a sink.



To set the head, adjust the reservoir holder on the back of the reservoir stand. It can be unscrewed and tightened using the screw on the back. Occasionally, the opening in the holder may need to be pried open to release it from the stand. Generally, the head will remain between the inflow and outflow holes, just above the outflow hole.



Water flow from the faucet should be adjusted to allow for the inflow and outflow to balance at a constant head before measurements are recorded. This should only take a few seconds once the water in the reservoir exceeds the height of the outflow valve.

If not done already, connect the reservoir to the sample chamber via the topmost valve. Turn on the faucet and allow water to flow. Open the brass valve and allow any air in the tube and chamber to escape. I typically wait until the water flowing out of the brass valve no longer sputters and remains a steady stream. Wipe up any water so any leaks are not hidden.



After closing the brass valve, the test is almost ready to begin. Generally, the finer the soil, the longer it will take for flow to develop. This can be a few seconds for sandier soils to the better part of an hour for finer soils. Once flow is relatively steady, data can be recorded.

To begin a trial, simultaneously start a stopwatch and place a container in the path of flow. After a measureable ($>10\text{mL}$) amount of water has accumulated, remove the collection container and stop the stopwatch. Measure the volume of water collected in a graduated cylinder (there are two in the lab, an 1000mL and a 100mL , depending on amount of flow collected).

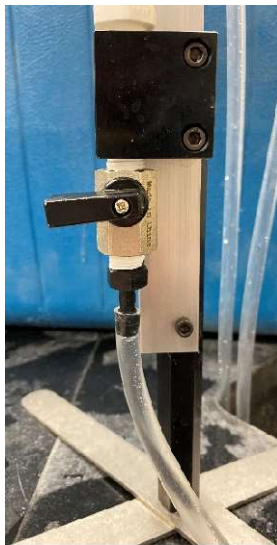


Record at least 4 consecutive trials. If these 4 trials result in values that are within 25% of the mean of the 4 trials, then flow is considered “steady.” If the trials are not within this range, continue to collect trials until any upward or downward trends are minimal and the values are within the acceptable range. For some low conductivity soils ($k < 1\text{E-}10$) this range is within 50% of the mean.

Falling Head Test

The falling head test is best for lower permeability sediments, providing hydraulic conductivity at an instant in time. It does not require the collection of water, nor does it require a reservoir. It instead uses the time it takes for a measurable change in head to occur to calculate hydraulic conductivity.

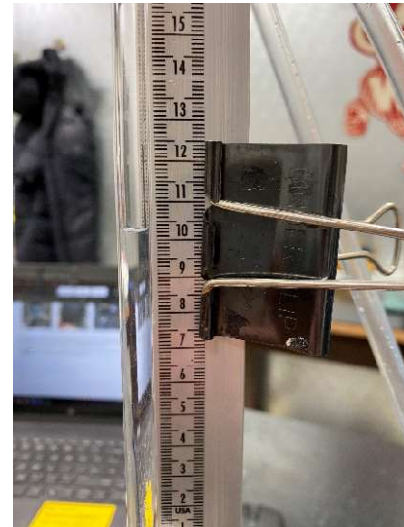
Close all valves on the chamber. Then fill the falling head pipe/manometer, which is attached to the stand. I have done this by attaching the manometer pipe directly to the sink and carefully filling it. I generally try to fill the tube as much as possible. If there are air bubbles in the manometer tube, continue to fill it until the bubbles bleed out the top of the tube. Once filled, close the valve and attach it to the chamber.



After attaching the manometer to the chamber, open its valve. If the water in the manometer rapidly drops, there is empty space above the sample that needs to be filled with water. This can be done by repeatedly filling and emptying the manometer into the chamber, or more easily by connecting the sink to the chamber with a bit of pipe. When connecting the chamber to the sink, slowly fill the rest of the chamber and use the brass valve to bleed out any air. While filling the chamber, it may be beneficial to stop flow out of the chamber using the outflow valve on the bottom plate of the chamber. Once the chamber is full, the water level in the manometer should drop and then rest after opening the manometer valve.



To begin the test, open all valves on the chamber (except the brass valve) and close the manometer valve. Pick two points on the manometer to start and stop a stopwatch. Generally, I use head values around 80cm and 20cm. ASTM standards require that the ending head, h_2 , must be less than 75% of the initial head, h_1 .



Open the manometer valve and wait for the head to drop to the selected h_1 value. Start the stopwatch when the head reaches this value. Then wait for the head to drop to the selected h_2 value. Stop the stopwatch once the head reaches this value. Record the change in time and the two head values.

Repeat the data collection cycle at least four times. The same rules apply as the constant head test, where 4 trials must result in values that are within 25% of the mean. If the trials are not within this range, continue to collect trials until any upward or downward trends are minimal and the values are within the appropriate range.

Concluding the Test and Cleanup

To end the test, detach all pipes connected to the chamber and turn off the faucet, if applicable. Drain all water to the sink. Carefully remove the upper plate and pour out any excess water.



To remove the porous stones and the soil sample, invert the chamber into a tray. Ideally, the porous stone will fall out, leaving the soil. Take a rubber mallet and tap the chamber to encourage the soil to relocate. Lifting the chamber should release the soil sample. Remove any filter paper and porous stones.



Next, collect the post-test soil sample. Record the mass of a tray and the tray with wet soil. Take roughly 100-200 grams. Place the tray into the oven at 110° C. Finally, disassemble and rinse out the pieces of the chamber.



After a day or so, record the mass of the dry soil and the tray. This allows for the calculation of the soils' water content prior to and following the test.

Store the porous stones on a paper towel to encourage drying. Store all permeameter components, glassware, and tools in the appropriate locations.



Notes

Appendices

Attached are several relevant ASTM standards, for reference.



Designation: D5856 – 15

Standard Test Method for Measurement of Hydraulic Conductivity of Porous Material Using a Rigid-Wall, Compaction-Mold Permeameter¹

This standard is issued under the fixed designation D5856; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ϵ) indicates an editorial change since the last revision or reapproval.

1. Scope

1.1 This test method covers laboratory measurement of the hydraulic conductivity (also referred to as *coefficient of permeability*) of laboratory-compacted materials with a rigid-wall, compaction-mold permeameter.

1.2 This test method may be used with laboratory-compacted specimens that have a hydraulic conductivity less than or equal to 1×10^{-5} m/s. The hydraulic conductivity of compacted materials that have hydraulic conductivities greater than 1×10^{-5} m/s may be determined by Test Method D2434.

1.3 *Units*—The values stated in SI units are to be regarded as the standard, unless other units are specifically given. By tradition in U.S. practice, hydraulic conductivity is reported in cm/s, although the common SI units for hydraulic conductivity are m/s.

1.4 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.*

2. Referenced Documents

2.1 ASTM Standards:²

D653 Terminology Relating to Soil, Rock, and Contained Fluids

D698 Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12 400 ft-lbf/ft³ (600 kN-m/m³))

D854 Test Methods for Specific Gravity of Soil Solids by Water Pycnometer

D1557 Test Methods for Laboratory Compaction Character-

istics of Soil Using Modified Effort (56,000 ft-lbf/ft³ (2,700 kN-m/m³))

D2216 Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass

D2434 Test Method for Permeability of Granular Soils (Constant Head) (Withdrawn 2015)³

D3740 Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

D4753 Guide for Evaluating, Selecting, and Specifying Balances and Standard Masses for Use in Soil, Rock, and Construction Materials Testing

D5084 Test Methods for Measurement of Hydraulic Conductivity of Saturated Porous Materials Using a Flexible Wall Permeameter

E145 Specification for Gravity-Convection and Forced-Ventilation Ovens

3. Terminology

3.1 Definitions:

3.1.1 *hydraulic conductivity, k*—the rate of discharge of water under laminar flow conditions through a unit cross-sectional area of a porous medium under a unit hydraulic gradient and standard temperature conditions (20°C).

3.1.1.1 *Discussion*—The term *coefficient of permeability* is often used instead of *hydraulic conductivity*, but *hydraulic conductivity* is used exclusively in this test method. A more complete discussion of the terminology associated with Darcy's law is given in the literature⁴.

3.1.2 For common definitions of technical terms in this standard, refer to Terminology D653.

3.2 Definitions of Terms Specific to This Standard:

3.2.1 *flux*—quantity of flow per unit area per unit time.

3.2.2 *pore volume of flow*—the cumulative quantity of outflow from a test specimen divided by the volume of pore space in the specimen.

¹ This test method is under the jurisdiction of ASTM Committee D18 on Soil and Rock and is the direct responsibility of Subcommittee D18.04 on Hydrologic Properties and Hydraulic Barriers.

Current edition approved June 1, 2015. Published July 2015. Originally approved in 1995. Last previous edition approved in 2007 as D5856–95(2007). DOI: 10.1520/D5856-15.

² For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

³ The last approved version of this historical standard is referenced on www.astm.org.

⁴ Olson, R. E., and Daniel, D. E., "Measurement of the Hydraulic Conductivity of Fine-Grained Soils," *Symposium on Permeability and Groundwater Contaminant Transport, ASTM STP 746*, ASTM, 1981, pp. 18–64.

4. Significance and Use

4.1 This test method applies to one-dimensional, laminar flow of water within laboratory-compacted, porous materials such as soil.

4.2 The hydraulic conductivity of porous materials generally decreases with an increasing amount of air in the pores of the material. This test method applies to porous materials containing little or no air. The test method is designed to minimize the amount of air in the test specimen. However, this test method does not ensure complete saturation of the test specimen with water. In cases where it is essential to saturate the test specimen fully with water, the compacted specimen may be tested using Test Method D5084.

4.3 This test method applies to permeation of porous materials with water. Permeation with other liquids, such as chemical wastes, can be accomplished using procedures similar to those described in this test method. However, this test method is only intended to be used when water is the permeant liquid.

4.4 It is assumed that Darcy's law is valid and that the hydraulic conductivity is essentially unaffected by hydraulic gradient. The validity of Darcy's law may be evaluated by measuring the hydraulic conductivity of the specimen at three hydraulic gradients; if all measured values are similar (within 25 %), then Darcy's law may be taken as valid. However, when the hydraulic gradient acting on a test specimen is changed, the state of stress will also change, and, if the specimen or pore fluid is compressible, the volume of the test specimen or pore fluid will change. Thus, some change in hydraulic conductivity may occur when the hydraulic gradient is altered, even in cases where Darcy's law is valid.

4.5 One potential problem with this method of testing is the possibility that water will flow along the interface between the test specimen and the compaction/permeameter ring. The problem tends to be of minimal significance for materials that swell when exposed to water (for example, compacted, clayey soils) but can be a very serious problem for materials that might tend to shrink and pull away from the walls of the permeameter. Test Method D5084 is recommended for any material that tends to shrink when exposed to the permeant liquid.

4.6 The correlation between results obtained with this test method and the hydraulic conductivities of in-place, compacted materials has not been fully investigated. Experience has sometimes shown that flow patterns in small, laboratory-prepared test specimens do not necessarily follow the same patterns on large field scales and that hydraulic conductivities measured on small test specimens are not necessarily the same as larger-scale values. Therefore, the results should be applied to field situations with caution and by qualified personnel.

NOTE 1—The quality of the result produced by this standard is dependent on the competence of the personnel performing it, and the suitability of the equipment and facilities used. Agencies that meet the criteria of Practice D3740 are generally considered capable of competent and objective testing/sampling/inspection/etc. Users of this standard are cautioned that compliance with Practice D3740 does not in itself ensure reliable results. Reliable results depend on many factors; Practice D3740

provides a means of evaluating some of those factors.

5. Apparatus

5.1 *Hydraulic System*—Constant head (Test Method A), falling head (Test Methods B, C, and D), or constant rate of flow (Test Method E) systems may be used provided they meet the criteria outlined as follows:

5.1.1 *Constant Head*—The system must be capable of maintaining a constant hydraulic pressure or head to within $\pm 5\%$ and shall include means to measure hydraulic pressures or heads to within the prescribed tolerance. In addition, the head loss across the test specimen must be held constant to within $\pm 5\%$ and shall be measured with the same accuracy or better. Pressures shall be measured by a pressure gauge, electronic pressure transducer, or any other device of suitable accuracy. Head of liquid in a standpipe may be measured with a graduated pipet, ruler, scale, or other device of suitable accuracy.

5.1.2 *Falling Head*—The system shall allow for measurement of the applied head loss, thus hydraulic gradient, to within $\pm 5\%$ or better at any time. In addition, the ratio of initial head loss divided by final head loss over an interval of time shall be measured such that this computed ratio is accurate to within $\pm 5\%$. The head loss shall be measured with a pressure gauge, electronic pressure transducer, engineer's scale, graduated pipet, or any other device of suitable accuracy. Falling head tests may be performed with either a constant tailwater elevation (Test Method B), rising tailwater elevation (Test Method C), or increasing tailwater elevation (Test Method D).

5.1.3 *Constant Rate of Flow*—The system must be capable of maintaining a constant rate of flow through the specimen to within $\pm 5\%$ or better. Flow measurement or control shall be by calibrated syringe, graduated pipet, or other device of suitable accuracy. The head loss across the specimen shall be measured to an accuracy of $\pm 5\%$ or better using an electronic pressure transducer or other device of suitable accuracy. A means to ensure that the head being measured is not affected by sidewall leakage should be included. More information on testing with a constant rate of flow is given in the literature⁵.

5.2 *Flow Measurement System*—Both inflow and outflow volumes shall be measured or controlled. Flow volumes shall be measured by a graduated accumulator, graduated pipet, graduated cylinder, vertical standpipe in conjunction with an electronic pressure transducer, marriott bottle, or other volume-measuring device of suitable accuracy. For long-term tests, evaporative losses may be significant and should be accounted for using a suitable correction procedure.

5.2.1 *Flow Accuracy*—Required accuracy for the quantity of flow measured over an interval of time is $\pm 5\%$ or better.

5.2.2 *Head Losses*—Head losses in the tubes, valves, porous end pieces, and filter paper may lead to error. To guard against such errors, the permeameter shall be assembled with no specimen inside (but with any porous end pieces or sheets of filter paper that will be used) and then the hydraulic system

⁵ Olsen, H. W., Gill, J. D., Willden, A. T., and Nelson, N. R., "Innovations in Hydraulic Conductivity Measurements," *Transportation Research Record No. 1309*, Transportation Research Board, National Research Council, Washington, DC, 1991.

filled. If a constant or falling head test is to be used, the hydraulic pressures or heads that will be used in testing a specimen shall be applied, and the rate of flow measured with an accuracy of $\pm 5\%$ or better. This rate of flow shall be at least ten times greater than the rate of flow that is measured when a specimen has been compacted inside the permeameter and the same hydraulic pressures or heads are applied. If a constant rate of flow test is to be used, the rate of flow to be used in testing a specimen shall be supplied to the permeameter and the head loss measured. The head loss without a specimen shall be less than 0.1 times the head loss when a specimen is present.

5.3 Permeameter Cell—The permeameter cell shall consist of a rigid-wall compaction mold into which the material to be tested is compacted and in which the compacted material is permeated; and two end plates to control flow into and out of the test specimen. A swell ring may be provided as discussed in 5.3.2. The permeameter shall be designed and operated so that permeant water flows downward through the test specimen, although upward flow may be used if the top of the specimen is protected from upward movement by a rigid porous element.

5.3.1 Compaction Mold/Permeameter Ring—The compaction mold/permeameter ring shall be constructed of a rigid material that will not be damaged during compaction of the test specimen and that will not undergo adverse chemical reactions with the test material or permeant water. Materials such as steel, aluminum, brass, plastic, and glass have been used. The mold shall be sufficiently rigid so that its expansion when the permeameter is pressurized is negligibly small. The mold can be any cylindrical shape so long as: the cross-sectional area along the direction of flow does not vary by more than $\pm 2\%$; height and diameter are each ≥ 25 mm; height does not vary by more than $\pm 1\%$, and the largest particle and clod size in the compacted specimen is $\leq \frac{1}{16}$ the lesser of the height or diameter.

5.3.2 Swell Ring—The top of the permeameter may be designed to function in one of three ways: (1) to allow no restraint against swelling (see Fig. 1), in which case a swell ring separates the compaction mold/permeameter ring from the top plate; (2) to allow no swelling of the test specimen (see Fig. 2), in which case no swell ring is provided; or (3) to control the vertical stress that is applied to the test specimen (see Fig. 3), in which case a swell ring may or may not be needed, depending upon how the top plate is designed and how the vertical stress is applied. If a swell ring is used, it shall be constructed of a rigid material that will not react adversely with the test material or permeant water, shall have the same diameter or width as the compaction mold/permeameter ring, and shall be sufficiently high to allow free swelling of the test specimen or to accommodate stress-control apparatus. Sand may be placed in the swell ring to minimize erosion of the specimen from influent flow provided that the sand is included in the measurement of head losses in the permeameter (see 5.2.2).

5.3.3 Stress-Control Apparatus—If the upper surface of the compacted test specimen is subjected to a controlled vertical stress, the stress shall be applied through a rigid plate using any means that maintains the stress within $\pm 5\%$ of the desired value.

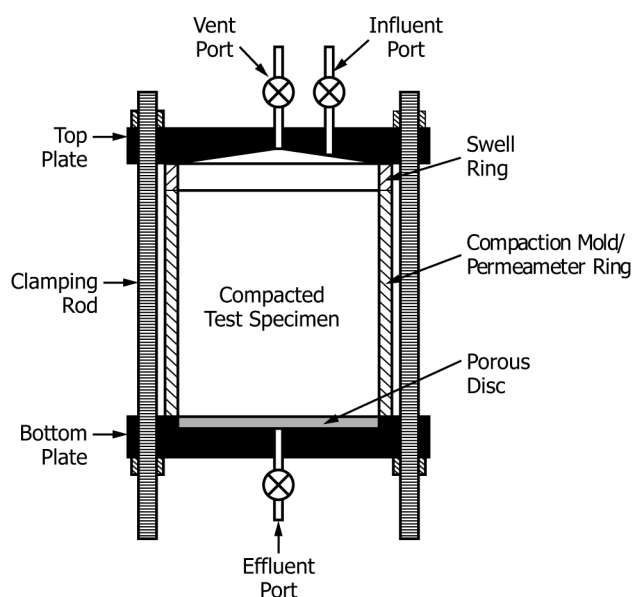


FIG. 1 Compaction-Mold Permeameter with No Restraint Against Swelling at Top of Test Specimen

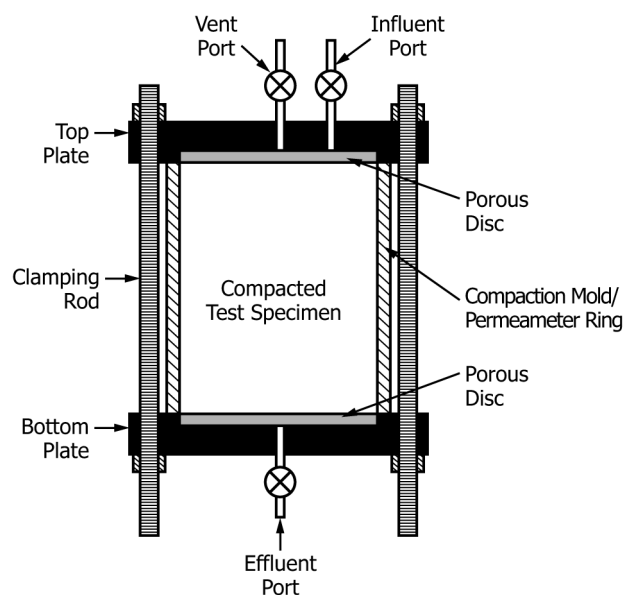


FIG. 2 Compaction-Mold Permeameter in Which Test Specimen Cannot Swell

5.3.4 Bottom Plate—The bottom plate shall be constructed of rigid material that does not react adversely with the test material or permeant liquid. The plate shall serve the purpose of preventing the test specimen from swelling downward, supporting the test specimen, collecting effluent liquid from the base of the test specimen, and ensuring one-dimensional flow near the effluent end of the test specimen. The base plate shall be sealed to the compaction mold/permeameter ring, for example, with an O-ring, to prevent leakage. Checks for leaks, conducted without soil in the cell, are helpful to ensure adequacy of the seals. Careful examination of the seal is particularly important when the apparatus is disassembled and re-assembled during the test. The bottom plate shall contain a

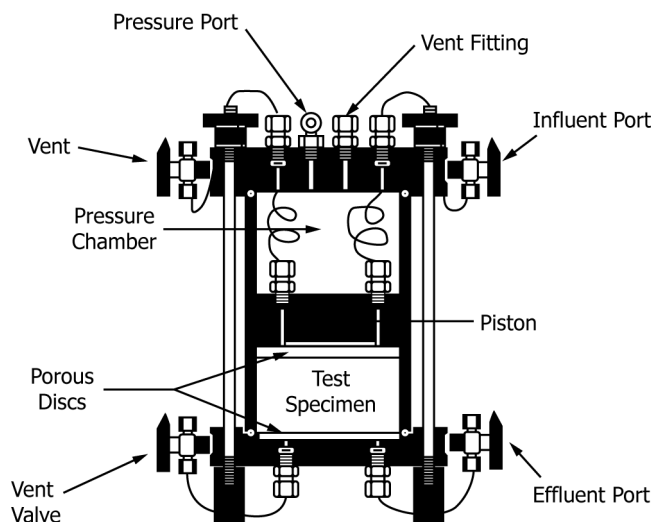


FIG. 3 Compaction-Mold Permeameter with a Controlled Vertical Stress Applied to the Top of the Test Specimen

porous material (such as porous plastic, porous metal, porous corundum, or well-compacted sand) that is far more permeable than the test specimen (5.2.2 ensures that this is the case) and that has the same overall diameter or width as the test specimen. Two base plate designs are acceptable:

5.3.4.1 *Single-Ring*—This design (Fig. 1, Fig. 2, and Fig. 3) consists of a single porous material with the same diameter or width as the test specimen.

5.3.4.2 *Double Ring*—With this design (see Fig. 4), outflow from the central part of the test specimen is separated from flow near the sidewalls in the base plate. The system shall be designed so that the collector ring (see Fig. 4) protrudes approximately 3 mm into the test specimen and completely separates flow entering the base plate through the inner zone

from flow entering the outer zone. A thin ring with a sharp edge should be used to prevent disturbance of the soil during installation.

NOTE 2—The double ring is preferred over the single ring because the double ring provides a means for identifying erroneous measurements caused by side-wall leakage. Unless the requestor specifically requests a single-ring base plate, the double-ring base plate shall be used.

5.3.5 *Top Plate*—The top plate shall be constructed from rigid materials that do not react adversely with the test material or permeant water. The top plate shall be sealed to the compaction mold/permeameter ring or swell ring (if present) in a way that prevents leakage. Tests for leaks, conducted without soil in the cell, are helpful to ensure the adequacy of the seals. The plate shall be designed to ensure that flow through the test specimen is one-dimensional.

5.4 *Filter Paper*—If necessary to prevent intrusion of material into the pores of the porous end pieces, one or more sheets of filter paper shall be placed between the top and bottom porous end pieces and the specimen. The paper shall have a negligibly small hydraulic impedance. The requirements outlined in 5.2.2 ensure that the impedance is small.

5.5 *Equipment for Compacting a Specimen*—Equipment (including compactor and mold) suitable for the method of compaction specified by the requestor shall be used. Examples of compaction equipment/procedures are Test Methods D698 and D1557.

5.6 *Devices for Measuring the Dimensions of the Specimen*—Devices used to measure the dimensions of the specimen shall be capable of measuring to the nearest 0.3 mm (0.01 in.) or better and shall be constructed such that their use will not disturb the specimen.

5.7 *Balances*—The balance shall be suitable for determining the mass of the specimen and shall be selected as discussed in Specification D4753. The mass of specimens less than 100 g shall be determined to the nearest 0.01 g. The mass of specimens 100 g or larger shall be determined to the nearest 0.1 g. The mass of specimens >1000 g shall be determined to the nearest 1.0 g.

5.8 *Temperature Maintaining Device*—The temperature of the permeameter, test specimen, and reservoir of permeant liquid shall not vary more than $\pm 3^{\circ}\text{C}$ ($\pm 5.7^{\circ}\text{F}$). Normally, this is accomplished by performing the test in a room with a relatively constant temperature. If such a room is not available, the apparatus shall be placed in a water bath, insulated chamber, or other device that maintains a temperature within the tolerance specified above. The temperature shall be periodically measured and recorded.

5.9 *Water Content Containers*—The containers shall be in accordance with Test Method D2216.

5.10 *Drying Oven*—The oven shall be in accordance with Specification E145.

6. Reagents

6.1 Permeant Water:

6.1.1 The permeant water is the liquid used to permeate the test specimen. Any liquid introduced into the test specimen,

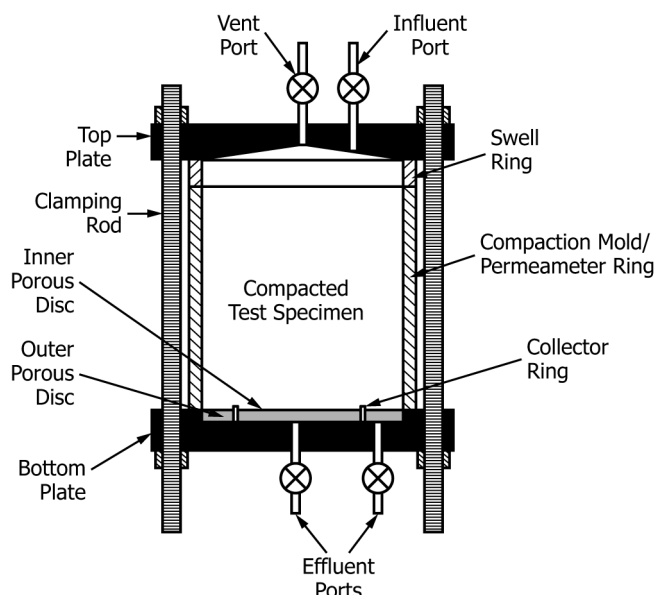


FIG. 4 Compaction-Mold Permeameter with Double Ring in Base Plate

porous elements, filter paper, or other portions of the permeameter shall be permeant water.

6.1.2 The type of permeant water shall be specified by the requestor. If no specification is made, de-aired tap water shall be used for the permeant liquid. The type of water used shall be indicated in the report. Regardless of the type of permeant water requested, the water shall be de-aired water (see 6.1.3), unless the requestor specifically requests that the water not be de-aired.

NOTE 3—Chemical interactions between a permeant liquid and the porous material may lead to variations in hydraulic conductivity. Distilled water can significantly lower the hydraulic conductivity of clayey soils.⁴ For this reason, distilled water is not usually recommended as a permeant liquid. A permeant liquid used by some is 0.005 *N* CaSO₄, that can be obtained, for example, by dissolving 6.8 g of non-hydrated, reagent-grade CaSO₄ in 10 L of de-aired, distilled (or de-ionized) water. This CaSO₄ solution is thought to neither increase nor decrease significantly the hydraulic conductivity of clayey soils. In areas with extremely brackish tap water, the CaSO₄ solution is recommended.

6.1.3 To aid in removing as much air from the test specimen as possible, de-aired water shall be used. The water is usually de-aired by boiling, by spraying a fine mist of water into an evacuated vessel attached to a vacuum source, or by forceful agitation of water in a container attached to a vacuum source. If boiling is used, care shall be taken not to evaporate an excessive amount of water, which can lead to a larger salt concentration in the permeant water than desired. To prevent dissolution of air back into the water, de-aired water shall not be exposed to air for prolonged periods.

7. Test Specimen

7.1 *Preparation of Compaction/Permeameter Ring*—Prior to compaction, determine the inside dimensions of the compaction/permeameter ring to the tolerances given in 5.3.1, and determine the mass of the compaction/permeameter ring to the tolerances given in 5.7. If desired, the inside of the permeameter may be coated with a thin layer of sodium bentonite paste to minimize the possibility of side-wall leakage.

7.2 *Compaction*—Prepare the material to be tested and compacted inside the compaction/permeameter ring in a manner specified by the requestor. If the specimen is placed and compacted in layers, lightly scarify the surface of each previously-compacted layer (roughen) with a fork, ice pick, or other suitable object, unless the requestor specifically states that scarification is not to be performed. Test Methods D698 and D1557 describe two methods of compaction, but any other method specified by the requestor may be used as long as the method is described in the report. Large clods of material should not be broken down prior to compaction unless it is known that they will be broken in field construction, as well, or the requestor specifically requests that the clod size be reduced. Neither hard clods nor individual particles of the material shall exceed 1/8 of either the height or diameter of the specimen. After compaction, determine the mass of the compaction/permeameter ring containing the compacted specimen and immediately mount the compaction/permeameter ring containing the test specimen in the permeameter. Determine the water content of the trimmings or excess material in accordance with

Test Method D2216 to the nearest 0.1 % or better. Determine the mass (*M*) of the test specimen to the tolerances given in 5.7 by subtracting the mass of the compaction/permeameter ring from the mass of the compaction/permeameter ring plus test specimen.

7.3 Calculate the total volume of the test specimen (*V_i*) to within 1 % or better from the length (*L_i*) and diameter (*D_i*) of the test specimen as follows:

$$V_i = \frac{\pi D_i^2 L_i}{4} \quad (1)$$

7.4 Calculate the initial dry density (ρ_{di}) of the compacted specimen to four significant digits and porosity (*n_i*) of the test specimen to three significant digits from the mass (*M*) and volume of the specimen (*V_i*) and from the approximate water content (*w*, in decimal form) of the specimen as follows:

$$\rho_{di} = \frac{M}{(1 + w)V_i} \quad (2)$$

and:

$$n_i = 1 - \frac{\rho_{di}}{G_s \rho_w} \quad (3)$$

where:

- V_i* = initial volume of the specimen, m³ or cm³,
- D_i* = initial diameter of the specimen or inside diameter of mold, m or cm,
- L_i* = initial length/height of the specimen, m or cm,
- ρ_{di} = initial dry density of the specimen, kg/m³ or g/cm³,
- M* = total mass of the specimen, g,
- w* = water content of the soil used in compacting the specimen, decimal format,
- n_i* = initial porosity of the specimen,
- G_s* = the specific gravity of solids, that shall either be measured in accordance with Test Method D854 or estimated, see Note 3, and
- ρ_w = density of water at 20°C, kg/m³ or g/cm³. This value versus temperature is given in Test Method D854 and is 998.2 kg/m³ or 0.9982 g/cm³.

NOTE 4—Accurate measurement of the specific gravity of solids of many materials is difficult. The specific gravity of solids shall be determined with an estimated accuracy of ±0.05. For many materials, it is possible to estimate *G_s* within this accuracy.

7.5 Calculate the volume of the pores in the test specimen (*V_p*) as follows:

$$V_p = n_i V_i \quad (4)$$

where:

- V_p* = volume of the pores in the specimen, m³ or cm³.

8. Procedure

8.1 Specimen Set-Up:

8.1.1 If filter paper is used to prevent clogging of porous end pieces, cut filter paper sheets to approximately the same shape as the cross section of the test specimen.

8.1.2 Soak any porous end pieces (discs) and filter paper sheets to be used in a container of permeant water and place the filter paper sheets on the porous end pieces.

8.1.3 Assemble the permeameter cell and fill it with de-aired permeant water. Flush permeant water through the influent line to remove any entrapped gas.

NOTE 5—Care should be exercised when assembling permeameters with a double ring in the base plate to ensure seating of the ring in the specimen. A restraining ring can be placed on top of the specimen during assembly to prevent slippage of the specimen as it is pressed into the ring. Alternatively, the specimen can be compacted directly on the base of the permeameter. If this latter procedure is used, then the base plate and ring should be sturdy enough to prevent breakage during compaction.

8.1.4 If a controlled stress is to be applied to the upper surface of a test specimen, apply it at this stage.

8.2 Permeation:

8.2.1 *Hydraulic Gradient*—When possible, the hydraulic gradient used for hydraulic conductivity measurements should be similar to that expected to occur in the field. In general, hydraulic gradients from <1 to 5 cover most field conditions. However, the use of small hydraulic gradients can lead to very long testing times for materials having low hydraulic conductivity (less than about 1×10^{-8} m/s). Somewhat larger hydraulic gradients are usually used in the laboratory to accelerate testing, but excessive gradients must be avoided because high seepage pressures may consolidate the material, material may be washed from the specimen, channeling along the sidewall may occur, or fine particles may be washed downstream and plug the effluent end of the test specimen. These effects could increase or decrease hydraulic conductivity. For high hydraulic gradients (>20), a base plate with a double ring (see Fig. 4) is recommended to avoid anomalous measurements caused by channeling along the sidewall. If no hydraulic gradient is specified by the requestor, the following guidelines may be followed:

Hydraulic Conductivity (m/s)	Recommended Maximum Hydraulic Gradient
1×10^{-5} to 1×10^{-6}	2
1×10^{-6} to 1×10^{-7}	5
1×10^{-7} to 1×10^{-8}	10
1×10^{-8} to 1×10^{-9}	20
$<1 \times 10^{-9}$	30

The constant flow rate method may be advantageous if low gradients and short testing times are desired for testing saturated materials with very low hydraulic conductivity.⁵

8.2.2 *Initialization*—Initiate permeation of the specimen by increasing the influent pressure. The pressure at the effluent end of the test specimen is normally atmospheric pressure (see Note 6). If restraint against upward movement of the specimen is provided, initially flowing the permeant upward may assist in saturation.

NOTE 6—Backpressure is normally not applied in this type of test. If backpressure is used, the flexible wall permeameter (see Test Method D5084) is recommended. If backpressure is specifically requested with this test method, the requestor shall specify the magnitude of the backpressure. If backpressure is used, use a double-ring effluent collector (see Fig. 5) or employ some other suitable methodology to ensure that the backpressure has not expanded the permeameter ring and caused spurious sidewall leakage.

8.2.3 *Constant Head Test (Test Method A)*—Measure and record the head loss across the test specimen and keep this head loss constant $\pm 5\%$. Measure and record periodically the quantity of inflow as well as the quantity of outflow. If a

double-ring effluent collector is used, record the quantity of outflow from both collector rings. Continue permeation until at least four values of hydraulic conductivity are obtained over an interval of time in which: the ratio of outflow to inflow rate is between 0.75 and 1.25, the hydraulic conductivity is steady, and if a double-ring collector is used, the ratio of the flux from the outer ring to that from the inner ring is between 0.75 and 1.25. Consider the hydraulic conductivity steady if four or more consecutive hydraulic conductivity determinations fall within $\pm 25\%$ of the mean value for $k \geq 1 \times 10^{-10}$ m/s or within $\pm 50\%$ for $k < 1 \times 10^{-10}$ m/s, and a plot of the hydraulic conductivity versus time shows no significant upward or downward trend. See Note 7.

NOTE 7—If the test specimen has been prepared using a dry, highly expandable material such as sodium bentonite or has very low hydraulic conductivity ($< 1 \times 10^{-10}$ m/s), hydration may not be complete even when the requirements in 8.2.3 are met. If incomplete hydration is of concern, a suitable method of soaking the sample can be conducted prior to permeation.

8.2.4 *Falling Head Tests (Test Methods B, C, and D)*—Measure and record the head loss across the test specimen. At no time shall the applied head loss across the specimen be less than 75 % of the initial (maximum) head loss during each individual hydraulic conductivity determination. Continue permeation until at least four values of hydraulic conductivity are obtained over an interval of time in which: the ratio of outflow to inflow rate is between 0.75 and 1.25, the hydraulic conductivity is steady (see 8.2.3), and if a double-ring collector is used, the ratio of the flux from the outer ring to that from the inner ring is between 0.75 and 1.25. See Note 7, Note 8, and Note 9.

NOTE 8—When the water pressure in a test specimen changes, the volume of the pore fluid (especially gas bubbles) can change and the test specimen itself may expand or contract. The requirement that the head loss not decrease very much during a falling head test is intended to minimize these potential sources of volume change.

NOTE 9—When the initial and final head losses across the test specimen do not differ by much, great accuracy is needed to comply with the requirement of 5.1.2 that the ratio of initial to final head loss be determined with an accuracy of $\pm 5\%$ or better. When the initial and final head losses over an interval of time do not differ very much, it may be possible to comply with the requirements of 8.2.3 and to treat the test as a constant head test.

8.2.4.1 *Test with Constant Tailwater Level (Test Method B)*—If the water pressure at the downstream (tailwater) end of the test specimen is kept constant, periodically measure and record either the quantity of inflow or the level of water in the influent standpipe; measure and record the quantity of outflow from the test specimen.

8.2.4.2 *Test with Constant Headwater and Rising Tailwater Level (Test Method C)*—If the water pressure at the upstream (headwater) end of the test specimen is kept constant, periodically measure and record either the quantity of outflow or the level of water in the effluent standpipe; measure and record the quantity of inflow into the test specimen. A marriott bottle is a convenient device to measure the quantity of inflow while maintaining a constant head.

8.2.4.3 *Test with Decreasing Headwater Level and Increasing Tailwater Level (Test Method D)*—If the water pressure at the downstream end of the test specimen rises during an

interval of time, periodically measure and record either the quantity of inflow and outflow or the changes in water levels in the influent and effluent standpipes.

8.2.5 Constant Rate of Flow Tests (Test Method E)—Initiate permeation of the specimen by imposing a constant flow rate. Choose the flow rate so the hydraulic gradient does not exceed the value specified, or if none is specified, the value recommended in 8.2.1. Periodically measure the head loss across the test specimen. Continue permeation until at least four values of hydraulic conductivity are obtained over an interval of time in which hydraulic conductivity is steady (see 8.2.3) and, if a double-ring collector is used, the ratio of the flux from the outer ring to that from the inner ring is between 0.75 and 1.25 (see Note 7). The user shall also ensure that the system is devoid of leaks and that rate of inflow and outflow are similar. One method to verify these conditions is to periodically compare the rate of inflow and outflow and determine if the ratio of inflow to outflow rates is between 0.75 and 1.25. Other suitable methods also can be used.

NOTE 10—Volume measurements should be made after sufficient time has elapsed to permit flow in the measurement reservoirs that is at least ten times the finest graduation on the reservoir. Measurements may be made at shorter time intervals only if it can be ensured that accurate measurements of rate of flow or change in head can be made, or both.

8.3 Final Dimensions of the Specimen—After completion of permeation, reduce the applied pressures, carefully disassemble the permeameter cell, and examine the specimen. If the specimen is free to swell, measure the height of the specimen. If the length of the specimen is more than 1.15 times the original length, trim the specimen and repeat the hydraulic conductivity measurement (see Note 11). Take care when trimming the specimen not to smear material across the surface of the test specimen. After trimming, scarify the surface of the specimen. Once the test is complete, measure and record the final height and diameter to the tolerances given in 5.6, and record the total mass of the specimen to the tolerances given in 5.7. Then determine the final water content of the specimen by the procedure of Test Method D2216 to the nearest 0.1 % or better.

NOTE 11—If the specimen swells into a swell ring, the boundary conditions at the influent end are uncertain. This problem is considered negligible if the amount of swell is less than or equal to 15 % of the original length of the specimen. If the swell is more than 15 %, the material that swelled into the swell ring must be trimmed away. If it is known prior to permeation that a large amount of swell will occur, the specimen may be trimmed at some intermediate point in the test, that is, it is not necessary to wait until all the requirements of Section 8 are satisfied prior to dismantling the cell and trimming away the material that swelled into the swell ring. However, all requirements must be met before a test is complete. Even if the height of the specimen is ≤ 1.15 times the original height, measure the entire length of specimen and then trim the specimen prior to determining the final length, mass, and water content (the final length, mass, and water content are to be determined on that portion of the test specimen contained within the compaction mold/permeameter ring). Use the length measured prior to trimming for the calculation of hydraulic conductivity.

NOTE 12—For some gap-graded materials, migration of particles may affect the measured hydraulic conductivity. Reversing the direction of flow may be useful in assessing the effect of particle migration.

9. Calculation

9.1 Constant Head and Constant Rate of Flow Tests (Test Methods A and E)—Calculate the hydraulic conductivity, k , as follows:

$$k = \frac{\Delta Q L_f}{A \Delta t \Delta h} \quad (5)$$

where:

- k = hydraulic conductivity, m/s or cm/s,
- ΔQ = quantity of flow, taken as the average of inflow and outflow, m^3 or cm^3 ,
- L_f = final length of specimen along path of flow, m or cm,
- A = cross-sectional area of specimen, m^2 or cm^2 ,
- Δt = interval of time, s, over which the flow ΔQ occurs, and
- Δh = difference in hydraulic head across the specimen, m or cm of water.

NOTE 13—If a double-ring effluent collector is used, the outflow quantity shall be taken as the total quantity from both rings.

9.2 Falling Head Tests:

9.2.1 Constant Tailwater Pressure (Test Method B) and Constant Headwater and Rising Tailwater (Test Method C)—Calculate the hydraulic conductivity, k , as follows:

$$k = \frac{a L_f}{A \Delta t} \ln \left(\frac{h_1}{h_2} \right) \quad (6)$$

where:

- a = cross-sectional area of the reservoir containing the influent liquid, m^2 or cm^2 ,
- L_f = final length of the specimen, m or cm,
- A = cross-sectional area of the specimen, m^2 or cm^2 ,
- Δt = elapsed time between determination of h_1 and h_2 , s,
- h_1 = head loss across the specimen, at time t_1 , m or cm, and
- h_2 = head loss across the specimen at time t_2 , m or cm.

NOTE 14—Test Method C only shall be used for tests conducted in a single ring permeameter (see Fig. 1 and Fig. 2) unless the set-up for a double ring test (see Fig. 4) ensures the tailwater elevations in the effluent reservoirs remain equal throughout the test. Provided the elevations remain equal, then $a = a_1 + a_2$.

where:

- a_1 = cross-sectional area of the reservoir containing the effluent liquid from the inner ring, m^2 or cm^2 and
- a_2 = cross-sectional area of the reservoir containing the effluent liquid from the outer ring, m^2 or cm^2 .

9.2.2 Decreasing Headwater and Increasing Tailwater Pressure (Test Method D)—Calculate the hydraulic conductivity, k , as follows:

$$k = \frac{a_{in} a_{out} L_f}{A \Delta t (a_{in} + a_{out})} \ln(h_1/h_2) \quad (7)$$

where:

- a_{in} = cross-sectional area of the reservoir containing the influent liquid, m^2 or cm^2 ,
- a_{out} = cross-sectional area of the reservoir containing the effluent liquid, m^2 or cm^2 ,
- L_f = final length of the specimen, m or cm,
- A = cross-sectional area of the specimen, m^2 or cm^2 ,
- Δt = elapsed time between determination of h_1 and h_2 , s,
- h_1 = head loss across the specimen at time t_1 , m or cm, and
- h_2 = head loss across the specimen at time t_2 , m or cm.

NOTE 15—For the case in Test Method D where $a_{out} = a_{in} = a$, the

equation for calculating k for a falling head test with a rising tailwater level is:

$$k = \frac{a L}{2 A \Delta t} \ln \left(\frac{h_1}{h_2} \right) \quad (8)$$

Also, Test Method D only shall be used for tests conducted in a single ring permeameter (see Fig. 1 and Fig. 2) unless the set-up for a double ring test (see Fig. 4) ensures the tailwater elevations in the effluent reservoirs remain equal throughout the test. Provided the elevations remain equal, then $a_{\text{out}} = a_1 + a_2$

where:

- a_1 = cross-sectional area of the reservoir containing the effluent liquid from the inner ring, m^2 or cm^2 and
- a_2 = cross-sectional area of the reservoir containing the effluent liquid from the outer ring, m^2 or cm^2 .

9.3 Hydraulic Conductivity at Standard Temperature—The measured hydraulic conductivity shall be corrected to that for 20°C , k_{20} , by multiplying the measured k by the ratio of the viscosity of liquid at test temperature to the viscosity of liquid at 20°C , R_T :

$$k_{20} = R_T \cdot k \quad (9)$$

where:

$$R_T = \frac{2.2902 \cdot 0.9842^T}{T^{0.1702}} \quad (10)$$

and:

- k_{20} = hydraulic conductivity at standard temperature, 20°C , (m/s or cm/s),
- R_T = ratio of the viscosity of liquid at test temperature to the viscosity of liquid at 20°C ,
- T = average test temperature $((T_1 + T_2)/2)$ during the permeation trial ($^\circ\text{C}$),
- T_1 = test temperature at start of permeation trial ($^\circ\text{C}$), and
- T_2 = test temperature at end of permeation trial ($^\circ\text{C}$).

9.3.1 The equation for R_T is only accurate to three significant digits between 5 and 50°C .

9.3.2 If the number of significant digits in the calculation of hydraulic conductivity can be one, then the test temperature can be measured to the nearest $^\circ\text{C}$.

9.4 Let M_s denote the oven dried mass (in kg or g) of material comprising the test specimen, V_f the final volume of the test specimen contained in the compaction mold/permeameter ring (in m^3 or cm^3), w_f (dimensionless) the final water content of the test specimen, and G_s the dimensionless specific gravity of solids (see Note 5). Determine the final dry density (ρ_{df} in kg/m^3 or g/cm^3) of the test specimen as follows:

$$\rho_{df} = \frac{M_s}{V_f} \quad (11)$$

and the final degree of saturation (S , expressed as a percentage):

$$S = \frac{w}{\frac{\rho_w}{\rho_{df}} - \frac{1}{G_s}} \times 100 \quad (12)$$

where:

- ρ_{df} = dry density of specimen after testing, kg/m^3 or g/cm^3 ,
- M_s = initial or final dry mass of the specimen, kg or g , and
- V_f = final volume of the specimen after testing, m^3 or cm^3 .

9.5 Determine the number of pore volumes of inflow (N_{PV}) at the end of the test as follows:

$$N_{PV} = \frac{\Sigma \Delta Q_{in}}{V_p} \quad (13)$$

where:

$\Sigma \Delta Q_{in}$ = cumulative quantity of inflow from the test specimen and V_p is the initial volume of the pore space within the test specimen (see 7.5).

10. Report: Test Data Sheet(s)/Form(s)

10.1 Record as a minimum the following general information (data):

10.1.1 Sample/specimen identifying information, such as Project Designation, Boring No., Sample No., Depth, Name of person conducting the test, etc.

10.1.2 Any special selection and preparation process, such as removal of stones or other materials, and

10.1.3 Descriptive information on method of compaction.

10.2 Record as a minimum the following test specimen data:

10.2.1 The initial dimensions of the specimen, and

10.2.2 The initial water content and dry density of the specimen.

10.3 Record as a minimum the following test data:

10.3.1 The type of permeant liquid used,

10.3.2 Type of permeameter used, including type (if any) of confinement against swelling, single- or double-ring base plate, and any non-standard test arrangement such as use of backpressure,

10.3.3 The test method (A, B, C, D or E) used,

10.3.4 Hydraulic gradient (report the range, if variable),

10.3.5 The total number of pore volumes of permeant liquid that came out of the test specimen and the duration of the test, measured from initiation of permeation to the end of the last set of readings used to determine hydraulic conductivity,

10.3.6 The final length, diameter, water content, dry density, and degree of saturation of the test specimen. Also report the specific gravity of solids used for computations and indicate whether this value was measured or assumed,

10.3.7 The amount of swelling that was measured and whether or not material was trimmed away from the swell ring (if applicable) prior to permeation,

10.4 Record as a minimum the following permeation data:

10.4.1 The date, time (or start and elapsed time), temperature (see 9.3), head loss reading(s), flow reading(s) (if applicable), and deformation gauge (if applicable) at the start and end of each trial/determination. Applicable measurements/readings and any averages/differences calculated using measurements/readings obtained shall have two or more significant digits, unless specified differently in Section 9.

10.4.2 The average hydraulic conductivity for the last four determinations of hydraulic conductivity, reported with two significant figures, for example, $7.1 \times 10^{-10} \text{ m/s}$, and reported in units of m/s (plus additional units, if requested or customary),

10.4.3 A graph or table of hydraulic conductivity versus time or pore volumes of flow and a graph or table of the ratio of outflow to inflow flow rate versus time or pore volume of flow, and

10.4.4 If a double-ring effluent collector was used, report a graph or table of the ratio of flux from the outer ring to that from the inner ring versus time or pore volumes of flow.

11. Precision and Bias

11.1 *Precision*—Data are being evaluated to determine the precision of this test method. In addition, Subcommittee

D18.04 on Hydrologic Properties of Soil and Rocks, is seeking pertinent data from users of the test method.

11.2 *Bias*—There is no accepted reference value for this test method; therefore, bias cannot be determined.

12. Keywords

12.1 coefficient of permeability; groundwater; hydraulic barriers; hydraulic conductivity; liner; permeameter

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Designation: D5084 – 24

Standard Test Methods for Measurement of Hydraulic Conductivity of Saturated Porous Materials Using a Flexible Wall Permeameter¹

This standard is issued under the fixed designation D5084; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ϵ) indicates an editorial change since the last revision or reapproval.

1. Scope*

1.1 These test methods cover laboratory measurement of the hydraulic conductivity (also referred to as *coefficient of permeability*) of water-saturated porous materials with a flexible wall permeameter at temperatures between about 15 and 30°C (59 and 86°F). Temperatures outside this range may be used; however, the user would have to determine the specific gravity of mercury and R_T (see 10.3) at those temperatures using data from *Handbook of Chemistry and Physics*. There are six alternate methods or hydraulic systems that may be used to measure the hydraulic conductivity. These hydraulic systems are as follows:

- 1.1.1 *Method A*—Constant Head
- 1.1.2 *Method B*—Falling Head, constant tailwater elevation
- 1.1.3 *Method C*—Falling Head, rising tailwater elevation
- 1.1.4 *Method D*—Constant Rate of Flow
- 1.1.5 *Method E*—Constant Volume—Constant Head (by mercury)
- 1.1.6 *Method F*—Constant Volume—Falling Head (by mercury), rising tailwater elevation

1.2 These test methods use water as the permeant liquid; see 4.3 and Section 6 on Reagents for water requirements.

1.3 These test methods may be utilized on all specimen types (intact, reconstituted, remolded, compacted, etc.) that have a hydraulic conductivity less than about 1×10^{-6} m/s (1×10^{-4} cm/s), providing the head loss requirements of 5.2.3 are met. For the constant-volume methods, the hydraulic conductivity typically has to be less than about 1×10^{-7} m/s.

1.3.1 If the hydraulic conductivity is greater than about 1×10^{-6} m/s, but not more than about 1×10^{-5} m/s; then the size of the hydraulic tubing needs to be increased along with the porosity of the porous end pieces. Other strategies, such as using higher viscosity fluid or properly decreasing the cross-sectional area of the test specimen, or both, may also be

possible. The key criterion is that the requirements covered in Section 5 have to be met.

1.3.2 If the hydraulic conductivity is less than about 1×10^{-11} m/s, then standard hydraulic systems and temperature environments will typically not suffice. Strategies that may be possible when dealing with such impervious materials may include the following: (a) controlling the temperature more precisely, (b) adoption of unsteady state measurements by using high-accuracy equipment along with the rigorous analyses for determining the hydraulic parameters (this approach reduces testing duration according to Zhang et al. (1)²), and (c) shortening the length or enlarging the cross-sectional area, or both, of the test specimen (with consideration to specimen grain size (2)). Other approaches, such as use of higher hydraulic gradients, lower viscosity fluid, elimination of any possible chemical gradients and bacterial growth, and strict verification of leakage, may also be considered.

1.4 The hydraulic conductivity of materials with hydraulic conductivities greater than 1×10^{-5} m/s may be determined by Test Method D2434.

1.5 All observed and calculated values shall conform to the guide for significant digits and rounding established in Practice D6026.

1.5.1 The procedures used to specify how data are collected, recorded, and calculated in this standard are regarded as the industry standard. In addition, they are representative of the significant digits that should generally be retained. The procedures used do not consider material variation, purpose for obtaining the data, special purpose studies, or any considerations for the user's objectives; and it is common practice to increase or reduce significant digits of reported data to be commensurate with these considerations. It is beyond the scope of this standard to consider significant digits used in analysis methods for engineering design.

1.6 This standard also contains a Hazards section (Section 7).

1.7 The time to perform this test depends on such items as the Method (A, B, C, D, E, or F) used, the initial degree of

¹ This standard is under the jurisdiction of ASTM Committee D18 on Soil and Rock and is the direct responsibility of Subcommittee D18.04 on Hydrologic Properties and Hydraulic Barriers.

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² The boldface numbers in parentheses refer to the list of references appended to this standard.

*A Summary of Changes section appears at the end of this standard

saturation of the test specimen and the hydraulic conductivity of the test specimen. The constant volume Methods (E and F) and Method D require the shortest period-of-time. Typically a test can be performed using Methods D, E, or F within two to three days. Methods A, B, and C take a longer period-of-time, from a few days to a few weeks depending on the hydraulic conductivity. Typically, about one week is required for hydraulic conductivities on the order of 1×10^{-9} m/s. The testing time is ultimately controlled by meeting the equilibrium criteria for each Method (see 9.5).

1.8 Units—The values stated in SI units are to be regarded as the standard. The inch-pound units given in parentheses are mathematical conversions, which are provided for information purposes only and are not considered standard, unless specifically stated as standard, such as 0.5 mm or 0.01 in.

1.9 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.10 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:³

- D653** Terminology Relating to Soil, Rock, and Contained Fluids
- D698** Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft³ (600 kN-m/m³))
- D854** Test Methods for Specific Gravity of Soil Solids by the Water Displacement Method
- D1140** Test Methods for Determining the Amount of Material Finer than 75- μ m (No. 200) Sieve in Soils by Washing
- D1557** Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft³ (2,700 kN-m/m³))
- D1587/D1587M** Practice for Thin-Walled Tube Sampling of Fine-Grained Soils for Geotechnical Purposes (Withdrawn 2024)⁴
- D2113** Practice for Rock Core Drilling and Sampling of Rock for Site Exploration (Withdrawn 2023)⁴
- D2216** Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
- D2434** Test Methods for Measurement of Hydraulic Conductivity of Coarse-Grained Soils
- D2435/D2435M** Test Methods for One-Dimensional Con-

- solidation Properties of Soils Using Incremental Loading
- D3550/D3550M** Practice for Thick Wall, Ring-Lined, Split Barrel, Drive Sampling of Soils
- D3740** Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
- D4220/D4220M** Practices for Preserving and Transporting Soil Samples (Withdrawn 2023)⁴
- D4318** Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
- D4753** Guide for Evaluating, Selecting, and Specifying Balances and Standard Masses for Use in Soil, Rock, and Construction Materials Testing
- D4767** Test Method for Consolidated Undrained Triaxial Compression Test for Cohesive Soils
- D5079** Practices for Preserving and Transporting Rock Core Samples (Withdrawn 2017)⁴
- D6026** Practice for Using Significant Digits and Data Records in Geotechnical Data
- D6151/D6151M** Practice for Using Hollow-Stem Augers for Geotechnical Exploration and Soil Sampling (Withdrawn 2024)⁴
- D6169/D6169M** Guide for Selection of Subsurface Soil and Rock Sampling Devices for Environmental and Geotechnical Investigations
- E177** Practice for Use of the Terms Precision and Bias in ASTM Test Methods
- E691** Practice for Conducting an Interlaboratory Study to Determine the Precision of a Test Method

3. Terminology

3.1 Definitions:

3.1.1 For common definitions of technical terms in this standard, refer to Terminology **D653**.

3.1.2 *head loss, Δh* —the change in total head of water across a given distance.

3.1.2.1 *Discussion*—In hydraulic conductivity testing, typically the change in total head is across the influent and effluent lines connected to the permeameter, while the given distance is typically the length of the test specimen.

3.1.3 *permeameter*—the apparatus (cell) containing the test specimen in a hydraulic conductivity test.

3.1.3.1 *Discussion*—The apparatus in this case is typically a triaxial-type cell with all of its components (top and bottom specimen caps, stones, and filter paper; membrane; chamber; top and bottom plates; valves; etc.).

3.1.4 *hydraulic conductivity, k* —the rate of discharge of water under laminar flow conditions through a unit cross-sectional area of porous medium under a unit hydraulic gradient and standard temperature conditions (20°C).

3.1.4.1 *Discussion*—In hydraulic conductivity testing, the term *coefficient of permeability* is often used instead of *hydraulic conductivity*, but *hydraulic conductivity* is used exclusively in this standard. A more complete discussion of the terminology associated with Darcy's law is given in the literature. (**3**, **4**)

³ For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at www.astm.org/contact. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

⁴ The last approved version of this historical standard is referenced on www.astm.org.

3.1.5 *pore volume of flow*—in hydraulic conductivity testing, the cumulative quantity of flow into a test specimen divided by the volume of voids in the specimen.

4. Significance and Use

4.1 These test methods apply to one-dimensional, laminar flow of water within porous materials such as soil and rock.

4.2 The hydraulic conductivity of porous materials generally decreases with an increasing amount of air in the pores of the material. These test methods apply to water-saturated porous materials containing virtually no air.

4.3 These test methods apply to permeation of porous materials with water. Permeation with other liquids, such as chemical wastes, can be accomplished using procedures similar to those described in these test methods. However, these test methods are only intended to be used when water is the permeant liquid. See Section 6.

4.4 Darcy's law is assumed to be valid and the hydraulic conductivity is essentially unaffected by hydraulic gradient.

4.5 These test methods provide a means for determining hydraulic conductivity at a controlled level of effective stress. Hydraulic conductivity varies with varying void ratio, which changes when the effective stress changes. If the void ratio is changed, the hydraulic conductivity of the test specimen will likely change, see Appendix X2. To determine the relationship between hydraulic conductivity and void ratio, the hydraulic conductivity test would have to be repeated at different effective stresses.

4.6 The correlation between results obtained using these test methods and the hydraulic conductivities of in-place field materials has not been fully investigated. Experience has sometimes shown that hydraulic conductivities measured on small test specimens are not necessarily the same as larger-scale values. Therefore, the results should be applied to field situations with caution and by qualified personnel.

4.7 In most cases, when testing high swell potential materials and using a constant-volume hydraulic system, the effective confining stress should be about 1.5 times the swell pressure of the test specimen or a stress which prevents swelling. If the confining stress is less than the swell pressure, anomalous flow conditions may occur; for example, mercury column(s) move in the wrong direction.

NOTE 1—The quality of the result produced by this standard is dependent on the competence of the personnel performing it and the suitability of the equipment and facility used. Agencies that meet the criteria of Practice D3740 are generally considered capable of competent and objective testing/sampling/inspection. Reliable results depend on many factors; Practice D3740 provides a means of evaluating some, but not all, of those factors.

5. Apparatus

5.1 *Hydraulic System*—Constant head (Method A), falling head (Methods B and C), constant rate of flow (Method D), constant volume-constant head (Method E), or constant volume-falling head (Method F) systems may be utilized provided they meet the following criteria:

5.1.1 *Constant Head*—The system must be capable of maintaining constant hydraulic pressures to $\pm 5\%$ or better and

shall include means to measure the hydraulic pressures to within the prescribed tolerance. In addition, the head loss across the permeameter must be held constant to $\pm 5\%$ or better and shall be measured with the same accuracy or better. A pressure gage, electronic pressure transducer, or any other device of suitable accuracy shall measure pressures to a minimum of three significant digits. The last digit may be due to estimation, see 5.1.1.1.

5.1.1.1 Practice D6026 discusses the use or application of estimated digits. When the last digit is estimated and that reading is a function of the eye's elevation/location, then a mirror or another device is required to reduce the reading error caused by parallax.

5.1.2 *Falling Head*—The system shall allow for measurement of the applied head loss, thus hydraulic gradient, to $\pm 5\%$ or better at any time. In addition, the ratio of initial head loss divided by final head loss over an interval of time shall be measured such that this computed ratio is accurate to $\pm 5\%$ or better. The head loss shall be measured with a pressure gage, electronic pressure transducer, engineer's scale, graduated pipette, or any other device of suitable accuracy to a minimum of three significant digits. The last digit may be due to estimation, see 5.1.1.1. Falling head tests may be performed with either a constant tailwater elevation (Method B) or a rising tailwater elevation (Method C), see Fig. 1. This schematic of a hydraulic system presents the basic components needed to meet the objectives of Method C. Other hydraulic systems or schematics that meet these objectives are acceptable.

5.1.3 *Constant Rate of Flow*—The system must be capable of maintaining a constant rate of flow through the specimen to $\pm 5\%$ or better. Flow measurement shall be by calibrated syringe, graduated pipette, or other device of suitable accuracy. The head loss across the permeameter shall be measured to a minimum of three significant digits and to an accuracy of $\pm 5\%$ or better using an electronic pressure transducer(s) or other device(s) of suitable accuracy. The last digit may be due to estimation, see 5.1.1.1. More information on testing with a constant rate of flow is given in the literature (5).

5.1.4 *Constant Volume-Constant Head (CVCH)*—The system, with mercury to create the head loss, must be capable of maintaining a constant head loss across the permeameter to $\pm 5\%$ or better and shall allow for measurement of the applied head loss to $\pm 5\%$ or better at any time. The head loss shall be measured to a minimum of three significant digits with an electronic pressure transducer(s) or equivalent device, (6) or based upon the pressure head caused by the mercury column, see 10.1.2. The last digit may be due to estimation, see 5.1.1.1.

5.1.4.1 Schematics of two CVCH systems are shown in Fig. 2 and Fig. 3. In each of these systems, the mercury-filled portion of the tubing may be continuous for constant head loss to be maintained. For the system showed in Fig. 2, the head loss remains constant provided the mercury column is vertical and is retained in only one half of the burette system (left burette in Fig. 2). If the mercury spans both columns, a falling head exists. In the system shown in Fig. 3, the head loss remains constant provided the water-mercury interface on the effluent end remains in the upper horizontal tube, and the water-mercury interface on the influent end remains in the

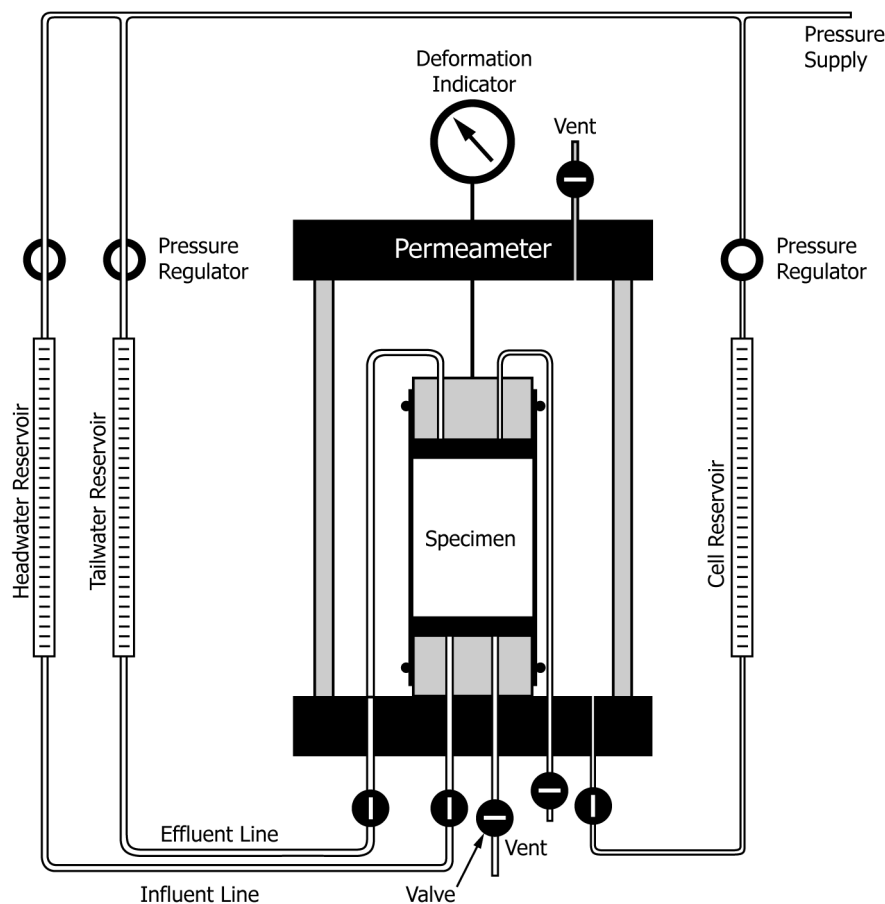


FIG. 1 Falling Head – Rising Tail System, Method C

lower horizontal tube. These schematics present the basic components needed to meet the objectives of Method E. Other hydraulic systems or schematics that meet these objectives are acceptable.

5.1.4.2 These types of hydraulic systems are typically not used to study the temporal or pore-fluid effect on hydraulic conductivity. The total volume of the specimen is maintained constant using this procedure, thereby significantly reducing effects caused by seepage stresses, pore fluid interactions, etc. Rather, these systems are intended for determining the hydraulic conductivity of a material as rapidly as possible.

5.1.4.3 *Hazards*—Since this hydraulic system contains mercury, special health and safety precautions have to be considered. See Section 7.

5.1.4.4 *Caution*—For these types of hydraulic systems to function properly, the separation of the mercury column has to be prevented. To prevent separation, the mercury and “constant head” tube have to remain relatively clean, and the inside diameter of this tube cannot be too large; typically a capillary tube is used. The larger diameter flushing tube (Fig. 2) is added to enable flushing clean water through the system without excessive mercury displacement. Traps to prevent the accidental flow of mercury out of the “Constant Head” tube or flushing tube are not shown in Fig. 2 and Fig. 3.

5.1.5 *Constant Volume-Falling Head (CVFH)*—The system, with mercury to create the head loss, shall meet the criteria given in 5.1.2. The head loss shall be measured to a minimum

of three significant digits with an electronic pressure transducer(s) or equivalent device(s), (6) or based upon the differential elevation between the top surfaces of the mercury level in the headwater and tailwater tubes. The last digit may be due to estimation, see 5.1.1.1.

5.1.5.1 A schematic drawing of a typical CVFH hydraulic system is shown in Fig. 4 (6). Typically, the tailwater tube has a smaller area than the headwater tube to increase the sensitivity of flow measurements, and to enable flushing clean water through the system without excessive mercury displacement in the headwater tube. The schematic of the hydraulic system in Fig. 4 presents the basic components needed to meet the objectives of Method F. Other hydraulic systems or schematics that meet these objectives are acceptable. The development of the hydraulic conductivity equation for this type of system is given in Appendix X1.

5.1.5.2 See 5.1.4.2.

5.1.5.3 *Hazards*—Since this hydraulic system contains mercury, special health and safety precautions have to be considered. See Section 7.

5.1.5.4 *Caution*—For these types of hydraulic systems to function properly, the separation of the mercury column and entrapment of water within the mercury column have to be prevented. To prevent such problems, the mercury and tubes have to remain relatively clean. In addition, if different size headwater and tailwater tubes are used, capillary head might have to be accounted for, see Appendix X1, X1.2.3.2, and

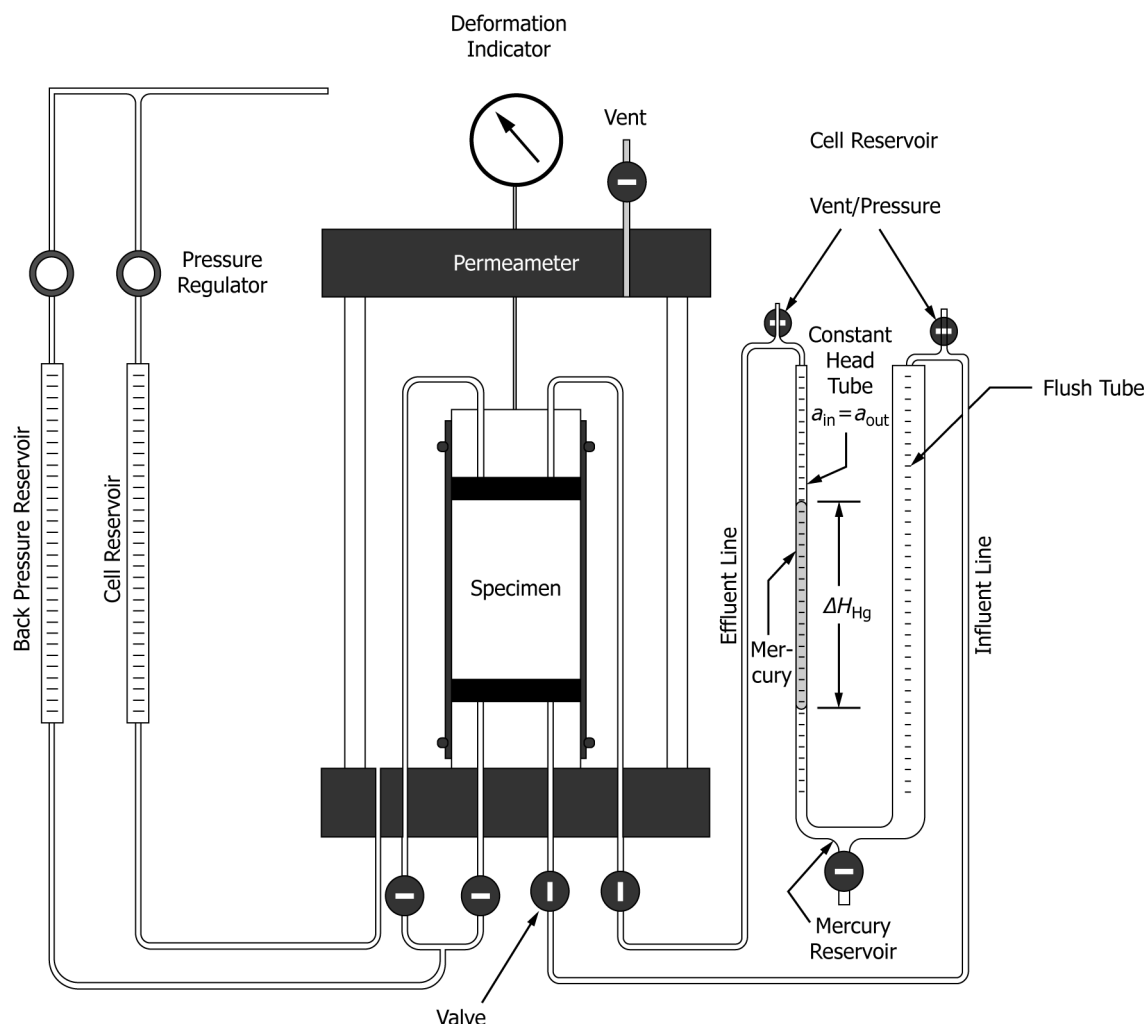


FIG. 2 Constant Volume – Constant or Falling Head System, Method E or F (6)

X1.4. Traps to prevent the accidental flow of mercury out of the tubes are not shown in Fig. 4.

5.1.6 System De-airing—The hydraulic system shall be designed to facilitate rapid and complete removal of free air bubbles from flow lines; for example, using properly sized tubing and ball valves and fittings without pipe threads. Properly sized tubing, etc., means they are small enough to prevent entrapment of air bubbles, but not so small that the requirements of 5.2.3 cannot be met.

5.1.7 Back Pressure System—The hydraulic system shall have the capability to apply back pressure to the specimen to facilitate saturation. The system shall be capable of maintaining the applied back pressure throughout the duration of hydraulic conductivity measurements. The back pressure system shall be capable of applying, controlling, and measuring the back pressure to $\pm 5\%$ or better of the applied pressure. The back pressure may be provided by a compressed gas supply, a deadweight acting on a piston, or any other method capable of applying and controlling the back pressure to the tolerance prescribed in this paragraph.

NOTE 2—Application of gas pressure directly to a fluid will dissolve gas in the fluid. A variety of techniques are available to minimize dissolution of gas in the back pressure fluid, including separation of gas and liquid phases with a bladder and frequent replacement of the liquid with de-aired water.

5.2 Flow Measurement System—Both inflow and outflow volumes shall be measured unless the lack of leakage, continuity of flow, and cessation of consolidation or swelling can be verified by other means. Flow volumes shall be measured by a graduated accumulator, graduated pipette, vertical standpipe in conjunction with an electronic pressure transducer, or other volume-measuring device of suitable accuracy.

5.2.1 Flow Accuracy—Required accuracy for the quantity of flow measured over an interval of time is $\pm 5\%$ or better.

5.2.2 De-airing and Compliance of the System—The flow-measurement system shall contain a minimum of dead space and be capable of complete and rapid de-airing. Compliance of the system in response to changes in pressure shall be

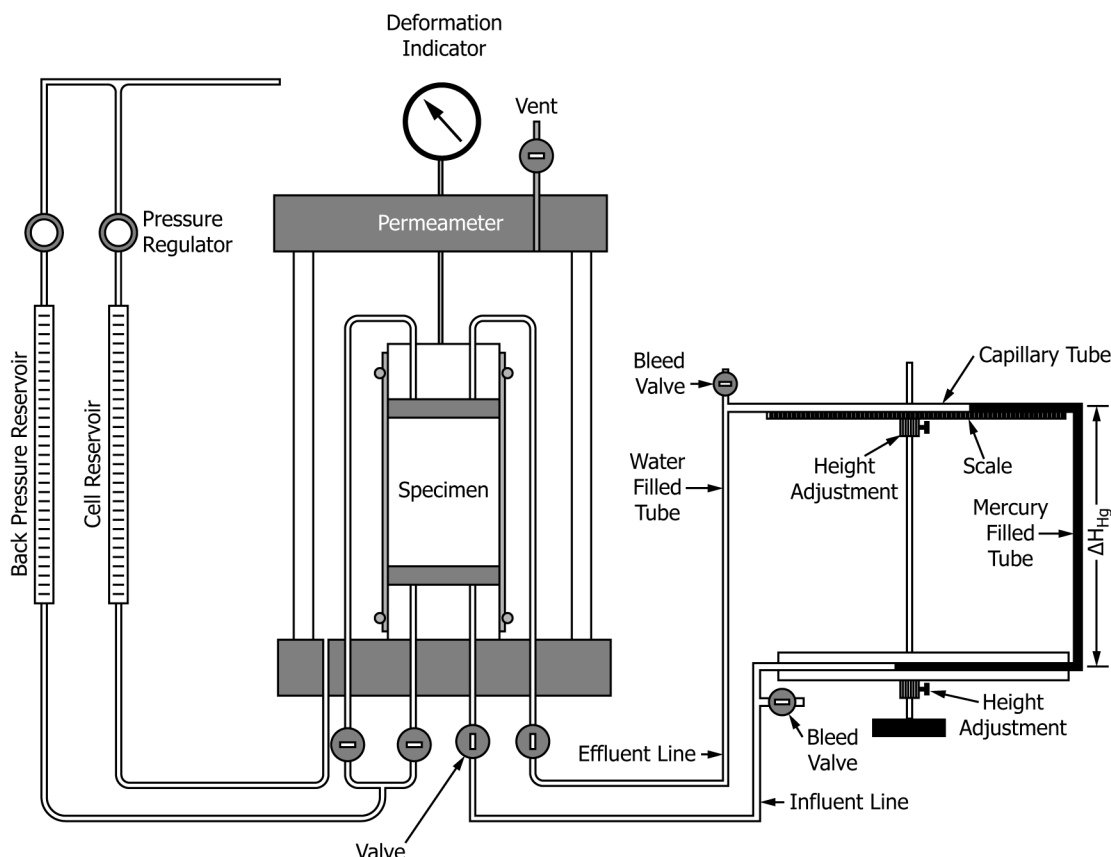


FIG. 3 Constant Volume—Constant Head System, Method E

minimized by using a stiff flow measurement system. Rigid tubing, such as metallic or rigid thermoplastic tubing, or glass shall be used.

5.2.3 Head Losses—Head losses in the tubes, valves, porous end pieces, and filter paper may lead to error. To guard against such errors, the permeameter shall be assembled with no specimen inside and then the hydraulic system filled.

5.2.3.1 Constant or Falling Head—If a constant or falling head test is to be used, the hydraulic pressures or heads that will be used in testing a specimen shall be applied, and the rate of flow measured with an accuracy of $\pm 5\%$ or better. This rate of flow shall be at least ten times greater than the rate of flow that is measured when a specimen is placed inside the permeameter and the same hydraulic pressures or heads are applied.

5.2.3.2 Constant Rate of Flow—If a constant rate of flow test is to be used, the rate of flow to be used in testing a specimen shall be supplied to the permeameter and the head loss measured. The head loss without a specimen shall be less than 0.1 times the head loss when a specimen is present.

5.3 Permeameter Cell Pressure System—The system for pressurizing the permeameter cell shall be capable of applying and controlling the cell pressure to $\pm 5\%$ or better of the applied pressure. However, the effective stress on the test specimen (which is the difference between the cell pressure and the pore water pressure) shall be maintained to the desired value with an accuracy of $\pm 10\%$ or better. The device for pressurizing the cell may consist of a reservoir connected to the

permeameter cell and partially filled with de-aired water, with the upper part of the reservoir connected to a compressed gas supply or other source of pressure (see [Note 3](#)). The gas pressure shall be controlled by a pressure regulator and measured by a pressure gage, electronic pressure transducer, or any other device capable of measuring to the prescribed tolerance. A hydraulic system pressurized by deadweight acting on a piston or any other pressure device capable of applying and controlling the permeameter cell pressure within the tolerance prescribed in this paragraph may be used.

NOTE 3—De-aired water is commonly used for the cell fluid to minimize potential for diffusion of air through the membrane into the specimen. Other fluids that have low gas solubilities such as oils, are also acceptable, provided they do not react with components of the permeameter. Also, use of a long (approximately 5 to 7 m) tube connecting the pressurized cell liquid to the cell helps to delay the appearance of air in the cell fluid and to reduce the flow of dissolved air into the cell.

5.4 Permeameter Cell—An apparatus shall be provided in which the specimen and porous end pieces, enclosed by a membrane sealed to the cap and base, are subjected to controlled fluid pressures. A schematic diagram of a typical permeameter cell and falling head (raising tailwater) hydraulic system is shown in [Fig. 1](#).

5.4.1 The permeameter cell may allow for observation of changes in height of the specimen, either by observation through the cell wall using a cathetometer or other instrument, or by monitoring of either a loading piston or an extensometer extending through the top plate of the cell bearing on the top

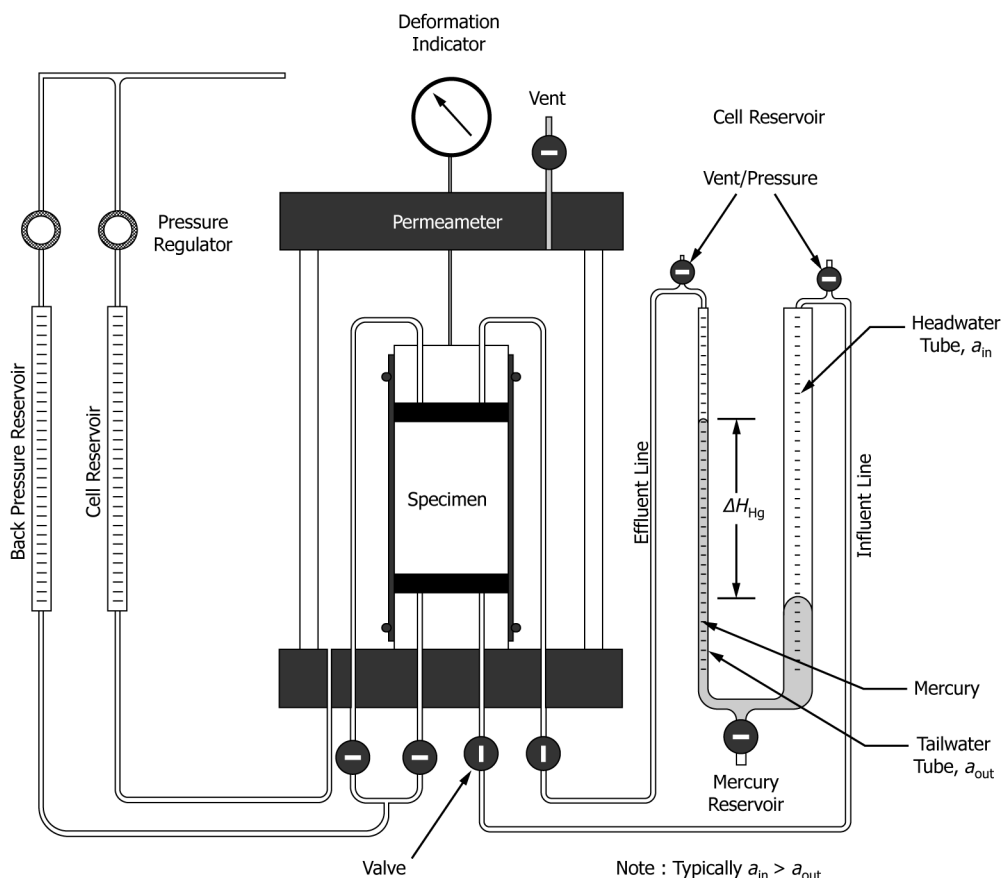


FIG. 4 Constant Volume – Falling Head System, Method F (6)

cap and attached to a dial indicator or other measuring device. The piston or extensometer should pass through a bushing and seal incorporated into the top plate and shall be loaded with sufficient force to compensate for the cell pressure acting over the cross-sectional area of the piston where it passes through the seal. If deformations are measured, the deformation indicator shall be a dial indicator or cathetometer graduated to 0.5 mm or 0.01 in. or better and having an adequate travel range. Any other measuring device meeting these requirements is acceptable.

5.4.2 In order to facilitate gas removal, and thus saturation of the hydraulic system, four drainage lines leading to the specimen, two each to the base and top cap, are recommended. The drainage lines shall be controlled by no-volume-change valves, such as ball valves, and shall be designed to minimize dead space in the lines.

5.4.3 *Top Cap and Base*—An impermeable, rigid top cap and base shall be used to support the specimen and provide for transmission of permeant liquid to and from the specimen. The diameter or width of the top cap and base shall be equal to the diameter or width of the specimen to $\pm 5\%$ or better. The base shall prevent leakage, lateral motion, or tilting, and the top cap shall be designed to receive the piston or extensometer, if used, such that the piston-to-top cap contact area is concentric with the cap. The surface of the base and top cap that contacts the membrane to form a seal shall be smooth and free of scratches.

5.4.4 *Flexible Membranes*—The flexible membrane used to encase the specimen shall provide reliable protection against leakage. The membrane shall be carefully inspected prior to use. If any flaws or pinholes are evident, the membrane shall be discarded. To minimize restraint to the specimen, the diameter or width of the non-stretched membrane shall be between 90 and 95 % of that of the specimen. The membrane shall be sealed to the specimen base and cap with rubber O-rings for which the unstressed, inside diameter or width is less than 90 % of the diameter or width of the base and cap, or by any other method that will produce an adequate seal.

NOTE 4—Membranes may be tested for flaws by placing them around a form sealed at both ends with rubber O-rings, subjecting them to a small air pressure on the inside, and then dipping them into water. If air bubbles come up from any point on the membrane, or if any visible flaws are observed, the membrane shall be discarded.

5.4.5 *Porous End Pieces*—The porous end pieces shall be of silicon carbide, aluminum oxide, or other material that is not attacked by the specimen or permeant liquid. The end pieces shall have plane and smooth surfaces and be free of cracks, chips, and discontinuities. They shall be checked regularly to ensure that they are not clogged.

5.4.5.1 The porous end pieces shall be the same diameter or width ($\pm 5\%$ or better) as the specimen, and the thickness shall be sufficient to prevent breaking.

5.4.5.2 The hydraulic conductivity of the porous end pieces shall be significantly greater than that of the specimen to be tested. The requirements outlined in 5.2.3 ensure this criterion is met.

5.4.6 *Filter Paper*—If necessary to prevent intrusion of material into the pores of the porous end pieces, one or more sheets of filter paper shall be placed between the top and bottom porous end pieces and the specimen. The paper shall have a negligibly small hydraulic impedance. The requirements outlined in 5.2.3 ensure that the impedance is small.

5.5 *Equipment for Compacting a Specimen*—Equipment (including compactor and mold) suitable for the method of compaction specified by the requester shall be used.

5.6 *Sample Extruder*—When the material being tested is a soil core, the soil core shall usually be removed from the sampler with an extruder. The sample extruder shall be capable of extruding the soil core from the sampling tube in the same direction of travel in which the sample entered the tube and with minimum disturbance of the sample. If the soil core is not extruded vertically, care should be taken to avoid bending stresses on the core due to gravity. Conditions at the time of sample extrusion may dictate the direction of removal, but the principal concern is to keep the degree of disturbance minimal.

5.7 *Trimming Equipment*—Specific equipment for trimming the specimen to the desired dimensions will vary depending on quality and characteristics of the sample (material). However, the following items listed may be used: lathe, wire saw with a wire about 0.3 mm (0.01 in.) in diameter, spatulas, knives, steel rasp for very hard clay specimens, cradle or split mold for trimming specimen ends, and steel straight edge for final trimming of specimen ends.

5.8 *Devices for Measuring the Dimensions of the Specimen*—Devices used to measure the dimensions of the specimen shall be capable of measuring to the nearest 0.5 mm or 0.01 in. or better (see 8.1.1) and shall be constructed such that their use will not disturb the specimen.

5.9 *Balances*—The balance shall be suitable for determining the mass of the specimen and shall be selected as discussed in Specification D4753. The mass of specimens less than 100 g shall be determined to the nearest 0.01 g. The mass of specimens between 100 g and 999 g shall be determined to the nearest 0.1 g. The mass of specimens equal to or greater than 1000 g shall be determined to the nearest gram.

5.10 *Equipment for Mounting the Specimen*—Equipment for mounting the specimen in the permeameter cell shall include a membrane stretcher or cylinder, and ring for expanding and placing O-rings on the base and top cap to seal the membrane.

5.11 *Vacuum Pump*—To assist with de-airing of permeant liquid (water) and saturation of specimens.

NOTE 5—For guidance or avoiding excessive consolidation in the use of vacuum for specimen saturation, consult 8.2 of Test Method D4767.

5.12 *Temperature Maintaining Device*—The temperature of the permeameter, test specimen, and reservoir of permeant liquid shall not vary more than $\pm 3^{\circ}\text{C}$ or $\pm 6^{\circ}\text{F}$ or better. Normally, this is accomplished by performing the test in a room with a relatively constant temperature. If such a room is

not available, the apparatus shall be placed in a water bath, insulated chamber, or other device that maintains a temperature within the tolerance specified above. The temperature shall be periodically measured and recorded.

5.13 *Water Content Containers*—The containers shall be in accordance with Method D2216.

5.14 *Drying Oven*—The oven shall be in accordance with Test Method D2216.

5.15 *Time Measuring Device(s)*—Devices to measure the duration of each permeation trial, such as either a clock with a second hand or a stopwatch (or equivalent), or both.

6. Reagents

6.1 Permeant Water:

6.1.1 The permeant water is the liquid used to permeate the test specimen and is also the liquid used in backpressuring the specimen.

6.1.2 The type of permeant water should be specified by the requestor. If no specification is made, one of the following shall be used: (i) potable tap water, (ii) a mixture of 0.0013 molar NaCl and 0.0010 molar CaCl_2 , or (iii) 0.01 molar CaCl_2 . The NaCl- CaCl_2 solution is representative of both typical tap waters and soil pore waters (7). The CaCl_2 solution has been used historically in areas with extremely hard or soft waters. The type of water used shall be indicated in the report.

6.1.2.1 The NaCl- CaCl_2 solution can be prepared by dissolving 0.76 g of reagent-grade NaCl and 1.11 g of reagent-grade CaCl_2 in 10 L of de-aired Type II deionized water.

6.1.2.2 The 0.01 CaCl_2 solution can be prepared by dissolving 11.1 g of reagent-grade CaCl_2 in 10 L of de-aired Type II deionized water.

6.1.2.3 Chemical interactions between a permeant liquid and the porous material may lead to variations in hydraulic conductivity. Distilled water can significantly lower the hydraulic conductivity of clayey soils (3). For this reason, distilled water is not usually recommended as a permeant liquid.

6.1.3 *Deaired Water*—To aid in removing as much air from the test specimen as possible, deaired water shall be used. The water is usually deaired by boiling, by spraying a fine mist of water into an evacuated vessel attached to a vacuum source, or by forceful agitation of water in a container attached to a vacuum source. If boiling is used, care shall be taken not to evaporate an excessive amount of water, which can lead to a larger salt concentration in the permeant water than desired. To prevent dissolution of air back into the water, deaired water shall not be exposed to air for prolonged periods.

7. Hazards

7.1 *Warning*—Mercury has been designated by many regulatory agencies as a hazardous material that can cause serious medical issues. Mercury, or its vapor, may be hazardous to health and corrosive to materials. Caution should be taken when handling mercury containing products. See the applicable product Safety Data Sheet (SDS) for additional information. Users should be aware that selling mercury or mercury containing products into your state or country may be prohibited by law.

7.1.1 Tubing composed of glass or other brittle materials may explode/shatter when under pressure, especially air. Therefore, such tubing should be enclosed. Establish allowable working pressures and make sure they are not exceeded.

7.2 *Precaution*—In addition to other precautions, store mercury in sealed shatterproof containers to control evaporation. When adding/subtracting mercury to/from the hydraulic system used in Method E or F, work in a well-ventilated area (preferably under a fume hood), and avoid contact with skin. Rubber gloves should be worn at all times when contact with mercury is possible.

7.2.1 Minimize uncontrolled flow of mercury out of the specialized hydraulic system by installing mercury traps or an inline check-valve mechanism. Minimize uncontrolled spills by using shatterproof materials or protective shields, or both.

7.2.2 If mercury comes into contact with brass/copper fittings, valves, etc., such items may rapidly become leaky. Therefore, where-ever practical use stainless steel fittings, etc.

7.2.3 Clean up spills immediately using a recommended procedure explicitly for mercury.

7.2.4 Dispose of contaminated waste materials containing mercury in a safe and environmentally acceptable manner.

8. Test Specimens

8.1 *Size*—Specimens shall have a minimum diameter of 25 mm (1.0 in.) and a minimum height of 25 mm. The height and diameter of the specimen shall be measured to three significant digits or better (see 8.1.1). The length shall vary by no more than $\pm 5\%$. The diameter shall vary by no more than $\pm 5\%$. The surface of the test specimen may be uneven, but indentations must not be so deep that the length or diameter vary by more than $\pm 5\%$. The diameter and height of the specimen shall each be at least 6 times greater than the largest particle size within the specimen. If, after completion of a test, it is found based on visual observation that oversized particles are present, that information shall be indicated on the data sheet(s)/form(s).

8.1.1 If the density or unit weight needs to be determined/recorded to four significant digits, or the void ratio to three significant digits; then the test specimens dimensions need to have four significant digits; that is, typically measured to the nearest 0.01 mm or 0.001 in.

8.1.2 Specimens of soil-cement and mixtures of cement, bentonite, and soils often have more irregular surfaces than specimens of soil. Thus, for these specimens the length and the diameter may vary by no more than $\pm 10\%$.

NOTE 6—Most hydraulic conductivity tests are performed on cylindrical test specimens. It is possible to utilize special equipment for testing prismatic test specimens, in which case reference to “diameter” in 8.1 applies to the least width of the prismatic test specimen.

8.2 *Intact Specimens*—Intact test specimens shall be prepared from a representative portion of intact samples secured in accordance with Practice D1587/D1587M, Practice D3550/D3550M, Practice D6151/D6151M, or Practice D2113. In addition, intact samples may be obtained by “block sampling” (8). Additional guidance on other drilling and sampling methods is given in Guide D6169/D6169M. Samples shall be preserved and transported in accordance with these require-

ments; for soils follow Group C in Practice D4220/D4220M, while for rock follow either “special care” or “soil-like care,” as appropriate in Practice D5079. Specimens obtained by tube sampling or coring may be tested without trimming except for cutting the end surfaces plane and perpendicular to the longitudinal axis of the specimen, provided soil characteristics are such that no significant disturbance results from sampling. Where the sampling operation has caused disturbance of the soil, the disturbed material shall be trimmed. Where removal of pebbles or crumbling resulting from trimming causes voids on the surface of the specimen that cause the length or diameter to vary by more than $\pm 5\%$, the voids shall be filled with remolded material obtained from the trimmings. The ends of the test specimen shall be cut and not troweled (troweling can seal off cracks, slickensides, or other secondary features that might conduct water flow). Specimens shall be trimmed, whenever possible, in an environment where changes in water content are minimized. A controlled high-humidity room is usually used for this purpose. The mass and dimensions of the test specimen shall be determined to the tolerances given in 5.8 and 5.9. The test specimen shall be mounted immediately in the permeameter. The water content of the trimmings shall be determined in accordance with Method D2216, to the nearest 0.1 % or better.

8.3 *Laboratory-Compacted Specimens*—The material to be tested shall be prepared and compacted inside a mold in a manner specified by the requester. If the specimen is placed and compacted in layers, the surface of each previously-compacted layer shall be lightly scarified (roughened) with a fork, ice pick, or other suitable object, unless the requester specifically states that scarification is not to be performed. Test Methods D698 and D1557 describe two methods of compaction, but any other method specified by the requester may be used as long as the method is described in the report. Large clods of material should not be broken down prior to compaction unless it is known that they will be broken in field construction, as well, or the requester specifically requests that the clod size be reduced. Neither hard clods nor individual particles of the material shall exceed $\frac{1}{4}$ of either the height or diameter of the specimen. After compaction, the test specimen shall be removed from the mold, the ends scarified, and the dimensions and weight determined within the tolerances given in 5.8 and 5.9. After the dimensions and mass are determined, the test specimen shall be immediately mounted in the permeameter. The water content of the trimmings shall be determined in accordance with Method D2216 to the nearest 0.1 % or better.

8.4 *Other Preparation Methods*—Other methods of preparation of a test specimen are permitted if specifically requested. The method of specimen preparation shall be identified in the data sheet(s)/form(s).

8.5 After the height, diameter, mass, and water content of the test specimen have been determined, the dry unit weight shall be calculated. Also, the initial degree of saturation shall be estimated (this information may be used later in the back-pressure stage).

8.6 In some cases, the horizontal hydraulic conductivity of a sample needs to be determined. In that case, the specimen may be trimmed such that its longitudinal axis is perpendicular to the longitudinal axis of the sample. Obtaining a specimen having a diameter of 36 mm (1.4 in.) typically requires a cylindrical sample with a diameter equal to or greater than about 70 mm (2.8 in.) or a rectangular sample with a minimum dimension of about 40 mm (1.6 in.).

9. Procedure

9.1 Specimen Setup:

9.1.1 Cut two filter paper sheets to approximately the same shape as the cross section of the test specimen. Soak the two porous end pieces and filter paper sheets, if used, in a container of permeant water.

9.1.2 Place the membrane on the membrane expander. Apply a thin coat of silicon high-vacuum grease to the sides of the end caps. Place one porous end piece on the base and place one filter paper sheet, if used, on the porous end piece, followed by the test specimen. Place the second filter paper sheet, if used, on top of the specimen followed by the second porous end piece and the top cap. Place the membrane around the specimen, and using the membrane expander or other suitable O-ring expander, place one or more O-rings to seal the membrane to the base and one or more additional O-rings to seal the membrane to the top cap.

9.1.3 Attach flow tubing to the top cap, if not already attached, assemble the permeameter cell, and fill it with de-aired water or other cell fluid. Attach the cell pressure reservoir to the permeameter cell line and the hydraulic system to the influent and effluent lines. Fill the cell pressure reservoir with deaired water, or other suitable liquid, and the hydraulic system with deaired permeant water. Apply a small confining pressure of 7 to 35 kPa (1 to 5 psi) to the cell and apply a pressure less than the confining pressure to both the influent and effluent systems, and flush permeant water through the flow system. After all visible air has been removed from the

flow lines, close the control valves. At no time during saturation of the system and specimen or hydraulic conductivity measurements shall the maximum applied effective stress be allowed to exceed that to which the specimen is to be consolidated.

9.2 *Specimen Soaking (Optional)*—To aid in saturation, specimens may be soaked under partial vacuum applied to the top of the specimen. Water under atmospheric pressure shall be applied to the specimen base through the influent lines, and the magnitude of the vacuum set to generate a hydraulic gradient across the specimen less than that which will be used during hydraulic conductivity measurements.

NOTE 7—Soaking under vacuum is applicable when there are continuous air voids in the specimen for example, specimens having a degree of saturation of less than about 85%. The specimen may swell when exposed to water; the effective stress will tend to counteract the swelling. However, for materials that tend to swell, unless the applied effective stress is greater than or equal to the swell pressure, the specimen will swell. In addition, see Note 5.

9.3 *Back-Pressure Saturation*—To saturate the specimen, back pressuring is usually necessary. Fig. 5 (9) provides guidance on back pressure required to attain saturation. Additional guidance on the back-pressure process is given by Black and Lee (10) and Head (11).

NOTE 8—The relationships presented in Fig. 5 are based on the assumption that the water used for back pressuring is deaired and that the only source for air to dissolve into the water is air from the test specimen. If air pressure is used to control the back pressure, pressurized air will dissolve into the water, thus reducing the capacity of the water used for back pressure to dissolve air located in the pores of the test specimen. The problem is minimized by using a long (>5 m) tube that is impermeable to air between the air-water interface and test specimen, by separating the back-pressure water from the air by a material or fluid that is relatively impermeable to air, by periodically replacing the back-pressure water with deaired water, or by other means.

9.3.1 During the saturation process, any change in the volume (swelling or compression of the void ratio, density, etc.) of the test specimen should be minimized. The easiest way

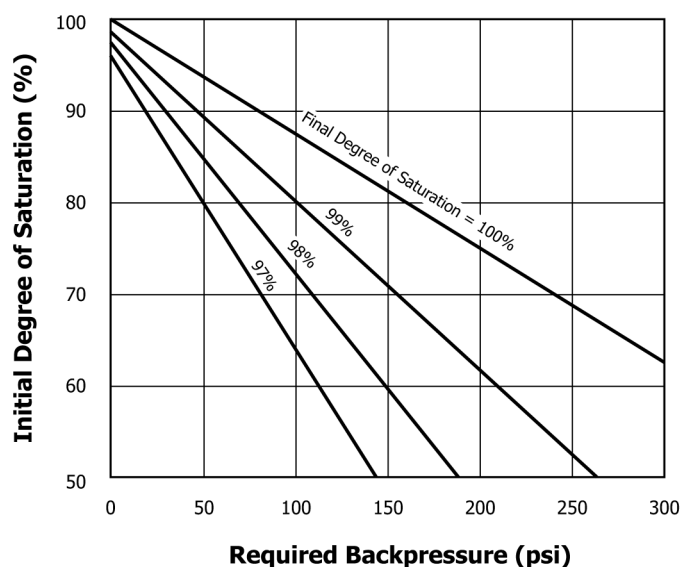


FIG. 5 Back Pressure to Attain Various Degrees of Saturation (9)

to verify that volume changes are minor is to measure the height of the specimen during the back-pressuring process. Volume changes are considered minor if the resulting change in hydraulic conductivity is less than about one-half the acceptable error of 25 % given in 9.5.4, unless more stringent control on density or hydraulic conductivity, or both, is required. For this to occur the axial strain should be less than about 0.4 % for normally consolidated soils, or about 0.1 % for overconsolidated soils. See Appendix X2.

9.3.2 Take and record an initial reading of specimen height, if being monitored. Open the flow line valves and flush out of the system any free air bubbles using the procedure outlined in 9.1.3. If an electronic pressure transducer or other measuring device is to be used during the test to measure pore pressures or applied hydraulic gradient, bleed any trapped air from the device.

9.3.3 Adjust the applied confining pressure to the value to be used during saturation of the specimen. Apply back pressure by simultaneously increasing the cell pressure and the influent and effluent pressures in increments. The maximum value of an increment in back pressure shall be sufficiently low such that no point in the specimen is exposed to an effective stress in excess of that to which the specimen will be subsequently consolidated. At no time shall a head be applied such that the effective confining stress is <7 kPa (1 psi) because of the danger of separation of the membrane from the test specimen. Maintain each increment of pressure for a period of a few minutes to a few hours, depending upon the characteristics of the specimen. To check for equalization after application of a back pressure increment or after the full value of back pressure has been applied, close the specimen drainage valves and measure the change in pore-pressure over a 1-min interval. If the change in pore pressure is less than 5 % of the difference between the chamber pressure and the back pressure, another back pressure increment may be added. To assist in removal of trapped air, a small hydraulic gradient may be applied across the specimen to induce flow.

9.3.4 Saturation shall be verified with one of the three following techniques:

9.3.4.1 Saturation may be verified by measuring the B coefficient as described in Test Method D4767 (see Note 9). The test specimen shall be considered to be adequately saturated if the B value is ≥ 0.95 , or for relatively incompressible materials, for example, rock, if the B value remains unchanged with application of larger values of back pressure. The B value may be measured prior to or after completion of the consolidation phase (see 9.4). An accurate B -value determination can only be made if no gradient is acting on the specimen and all pore-water pressure induced by consolidation has dissipated. That is, conform completion of primary consolidation before this determination; see Test Method D2435/D2435M or D4767 on how to confirm completion of primary consolidation.

NOTE 9—The B coefficient is defined for this type of test as the change in pore-water pressure in the porous material divided by the change in confining pressure. Compressible materials that are fully saturated with water will have a B value of 1.0. Relatively incompressible, saturated materials have B values that are somewhat less than 1.0 (12).

9.3.4.2 Saturation of the test specimen may be confirmed at the completion of the test by calculation of the final degree of saturation. The final degree of saturation shall be 100 ± 5 %. However, measurement of the B coefficient as described in 9.3.4.1 or use of some other technique (9.3.4.3) is strongly recommended because it is much better to confirm saturation prior to permeation than to wait until after the test to determine if the test was valid.

9.3.4.3 Other means for verifying saturation, such as observing the flow of water into the specimen when the back pressure is increased, can be used for verifying saturation provided data are available for similar materials to establish that the procedure used confirms saturation as required in 9.3.4.1 or 9.3.4.2.

9.4 Consolidation—The specimen shall be consolidated to the effective stress specified by the requester. Consolidation shall be accomplished in stages, with the increase in cell pressure minus back pressure (effective stress) in each new stage equal to or less than the effective stress in the previous stage that is, consolidation increment ratio of one or less.

NOTE 10—The test specimen may be consolidated prior to application of back pressure. Also, the back pressure and consolidation phases may be completed concurrently if back pressures are applied sufficiently slowly to minimize potential for overconsolidation of the specimen.

9.4.1 Record the specimen height, if being monitored, prior to application of consolidation pressure and periodically during consolidation.

9.4.2 Increase the cell pressure to the level necessary to develop the desired effective stress, and begin consolidation. Drainage may be allowed from the base or top of the specimen, or simultaneously from both ends.

9.4.3 (Optional) Record outflow volumes to confirm that primary consolidation has been completed prior to initiation of the hydraulic conductivity test. Alternatively, measurements of the change in height of the test specimen can be used to confirm completion of consolidation.

NOTE 11—The procedure in 9.4.3 is optional because the requirements of 9.5 ensure that the test specimen is adequately consolidated during permeation because if it is not, inflow and outflow volumes will differ significantly. However, for accurate B -value determination, saturation should be confirmed at the completion of consolidation (see 9.3.4.1). Recording outflow volumes or height changes is recommended as a means for verifying the completion of consolidation prior to initialization of permeation. Also, measurements in the change in height of the test specimen, coupled with knowledge of the initial height, provide a means for checking the final height of the specimen.

9.5 Permeation:

9.5.1 Hydraulic Gradient—When possible, the hydraulic gradient ($i = \Delta h/L$, for definitions of notation see 10.1) used for hydraulic conductivity measurements should be similar to that expected to occur in the field. In general, hydraulic gradients from <1 to 5 cover most field conditions. However, the use of small hydraulic gradients can lead to very long testing times for materials having low hydraulic conductivity (less than about 1×10^{-8} m/s). Somewhat larger hydraulic gradients are usually used in the laboratory to accelerate testing, but excessive gradients must be avoided because high seepage pressures may consolidate the material, material may be washed from the specimen, or fine particles may be washed

downstream and plug the effluent end of the test specimen. These effects could increase or decrease hydraulic conductivity. If no gradient is specified by the requestor, the following guidelines may be followed:

Hydraulic Conductivity, m/s	Recommended Maximum Hydraulic Gradient
1×10^{-5} to 1×10^{-6}	2
1×10^{-6} to 1×10^{-7}	5
1×10^{-7} to 1×10^{-8}	10
1×10^{-8} to 1×10^{-9}	20
less than 1×10^{-9}	30

9.5.1.1 A higher gradient than given above may be used if the higher gradient can be shown not to change the hydraulic conductivity. For example, on a representative specimen, perform a hydraulic conductivity determination at $i = 30$ than at $i = 50$ or 100, or more. Determine which, if any, of the hydraulic conductivities (k) determined at these gradients are similar (that is, within the acceptable steady-state range given for the Method (A, B, C, D, E, or F). Any gradient equal to or less than the highest gradient yielding a similar hydraulic conductivity may be used for testing.

NOTE 12—Seepage pressures associated with large hydraulic gradients can consolidate soft, compressible specimens and reduce their hydraulic conductivity. Smaller hydraulic gradients (<10) may be necessary for such specimens.

9.5.2 *Initialization*—Initiate permeation of the specimen by increasing the influent (headwater) pressure (see 9.3.3). The effluent (tailwater) pressure shall not be decreased because air bubbles that were dissolved by the specimen water during backpressuring may come out of solution if the pressure is decreased. The back pressure shall be maintained throughout the permeation phase.

9.5.2.1 The maximum increase in headwater pressure cannot exceed 95 % of the effective consolidation stress. Alternatively, the difference between the cell pressure and the total headwater pressure cannot be less than 5 % of the effective consolidation stress.

9.5.2.2 At the start and end of each permeation trial, at t_1 and t_2 , read and record the test temperature to the nearest 0.1°C. See Section 10. If the number of significant digits in the calculation of hydraulic conductivity at 20°C can be one, then the test temperature can be measured to the nearest degree Celsius.

9.5.3 *Time Measurements*—Measure and record the time at the start and end of each permeation trial (or its interval) to two or more significant digits. That is the time interval has to be greater than 9 s unless the time is recorded to the nearest 0.1 s.

9.5.4 *Constant Head Tests:*

9.5.4.1 (*Method A*)—Measure and record the required head loss across the tolerances and significant digits stated in 5.1.1 and 5.2.3 at the start and end of each permeation trial (as a minimum). The head loss across the permeameter shall be kept constant to ± 5 % or better. Measure and record periodically the quantity of inflow as well as the quantity of outflow to a minimum of three significant digits. Also measure and record any changes in height of the test specimen, if being monitored (see Note 12). Continue permeation until at least four values of hydraulic conductivity are obtained over an interval of time in which: (1) the ratio of outflow to inflow rate is between 0.75

and 1.25, and (2) the hydraulic conductivity is steady. The hydraulic conductivity shall be considered steady if four or more consecutive hydraulic conductivity determinations fall within ± 25 % or better of the mean value for $k \geq 1 \times 10^{-10}$ m/s or within ± 50 % or better for $k < 1 \times 10^{-10}$ m/s, and a plot or tabulation of the hydraulic conductivity versus time shows no significant upward or downward trend.

9.5.4.2 *Method E (Constant Volume)*—Measure and record the required head loss across the permeameter to the tolerances and significant digits stated in 5.1.4. The head loss across the permeameter shall be kept constant to ± 5 % or better. Measure and record, to a minimum of three significant digits, the quantity of either inflow (influent) or outflow (effluent). In this measurement the last digit may be due to estimation, see 5.1.1.1. In addition, measure and record any changes in the height of the test specimen, if being monitored (see Note 12). Continue permeation until at least two or more values of hydraulic conductivity (k) are steady. The hydraulic conductivity shall be considered steady if two or more consecutive k determinations fall within ± 15 % or better of the mean value (two or more determinations) for $k \geq 1 \times 10^{-10}$ m/s or within ± 50 % or better for $k < 1 \times 10^{-10}$ m/s.

9.5.5 *Falling-Head Tests (Methods B, C, and F)*—Measure and record the required head loss across the permeameter to the tolerances and significant digits stated in 5.1.2. Measure and record these head losses at the start and end of each permeation trial (as a minimum). At no time shall the applied head loss across the specimen be less than 75 % of the initial (maximum) head loss during the hydraulic conductivity determination (see Note 13). At the start and end of each trial, as a minimum, measure and record any changes in the height of the test specimen, if being monitored. To meet these requirements, especially for Method F, the initial head loss in each trial will most likely have to be reset to the same value (± 5 %) used in the first trial. In addition, the “75 % criterion” mentioned above has to be adhered to closely.

9.5.5.1 *Methods B and C*—The volumes of outflow and inflow shall be measured and recorded to three significant digits (the last digit may be due to estimation, see 5.1.1.1). Measure and record these volumes at the start and end of each permeation trial (as a minimum). Continue permeation until at least four values of hydraulic conductivity are obtained over an interval of time in which: the ratio of outflow to inflow rate is between 0.75 and 1.25, and the hydraulic conductivity is steady (see 9.5.4.1).

NOTE 13—When the water pressure in a test specimen changes and the applied total stress is constant, the effective stress in the test specimen changes, which can cause volume changes that can invalidate the test results. The requirement that the head loss not decrease very much is intended to keep the effective stress from changing too much. For extremely soft, compressible test specimens, even more restrictive criteria may be needed. Also, when the initial and final head losses across the test specimen do not differ by much, great accuracy is needed to comply with the requirement of 5.1.2 that the ratio of initial to final head loss be determined with an accuracy of ± 5 % or better. When the initial and final head loss over an interval of time do not differ very much, it may be possible to comply with the requirements for a constant head test (9.5.4) in which the head loss must not differ by more than ± 5 % and to treat the test as a constant head test.

9.5.5.2 *Method F (Constant Volume)*—Continue permeation until at least two or more values of hydraulic conductivity (k) meet the requirements stated in 9.5.4.2.

9.5.6 *Constant Rate of Flow Tests (Method D)*—Initiate permeation of the specimen by imposing a constant flow rate. Choose the flow rate so the hydraulic gradient does not exceed the value specified, or if none is specified, the value recommended in 9.5.1. Periodically measure the rate of inflow, the rate of outflow, and head loss across the test specimen to the tolerances and significant digits given in 5.1.3. Also, measure and record any changes in specimen height, if being monitored. Continue permeation until at least four values of hydraulic conductivity are obtained over an interval of time in which the ratio of inflow to outflow rates is between 0.75 and 1.25, and hydraulic conductivity is steady (see 9.5.4.1).

9.6 *Final Dimensions of the Specimen*—After completion of permeation, reduce the applied confining, influent, and effluent pressures in a manner that does not generate significant volume change of the test specimen. Then carefully disassemble the permeameter cell and remove the specimen. Measure and record the final height, diameter, and total mass of the specimen. Then determine the final water content of the specimen by the procedure of Method D2216. Dimensions and mass of the test specimen shall be measured to the tolerances specified in 5.8 and 5.9.

NOTE 14—The specimen may swell after removal of back pressure as a result of air coming out of solution. A correction may be made for this effect, provided that changes in the length of the specimen are monitored during the test. The strain caused by dismantling the cell is computed from the length of the specimen before and after dismantling the cell. The same strain is assumed to have occurred in the diameter. The corrected diameter and actual length before the back pressure was removed are used to compute the volume of the test specimen prior to dismantling the cell. The volume prior to dismantling the cell is used to determine the final dry density and degree of saturation.

10. Calculation

10.1 Constant Head and Constant Rate of Flow Tests:

10.1.1 *Methods A and D*—Calculate the hydraulic conductivity, k , as follows:

$$k = \frac{\Delta Q \cdot L}{A \cdot \Delta h \cdot \Delta t} \quad (1)$$

where:

- k = hydraulic conductivity, m/s,
- ΔQ = quantity of flow for given time interval Δt , taken as the average of inflow and outflow, m^3 ,
- L = length of specimen, m,
- A = cross-sectional area of specimen, m^2 ,
- Δt = interval of time, s , over which the flow ΔQ occurs ($t_2 - t_1$),
- t_1 = time at start of permeation trial, date: hr:min:sec,
- t_2 = time at end of permeation trial, date:hr:min:sec,
- Δh = average head loss across the permeameter/specimen $((\Delta h_1 + \Delta h_2)/2)$, m of water,
- Δh_1 = head loss across the permeameter/specimen at t_1 , m of water, and
- Δh_2 = head loss across the permeameter/specimen at t_2 , m of water.

NOTE 15—The interval of time, Δt , can be measured directly using a

stop watch or equivalent device, see 11.5.1. Units other than second(s), meters (m), etc., may be used providing an appropriate unit conversion factor (UCF) is used so k is in m/s or other units, if requested or customary (see Section 11).

10.1.2 *Method E*—Use the above Eq 1. If the height of the mercury column in the “Constant Head” tube is used to determine the head loss, Δh , use the following equation.

TABLE 1 Specific Gravity of Mercury (G_{Hg})

Temperature ($^{\circ}\text{C}$)	$G_{\text{Hg}} = (\rho_{\text{Hg}}/\rho_w)$
15	13.570
16	13.570
17	13.570
18	13.570
19	13.570
20	13.570
21	13.571
22	13.571
23	13.572
24	13.573
25	13.574
26	13.575
27	13.576
28	13.577
29	13.579
30	13.580

$$\Delta h = \Delta H_{\text{Hg}} \cdot \left(\frac{\rho_{\text{Hg}}}{\rho_w} - 1 \right) = \Delta H_{\text{Hg}} \cdot (G_{\text{Hg}} - 1) \quad (2)$$

where:

- ΔH_{Hg} = the peak to peak height of mercury column (see Fig. 2), m, and
- ρ_{Hg} = the density of mercury, g/cm^3 ,
- ρ_w = the density of water, g/cm^3 ,
- G_{Hg} = the ratio of the density of mercury to the density of water (specific gravity of mercury) at the test/trial temperature. See Table 1.

NOTE 16—For the constant-volume hydraulic systems, there is no head loss across the permeameter/specimen due to elevation head. Units other than seconds (s), meters (m), etc., may be used providing an appropriate UCF is used so k is in m/s or other units, if requested or customary (see Section 11).

10.2 Falling-Head Tests:

10.2.1 *Constant Tailwater Pressure (Method B)*—Calculate the hydraulic conductivity, k , as follows:

$$k = \frac{a \cdot L}{A \cdot \Delta t} \ln \left(\frac{\Delta h_1}{\Delta h_2} \right) \quad (3)$$

where:

- a = cross-sectional area of the reservoir containing the influent liquid, m^2 , and
- $\ln$ = natural logarithm (base $e = 2.71828$).

See Note 15.

10.2.2 *Increasing Tailwater Pressure (Method C)*—Calculate the hydraulic conductivity, k , as follows:

$$k = \frac{a_{\text{in}} \cdot a_{\text{out}} \cdot L}{(a_{\text{in}} + a_{\text{out}}) \cdot A \cdot \Delta t} \ln \left(\frac{\Delta h_1}{\Delta h_2} \right) \quad (4)$$

where:

- a_{in} = cross-sectional area of the reservoir containing the influent/inflow liquid, m^2 , and

a_{out} = cross-sectional area of the reservoir containing the effluent/outflow liquid, m^2 .

See **Note 15**.

NOTE 17—For the case in which $a_{\text{out}} = a_{\text{in}} = a$, the equation for calculating k for a falling head test with a rising tailwater level is:

$$k = \frac{a \cdot L}{2 \cdot A \cdot \Delta t} \ln \left(\frac{\Delta h_1}{\Delta h_2} \right) \quad (5)$$

where:

a = area of the reservoirs containing either the influent/inflow or effluent/outflow liquid, m^2

10.2.3 *Constant-Volume System (Method F)*—Calculate the hydraulic conductivity, k , as follows:

$$k = \left(\frac{a_{\text{in}} \cdot a_{\text{out}}}{(a_{\text{out}} + a_{\text{in}})} \cdot \frac{1}{(G_{\text{Hg}} - 1)} \right) \cdot \frac{L}{A} \cdot \frac{1}{\Delta t} \cdot \ln \left(\frac{\Delta h_1}{\Delta h_2} \right) \quad (6)$$

10.2.3.1 If the differential elevation between the top surfaces of the mercury level in the headwater and tailwater tubes is used to determine the head loss, Δh , use the following equations.

a) For the head loss at the start of the permeation trial, h_1 :

$$h_1 = (\Delta H_{\text{Hg},1} + \Delta H_{\text{Hg},c}) \cdot \left(\frac{\rho_{\text{Hg}}}{\rho_w} - 1 \right) = (\Delta H_{\text{Hg},1} + \Delta H_{\text{Hg},c}) \cdot (G_{\text{Hg}} - 1) \quad (7)$$

where:

$\Delta H_{\text{Hg},1}$ = difference in elevation between the top surfaces of the mercury level in the tailwater and headwater tubes at the start of the permeation trial,

t_1 (see **Fig. 3**), m, and

$\Delta H_{\text{Hg},c}$ = difference in elevation of mercury in the headwater and tailwater tubes of the manometer with equal pressures applied to both tubes, m. This value is positive if the inside diameter (ID) of the headwater tube is larger than the ID of the tailwater tube, and negative if the opposite is true. A discussion on capillary head is given in **Appendix X1, X1.2.3.2** and **X1.4**. See **Note 16**.

b) For the head loss at the end of the permeation trial, Δh_2 :

$$\Delta h_2 = \Delta h_1 + \left((-\Delta H_{g,\text{tail}}) \cdot \left(\frac{a_{\text{out}}}{a_{\text{in}}} + 1 \right) \cdot (G_{\text{Hg}} - 1) \right) \quad (8)$$

where:

$-\Delta H_{g,\text{tail}}$ = the negative change in elevation of the mercury levels in the tailwater tube during the permeation trial, m.

The reason why $\Delta H_{g,\text{tail}}$ is used instead of $\Delta H_{g,\text{Hg},2}$ (difference in mercury levels at end of trial) is explained in **Appendix X1, X1.3.2.1**.

10.3 *Hydraulic Conductivity at Standard Temperature*—Correct the hydraulic conductivity to that for 20°C (68°F), k_{20} , by multiplying k by the ratio of the viscosity of water at test temperature to the viscosity of water at 20°C (68°F), R_T :

$$k_{20} = R_T \cdot K \quad (9)$$

with

$$R_T = 2.2902 (0.9842^T) / T^{0.1702} \quad (10)$$

where:

k_{20} = hydraulic conductivity corrected to 20°C, m/s

R_T = ratio of the viscosity of water at test temperature to the viscosity of water at 20°C

T = average test temperature during the permeation trial $((T_1 + T_2)/2)$, to the nearest 0.1°C.

T_1 = test temperature at start of permeation trial, to nearest 0.1°C, and

T_2 = test temperature at end of permeation trial, to nearest 0.1°C

10.3.1 The equation for R_T is only accurate to three significant digits between 5 and 50°C (41 and 122°F), see **1.1**.

10.3.2 If the number of significant digits in the calculation of hydraulic conductivity at 20°C can be one, then the test temperature can be measured to the nearest °C.

11. Report: Test Data Sheet(s)/Form(s)

11.1 The methodology used to specify how data are recorded on the test data sheet(s)/form(s), as given below, is covered in **1.5** and in Practice **D6026**.

11.2 Record as a minimum the following general information (data):

11.2.1 Sample/specimen identifying information, such as Project No., Boring No., Sample No., Depth, etc.

11.2.2 Any special selection and preparation process, such as removal of gravel or other materials, or identification of their presence, if “intact” specimen.

11.2.3 If the specimen is reconstituted, remolded or trimmed in a specialized manner (determine horizontal hydraulic conductivity, see **8.6**), provide information on method of reconstitution, remolding, etc.

11.2.4 Name or initials of the person performing the test.

11.3 Record as a minimum the following test specimen data:

11.3.1 The measured specific gravity test (Test Method **D854**) or assumed value.

11.3.2 The initial mass, dimensions (length and diameter), area, and volume of the specimen, to either three or four significant digits (see **8.1** and **8.1.1**).

11.3.3 The initial water content (nearest 0.1 percent), dry unit weight (three or four significant digits, see **8.1.1**) and saturation (nearest percent) of the test specimen.

11.3.4 The final mass, dimensions (length and diameter), area, and volume of the specimen, to either three or four significant digits (see **8.1** and **8.1.1**).

11.3.5 The final water content (nearest 0.1 percent), dry unit weight (three or four significant digits, see **8.1.1**) and saturation (nearest percent) of the test specimen.

11.4 Record as a minimum the following test boundary conditions:

11.4.1 The type of permeant liquid used.

11.4.2 The magnitude of total back pressure (two significant digits or three if used in the head loss determination).

11.4.3 The effective consolidation stress (two or more significant digits).

11.4.4 The area of the headwater and tailwater tubes (such as burettes, reservoirs, U-tube manometers, etc.), as applicable (three or more significant digits).

11.4.5 The length (L) and area (A) of the test specimen during permeation (minimum of three significant digits).

11.4.5.1 These values can be determined based on either *a*) the initial dimensions of specimen plus any length/height and volume changes occurring during saturation and consolidation; or *b*) final dimensions of the test specimen, see 11.3.4.

11.5 Record as a minimum the following permeation data:

11.5.1 The date, time (or start and elapsed time), temperature (nearest 0.1°C see 10.3.2), head loss reading(s), flow reading(s) (if applicable), and deformation gage (if applicable) at the start and end of each trial/determination. Applicable measurements/readings and any averages/differences calculated using measurements/readings obtained shall have two or more significant digits, unless specified differently in Section 9.

11.5.2 The calculated initial hydraulic gradient and ending value if falling head Method B, C, or F is being used, and the hydraulic conductivity to two or more significant digits.

11.5.3 The average corrected hydraulic conductivity (k_{20} , see 10.3) for the values meeting the applicable requirements in 9.5.4 to 9.5.6. Record this value to two or three significant digits in units of m/s or other units, if requested or customary, for example, 7.1×10^{-10} or 7.13×10^{-10} m/s.

11.5.4 A graph or table of hydraulic conductivity versus time or pore volumes of flow is recommended, unless a constant-volume hydraulic system is used.

12. Precision and Bias

12.1 *Precision*—The precision of this test method is based on an interlaboratory study of D5084, Standard Test Methods for Measurement of Hydraulic Conductivity of Saturated Porous Materials Using a Flexible Wall Permeameter, conducted in 2008. Each of twelve laboratories tested three different soil types:

ML-1: Vicksburg silt (ASTM Reference Soil ML-1)

CH-1: Vicksburg clay (ASTM Reference Soil CH-1)

CL-1: Annapolis clay (ASTM Reference Soil CL-1)

All three soils are from the D18 ISR Reference Soils and Testing Program. Index properties for the soils are shown in Table 2. These properties are from the ASTM Reference Soils and Testing Program.

Every “test result” represents an individual determination. Each laboratory reported three replicate test results for the analyses. Practice E691 was followed for the design and analysis of the data; the details are given in ASTM Research Report RR:D18-D1018.⁵

⁵ Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D18-1018.

TABLE 2 Index Properties for ASTM Reference Soils Used in Interlaboratory Study on Saturated Hydraulic Conductivity

ASTM Reference Soil ID	Liquid Limit per D4318 (%)	Plasticity Index per D4318 (%)	Specific Gravity of Soils per D854 (-)	Percent Finer than No. 200 Sieve per D1140 (%)
ML-1	27.3 ± 2.5	3.9 ± 4.5	2.725 ± 0.043	99.0 ± 0.3
CL-1	33.2 ± 1.4	13.4 ± 3.7	2.675 ± 0.030	88.5 ± 0.8
CH-1	59.7 ± 2.8	39.3 ± 7.0	2.726 ± 0.032	98.8 ± 0.4

12.1.1 *Repeatability Limit (r)*—Two test results obtained within one laboratory shall be judged not equivalent if they differ by more than the “r” value for that material; “r” is the interval representing the critical difference between two test results for the same material, obtained by the same operator using the same equipment on the same day in the same laboratory.

12.1.1.1 Repeatability limits are listed in Table 3.

12.1.2 *Reproducibility Limit (R)*—Two test results shall be judged not equivalent if they differ by more than the “R” value for that material; “R” is the interval representing the critical difference between two test results for the same material, obtained by different operators using different equipment in different laboratories.

12.1.2.1 Reproducibility limits are listed in Table 3.

12.1.3 The above terms (repeatability limit and reproducibility limit) are used as specified in Practice E177.

12.1.4 Any judgment in accordance with statements 12.1.1 and 12.1.2 would have an approximately 95 % probability of being correct.

12.2 *Bias*—At the time of the study, there was no accepted reference material suitable for determining the bias for this test method. Therefore no statement on bias is being made.

12.3 The precision statement was determined through statistical examination of 104 results, from twelve laboratories, on the three soils described in 12.1.

13. Keywords

13.1 coefficient of permeability; constant head; constant rate of flow; constant volume; falling head; hydraulic barriers; hydraulic conductivity; liner; permeability; permeameter

TABLE 3 Hydraulic Conductivity Statistics from ILS Report RR:D18-D1018 (All Units in cm/s)

ILS Soil	Average	Repeatability Standard Deviation	Reproducibility Standard Deviation	Repeatability Limit r	Reproducibility Limit R
		S_r	S_R		
ML-1	1.2×10^{-6}	3.3×10^{-7}	4.4×10^{-7}	9.3×10^{-7}	1.2×10^{-6}
CL-1	3.8×10^{-8}	4.4×10^{-9}	6.2×10^{-9}	1.2×10^{-8}	1.8×10^{-8}
CH-1	3.6×10^{-9}	2.9×10^{-9}	4.7×10^{-9}	8.2×10^{-9}	1.3×10^{-8}

APPENDIXES

(Nonmandatory Information)

X1. DEVELOPMENT OF HYDRAULIC CONDUCTIVITY EQUATION FOR THE MERCURY CONSTANT VOLUME-FALLING HEAD HYDRAULIC SYSTEM

X1.1 Introduction—A schematic of a mercury constant volume-falling head hydraulic system is given in Fig. 4. In this figure, the falling head is applied by the difference in elevation between the mercury levels in the tailwater and headwater tubes of the mercury U-tube manometer. In designing this type of hydraulic system, the area of the tailwater tube (a_{out}) is made significantly smaller than that of the headwater tube (a_{in}). This is done for three reasons:

First, to increase the sensitivity of the flow/volume measurement;

Second, to decrease the time required to measure the hydraulic conductivity; and,

Third, so one can clean the hydraulic system by flushing water through the tailwater tube and out the headwater tube of the mercury U-tube manometer without losing mercury.

X1.1.1 The tubing lines leading from the test specimen to mercury U-tube manometer are filled with water, as well as the spaces above the mercury in the manometer. Therefore, the volume of the saturated test specimen remains constant during permeation. This occurs because the components (water, tubing, and manometer) of the hydraulic system are relatively incompressible compared to soil. In addition, there is continuity of inflow and outflow of permeant water during permeation.

X1.1.2 The presentation of determining heads and head losses in a mercury constant volume-falling head hydraulic system is presented before the development of the hydraulic conductivity equation. This allows one to become familiar with the notation and required parameters before addressing that relatively complicated equation development process. The heads involved are the total head (H), pressure head (H_p), elevation head (H_e), capillary head (H_c), velocity head (H_v), and total head loss (Δh).

X1.2 Determination of Total Head and Total Head Loss

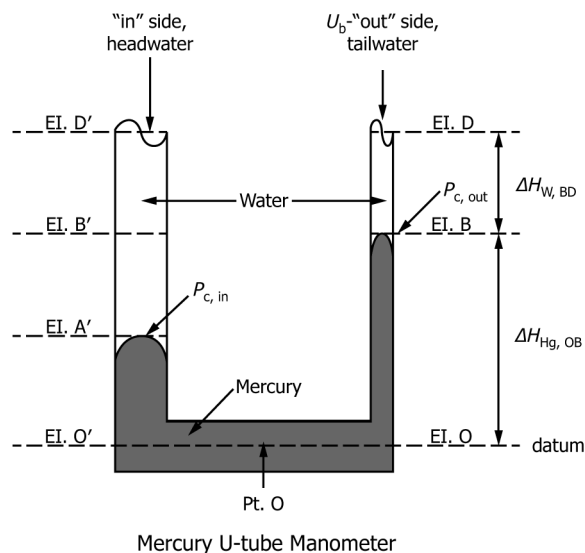
X1.2.1 Total Head—The total head is equal to the sum of the pressure head (H_p), elevation head (H_e), velocity head (H_v), and capillary head (H_c). Usually it is expressed in height of water, such as m or cm of water. In addition, the velocity head is assumed to be zero/insignificant providing the conditions specified in 4.4 through 5.2.3.2 are met. The pressure head is the height of a vertical column of static water that can be supported by the static pressure (p) at a given point. It may be expressed as

$$H_p = \text{UCF} \cdot \frac{p}{\rho_w \cdot g} \quad (\text{X1.1})$$

where:

- H_p = the pressure head at given point (units of vertical height of water column, m),
- UCF = a unit conversion factor,
- p = the static pressure at a given point (units of force per unit area, kPa),
- ρ_w = the density of the water (units of mass per unit volume, Mg/m^3), and
- g = the acceleration of gravity, convert mass to force (9.80665 m/s^2 or 980.665 cm/s^2).

X1.2.2 The static pressure at any point within a confined fluid may be calculated as shown in Fig. X1.1, assuming there



Determination of pressure for Pt. O at EL. O on the "out" or tailwater side, $p_{O, out}$.

$$p_{O, out} = \text{UCF} \cdot (\text{OB} \cdot \rho_{Hg} \cdot g + \Delta P_{C, out} + \text{BD} \cdot \rho_w \cdot g + U_b)$$

or

$$= \text{UCF} \cdot (\Delta H_{Hg, OB} \cdot \rho_{Hg} \cdot g + \Delta P_{C, out} + \Delta H_{W, BD} \cdot \rho_w \cdot g + U_b)$$

where:

- UCF = a unit conversion factor,
- ρ_{Hg} = the density of mercury, g/cm^3 or Mg/m^3 ,
- ρ_w = the density of water, g/cm^3 or Mg/m^3 ,
- g = the acceleration of gravity, m/s^2 or cm/s^2 ,
- $\Delta P_{C, out}$ = the change in pressure due to capillarity at the tube-water-mercury interface in the outflow or tailwater tube, and
- U_b = the applied back pressure, kPa or kN/m^2

FIG. X1.1 Static Pressure Calculations

is no drop in pressure due to velocity head loss. This figure shows that fluid pressure for Point O at elevation El. O on the “out” or tailwater side may be expressed as:

$$p_{o, out} \approx UCF \cdot (\Delta H_{Hg, OB} \cdot \rho_{Hg} \cdot g + \Delta p_{c, out} + \Delta H_{w, BD} \cdot \rho_w \cdot g + U_b) \quad (X1.2)$$

or

$$p \approx UCF \cdot (\Delta H_{Hg} \cdot \rho_{Hg} \cdot g + \Delta p_c + \Delta H_w \cdot \rho_w \cdot g + U_b)$$

where:

- ΔH_{Hg} = the differential height of the mercury column, m,
- ρ_{Hg} = the density of the mercury, Mg/m³,
- ΔH_w = the differential height of the water column, m,
- Δp_c = the change in pressure due to capillarity at the tube-water-mercury interface, kPa, and
- U_b = the applied back pressure, kPa.

X1.2.3 For the above case, the pressure head (in height of water) is

$$H_p \approx UCF \cdot \left(H_{Hg} \times \frac{\rho_{Hg} \cdot g}{\rho_w \cdot g} + H_w \times \frac{\rho_w \cdot g}{\rho_w \cdot g} + \frac{\Delta p_c}{\rho_w \cdot g} + \frac{U_b}{\rho_w \cdot g} \right) \quad (X1.3)$$

or

$$H_p \approx UCF \cdot \left(H_{Hg} \times G_{Hg} + \Delta H_w + \Delta H_c + \frac{U_b}{\rho_w \cdot g} \right)$$

where:

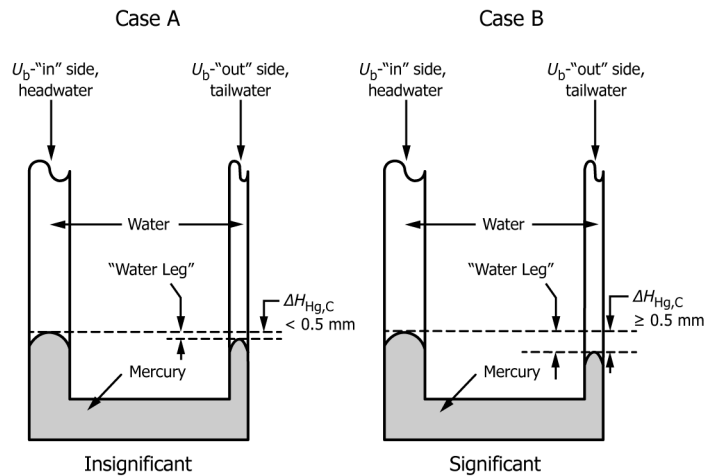
G_{Hg} = the specific gravity of mercury at a given temperature, and

ΔH_c = the change in head due to capillarity, m of water, see X1.2.3.2

X1.2.3.1 *Velocity Head*—In most cases, the velocity head or velocity head loss is assumed equal to zero or insignificant, providing the requirements specified in 5.2.3 are met.

X1.2.3.2 *Capillary Head*—In most cases, the capillary head or capillary head loss is assumed equal to zero. However, in some mercury U-tube manometers, the difference in capillary head between the headwater and tailwater tubes; that is, the capillary head loss, ΔH_c is significant. Therefore, it has to be accounted for as shown in Fig. X1.2. To help one understand the derivation of ΔH_c in this figure, one has to remember to account for the “water leg” in the mercury U-tube manometer containing water instead of air. Subtracting away the “water leg” pressure does this. Also, the pressure difference measured

Capillary Head for Mercury U-tube Manometer at No Flow or Equilibrium



$$\text{Capillary Head} = \Delta H_c = UCF \cdot \left(\frac{\Delta H_{Hg, c} \cdot \rho_{Hg} \cdot g - \Delta H_{Hg, c} \cdot \rho_w \cdot g}{\rho_w \cdot g} \right)$$

or

$$\Delta H_c = UCF \cdot \Delta H_{Hg, c} \cdot \left(\frac{\rho_{Hg}}{\rho_w} - 1 \right) = UCF \cdot \Delta H_{Hg, c} \cdot (G_{Hg} - 1)$$

Notes:

- 1) For definitions of notation, see X1.2.1 through X1.2.3.2.
- 2) For this case, the capillary head loss is a positive value since the total head on the headwater side would have to be increased to make the mercury levels equal.

NOTE 1—For this case capillary head loss is a positive value since the total head on the headwater side would have to be increased to make the mercury levels equal.

FIG. X1.2 Difference in Capillary Head in Mercury U-tube Manometer

by the manometer has to be converted to a pressure head by dividing it by $\rho_w \cdot g$. As shown in Fig. X1.2, the capillary head loss is

$$\Delta H_c = \Delta H_{Hg,c} \cdot (G_{Hg} - 1) \quad (X1.4)$$

where:

- ΔH_c = capillary head loss in hydraulic system, m of water,
- $\Delta H_{Hg,c}$ = differential height of mercury in the tailwater/outflow and headwater/inflow tubes of the manometer with equal pressure applied to each tube, m of mercury, and
- G_{Hg} = specific gravity of mercury at test/trial temperature, see Table 1.

X1.2.3.2.1 As shown in Fig. X1.2, the application of either ΔH_c or $\Delta H_{Hg,c}$ is only necessary when $\Delta H_{Hg,c}$ is equal to or greater than 0.0005 m or 0.5 mm (0.02 in.). An explanation of how to measure ΔH_c is given in X1.4.

X1.2.4 **Total Head**—As stated above, the total head equals the sum of the pressure, elevation, velocity, and capillary heads. Since the change in velocity head is assumed to be zero and the change in capillary head is included in the pressure head calculation given above (Eq X1.3), the total head relationships at various points/elevations, as shown in Fig. X1.3, may be expressed as follows.

X1.2.4.1 Assuming there are no head losses in the tubing, the pressure head at Point Z ($H_{p,Z}$) equals the pressure head just above Point B ($H_{p,B}$), before the effect of capillary head; therefore

$$H_{p,Z} = H_{p,B} \text{ and } H_{p,X} = H_{p,B'}$$

X1.2.4.2 By definition, the total head just above Point B equals the pressure plus elevation heads at that point, therefore

$$H_B = H_{p,B} + H_{e,OB} = \frac{U_b}{\rho_w \cdot g} + H_{e,OB}$$

and

$$H_{B'} = H_{p,B'} + H_{e,O'B'}$$

Assuming continuity in hydraulics, the pressure head at Point B' equals

$$H_{p,B'} = \frac{U_b}{\rho_w \cdot g} + \Delta H_{p,c,B} + H_{p,BA} + H_{p,AO} - H_{p,O'A'} - \Delta H_{p,c,A'} - H_{p,A'B'}$$

where:

- $\Delta H_{p,c,B}$ = the change in capillary pressure head going from just above Point B to just below it, and
- $\Delta H_{p,c,A'}$ = the change in capillary pressure head going from just below Point A' to just above it.

X1.3 **Total Head Loss**—Based on a detailed review of the hydraulic systems shown in Fig. X1.1, Fig. X1.2, and Fig. X1.3, one can come to the conclusion that the flow of fluid (permeant) will only occur when the difference in the mercury heights in the U-tube manometer is greater than equilibrium value, as shown in Fig. X1.2.

X1.3.1 **Initial Head Loss**—Using the total head discussion given above and the notation given in Fig. X1.3; the initial total-head loss (Δh_1) across the specimen is:

$$\Delta h_1 = H_X - H_Z = H_{B'} - H_B \quad (X1.5)$$

or

$$\Delta h_1 = H_{p,B'} - H_{p,B} =$$

$$= \frac{U_b}{\rho_w \cdot g} + \Delta H_{p,c,B} + H_{p,BA} + H_{p,AO}$$

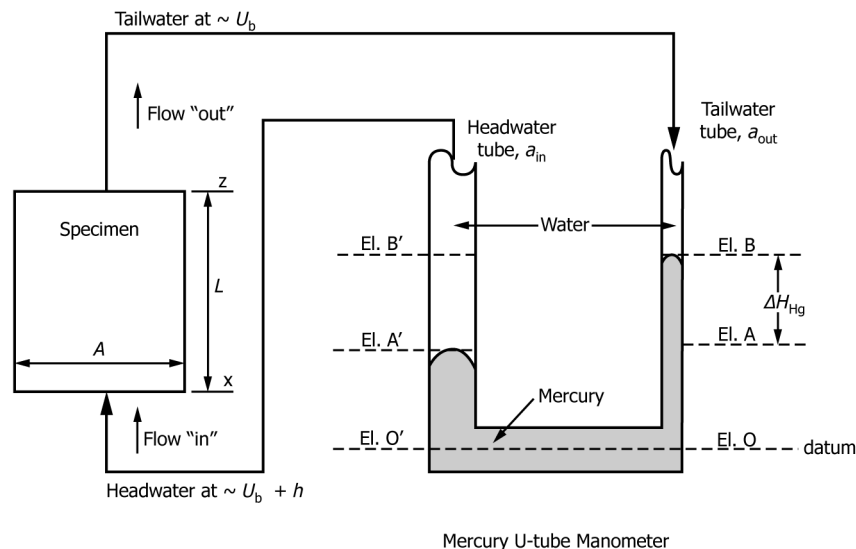


FIG. X1.3 Schematic of Mercury Constant Volume – Falling Head Hydraulic System for Head and Head Loss Equations

$$-H_{p,O'A'} - \Delta H_{p,c,A'} - H_{p,A'B'} - H_{p,B}$$

Since

$$H_{p,b} = \frac{U_b}{\rho_w \cdot g}, \quad (X1.6)$$

$$H_{p,AO} = H_{p,O'A'}, \text{ and}$$

$$\Delta H_c = \Delta H_{p,c,B} - \Delta H,$$

therefore

$$\Delta h_1 = H_{p,BA} - H_{p,A'B'} + \Delta H_c \quad (X1.7)$$

$$\Delta h_1 = \frac{\Delta H_{Hg,1} \cdot g}{\rho_w} - \Delta H_{Hg,1} + \Delta H_{Hg,c} \times \left(\frac{\rho_{Hg}}{\rho_w} - 1 \right) \quad (X1.8)$$

and rearranging, therefore

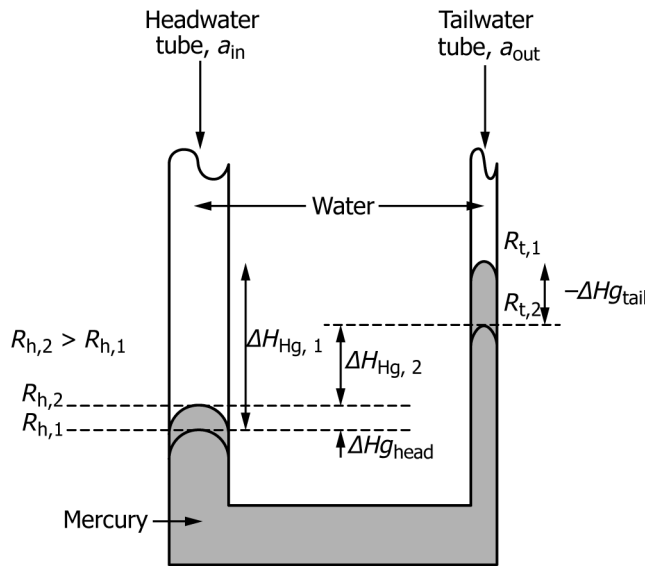
$$\Delta h_1 = (\Delta H_{Hg} + \Delta H_{Hg,c}) \cdot \left(\frac{\rho_{Hg}}{\rho_w} - 1 \right) \quad (X1.9)$$

$$\Delta h_1 = (\Delta H_{Hg,1} + \Delta H_{Hg,c}) \cdot (G_{Hg} - 1)$$

where:

Δh_{Hg} = the initial total-head loss at the start (t_1) of a given permeation trial, in m of water,

Using generic notation instead of specific notation as given in Fig. X1.1 to Fig. X1.4, therefore



"Flow" Functions:

$$\text{Flow "in"} = \Delta Q_{in} = \Delta Hg_{head} \cdot a_{in}$$

$$\text{Flow "out"} = -\Delta Q_{out} = -\Delta Hg_{tail} \cdot a_{out}$$

$$\text{Based on continuity, } \Delta Q_{in} = -\Delta Q_{out}$$

$$\text{Therefore, } \Delta Hg_{head} \cdot a_{in} = -\Delta Hg_{tail} \cdot a_{out}$$

or

$$\Delta Hg_{head} = -\Delta Hg_{tail} \cdot \frac{a_{out}}{a_{in}}$$

"Head" Functions:

$$\Delta H_{Hg,2} = \Delta H_{Hg,1} + (-\Delta Hg_{tail}) - \Delta Hg_{head}$$

Substituting for ΔHg_{head} ,

$$\Delta H_{Hg,2} = \Delta H_{Hg,1} + (-\Delta Hg_{tail}) + \left(-\Delta Hg_{tail} \cdot \frac{a_{out}}{a_{in}} \right)$$

$$\Delta H_{Hg,2} = \Delta H_{Hg,1} + (-\Delta Hg_{tail}) \cdot \left(\frac{a_{out}}{a_{in}} + 1 \right)$$

Substituting for $\Delta H_{Hg,2}$ in the final head loss equation (X1.10) and rearranging,

$$\Delta h_2 = \left(\Delta H_{Hg,1} + \Delta H_{Hg,c} + (-\Delta Hg_{tail}) \cdot \left(\frac{a_{out}}{a_{in}} + 1 \right) \right) \cdot (G_{Hg} - 1)$$

or

$$\Delta h_2 = \Delta h_1 + \left((-\Delta Hg_{tail}) \cdot \left(\frac{a_{out}}{a_{in}} + 1 \right) \right) \cdot (G_{Hg} - 1)$$

FIG. X1.4 Relationship Between Change in Flow and Total Head Loss

ΔH_{Hg} = the initial differential height of mercury in the tailwater and headwater tubes of the manometer at the start (t_1) of a given permeation trial, in m,
 $\Delta H_{\text{Hg},c}$ = the positive differential height of mercury in the tailwater and headwater tubes of the manometer with equal pressures applied to both tubes, in m. This height differential is caused by the difference in capillary pressure heads within the two tubes making up the mercury U-tube manometer, see X1.2.3.2 and X1.4

ΔH

X1.3.2 *Final Head Loss*—The final total-head loss (Δh_2) across the specimen is

$$\Delta h_2 = (\Delta H_{\text{Hg},2} + \Delta H_{\text{Hg},c}) \cdot (G_{\text{Hg}} - 1) \quad (\text{X1.10})$$

where:

Δh_2 = the final total-head loss at the end (t_2) Of a given permeation trial, in m of water,
 $\Delta H_{\text{Hg},2}$ = the final differential height of mercury in the tailwater and headwater tubes of the manometer at the end (t_2) of a given permeation trial, in m.

X1.3.2.1 The determination of $\Delta H_{\text{Hg},2}$ requires two readings; that is, the elevation of the top surfaces of the mercury (meniscus) in the tailwater and headwater tubes. Each of these readings will have some error, especially the headwater reading. In addition, the change in the headwater readings between t_1 and t_2 is typically very small and at about the sensitivity to which readings can be made/estimated. Because of these factors, it is assumed that the accuracy of Δh_2 can be increased by just measuring the change in elevation of the top surface mercury level in the tailwater tube and calculating what $\Delta H_{\text{Hg},2}$ should be based on the area relationships between the tailwater and headwater tubes. As shown in Fig. X1.4, the following flow relationships can be established.

$$\Delta Q_{\text{in}} = \Delta H_{g_{\text{head}}} \cdot a_{\text{in}}$$

$$-\Delta Q_{\text{out}} = -\Delta H_{g_{\text{tail}}} \cdot a_{\text{out}}$$

NOTE X1.1—the symbol for height (H) has been omitted to keep the notation simpler, and

ΔQ_{in} = inflow of permeant water for given time interval (positive units of volume),
 $-\Delta Q_{\text{out}}$ = outflow of permeant water for given time interval (negative units of volume),
 $\Delta H_{g_{\text{head}}}$ = positive change in elevation of the mercury level (top of meniscus) in the headwater tube (units of distance)
 $-\Delta H_{g_{\text{tail}}}$ = negative change in elevation of the mercury level (top of meniscus) in the tailwater tube (units of distance),
 a_{in} = area of the headwater/inflow tube containing mercury (units of area), and
 a_{out} = area of the tailwater/outflow tube containing mercury (units of area)

Based on continuity of flow in a saturated specimen at constant volume,

$$\Delta Q_{\text{in}} = -\Delta Q_{\text{out}}$$

therefore,

$$\Delta H_{g_{\text{head}}} \cdot a_{\text{in}} = -\Delta H_{g_{\text{tail}}} \cdot a_{\text{out}} \quad (\text{X1.11})$$

or

$$\Delta H_{g_{\text{head}}} = -\Delta H_{g_{\text{tail}}} \cdot \frac{a_{\text{out}}}{a_{\text{in}}}$$

therefore,

$$\Delta H_{\text{Hg},2} = \Delta H_{\text{Hg},1} + (-\Delta H_{g_{\text{tail}}}) + \left(\Delta H_{g_{\text{tail}}} \cdot \frac{a_{\text{out}}}{a_{\text{in}}} \right) \quad (\text{X1.12})$$

or

$$\Delta H_{\text{Hg},2} = \Delta H_{\text{Hg},1} + (-\Delta H_{g_{\text{tail}}}) \cdot \left(\frac{a_{\text{out}}}{a_{\text{in}}} + 1 \right)$$

Substituting for $\Delta H_{\text{Hg},2}$ and rearranging

$$\Delta h_2 = \left(\Delta H_{\text{Hg},1} + \Delta H_{\text{Hg},c} + (-\Delta H_{g_{\text{tail}}}) \cdot \left(\frac{a_{\text{out}}}{a_{\text{in}}} + 1 \right) \right) \cdot (G_{\text{Hg}} - 1) \quad (\text{X1.13})$$

or

$$\Delta h_2 = \Delta h_1 + \left((-\Delta H_{g_{\text{tail}}}) \cdot \left(\frac{a_{\text{out}}}{a_{\text{in}}} + 1 \right) \right) \cdot (G_{\text{Hg}} - 1)$$

X1.4 *Capillary Head Measurements*—The key to measuring the difference in capillary head (ΔH_c) between the headwater and tailwater tubes of the mercury U-tube manometer is to ensure that an equal water pressure is applied to both tubes. In addition, flow of water can occur under that equal water pressure. This can be accomplished by individually connecting the tailwater and headwater tubing lines to clean burettes containing water at equal elevation. These lines can not have any air bubbles in them. Then, apply the same air pressure to these two burettes. This air pressure should be similar to the back pressure applied during testing. Finally, adjust the height of one burette (typically the one connected to the headwater line) so the water level within each burette is equal. In making this height adjustment, make sure the water level in the headwater burette starts out below that of the tailwater burette. This simulates the direction of fluid flow during the test.

X1.4.1 Once the water level in the two burettes are level, determine the difference in elevation of the two mercury columns at the tops of their menisci. The mercury level in the tailwater tube (one with a smaller ID) should be below that in the headwater tube. If it is not, there is an error in applying equal pressures to the two tubes of the U-tube manometer, check for air in the lines, external pressure source, etc..

X1.4.2 If the mercury U-tube manometer being used is the version in which the tailwater tube is contained within the headwater tube, a different approach has to be used. A different approach is required since the mercury level in the tailwater tube is not visible at equilibrium. One approach would be to raise the headwater burette until the mercury levels (top of menisci) in the U-tube manometer (headwater and tailwater columns) are equal. Then determine the difference in elevation of the water levels in the headwater and tailwater burettes in m of water,

Next, convert ΔH_c to $\Delta H_{\text{Hg},c}$ with

$\Delta H_{\text{Hg,c}} = \Delta H_{\text{c}}(G_{\text{Hg}} - 1) = \Delta H_{\text{c}}12.74$, in m of mercury. The value of 12.57 is good for temperatures ranging between 15°C and 25°C.

X1.5 Falling-Head Hydraulic Conductivity Equation—Darcy's law for hydraulic conductivity in a saturated medium requires that:

$$q = k \cdot i \cdot A = k \cdot \frac{\Delta h}{L} \cdot A \quad (\text{X1.14})$$

or

$$\Delta Q = k \cdot \frac{\Delta h}{L} \cdot A \cdot \Delta t$$

where:

- q = rate of flow of the fluid (units of volume over time, m³/s),
- k = hydraulic conductivity or coefficient of permeability (units of length over time, m/s),
- i = hydraulic gradient (no unit),
- Δh = total head loss across a given length/test specimen (unit of height of water, m),
- L = given length (test specimen) over which the total head loss occurs (unit of distance, m),
- ΔQ = volume of flow for a given time interval (unit of volume, m³), and
- Δt = time interval (unit of time, s).

X1.5.1 For a differential volume of flow and time period, this equation becomes

$$d\Delta Q = k \cdot \frac{\Delta h}{L} \cdot A \cdot d\Delta t \quad (\text{X1.15})$$

where:

- $d\Delta Q$ = differential volume of flow in a differential time period, and
- $d\Delta t$ = differential time period.

X1.5.2 It can be demonstrated that the differential volume of flow is a function of the differential head loss, as shown below.

From **Fig. X1.4** or **X1.3.2.1**;

$$-\Delta Q = -\Delta H_{\text{g,tail}} \cdot a_{\text{out}} \quad (\text{X1.16})$$

or

$$-d\Delta Q = -d\Delta H_{\text{g,tail}} \cdot a_{\text{out}}$$

and from **Fig. X1.4**,

$$-d\Delta H_{\text{g,tail}} = \frac{dh}{1} \cdot \frac{a_{\text{in}}}{a_{\text{out}} + a_{\text{in}}} \cdot \frac{1}{(G_{\text{Hg}} - 1)} \quad (\text{X1.17})$$

By substituting for $-d\Delta Q$ and $-d\Delta H_{\text{g,tail}}$ from the above equations in **Eq X1.15** we get,

$$-\frac{dh}{1} \cdot \frac{a_{\text{in}} \cdot a_{\text{out}}}{a_{\text{out}} + a_{\text{in}}} \cdot \frac{1}{(G_{\text{Hg}} - 1)} = k \cdot \frac{\Delta h}{L} \cdot A \cdot d\Delta t \quad (\text{X1.18})$$

or

$$d\Delta t = -\frac{L}{A} \cdot \frac{1}{k} \cdot \frac{1}{\Delta h} \cdot \frac{1}{(G_{\text{Hg}} - 1)} \cdot \frac{a_{\text{in}} \cdot a_{\text{out}}}{a_{\text{out}} + a_{\text{in}}} \cdot dh$$

By integrating between times t_1 and t_2 and h_1 and h_2 we get,

$$\int_{t_1}^{t_2} d\Delta t = \int_{\Delta h_1}^{\Delta h_2} -\frac{L}{A} \cdot \frac{1}{k} \cdot \frac{1}{\Delta h} \cdot \frac{1}{(G_{\text{Hg}} - 1)} \cdot \frac{a_{\text{in}} \cdot a_{\text{out}}}{a_{\text{out}} + a_{\text{in}}} \cdot dh$$

which yields the general constant volume-falling head equation,

$$\Delta t = -\frac{L}{A} \cdot \frac{1}{k} \cdot \frac{1}{(G_{\text{Hg}} - 1)} \cdot \frac{a_{\text{in}} \cdot a_{\text{out}}}{a_{\text{out}} + a_{\text{in}}} \ln \left(\frac{\Delta h_2}{\Delta h_1} \right) \quad (\text{X1.19})$$

or

$$k = -\frac{L}{A} \cdot \frac{1}{\Delta t} \cdot \frac{1}{(G_{\text{Hg}} - 1)} \cdot \frac{a_{\text{in}} \cdot a_{\text{out}}}{a_{\text{out}} + a_{\text{in}}} \ln \left(\frac{\Delta h_2}{\Delta h_1} \right)$$

Noting that,

$$-\ln \frac{\Delta h_2}{\Delta h_1} = \ln \frac{\Delta h_1}{\Delta h_2}$$

the equation becomes,

$$k = \left(\frac{a_{\text{in}} \cdot a_{\text{out}}}{(a_{\text{out}} + a_{\text{in}})} \cdot \frac{1}{(G_{\text{Hg}} - 1)} \right) \cdot \frac{L}{A} \cdot \frac{1}{\Delta t} \cdot \ln \left(\frac{\Delta h_1}{\Delta h_2} \right) \quad (\text{X1.20})$$

where:

- k = hydraulic conductivity of the test specimen at the test temperature, m/s,
- Δh_1 = total head loss across length L at the start of a permeation trial; that is, initial total head loss, m of water,
- Δh_2 = final total head loss across length L at the end of a permeation trial; that is, final total head loss, m of water,
- Δt = elapsed time during a permeation trial; that is, $\Delta t = t_2 - t_1$, s,
- a_{out} = area of the tailwater tube (tube with smaller ID), m²,
- a_{in} = area of the headwater tube, in m²,
- G_{Hg} = specific gravity of mercury ($\rho_{\text{Hg}}/\rho_{\text{w}}$) at the test temperature,
- ρ_{Hg} = density of mercury at the test temperature, in Mg/m³,
- ρ_{w} = density of water at the test temperature, Mg/m³,
- $(G_{\text{Hg}} - 1)$ = constant equal to 12.57 between 15 and 25°C,
- L = length/height of the test specimen, in m,
- A = area of the test specimen, in m²,
- $\ln$ = natural logarithm (base e),

and head loss equations from **X1.3.1** and **X1.3.2** are

$$\Delta h_1 = (\Delta H_{\text{Hg,1}} + \Delta H_{\text{Hg,c}}) \cdot (G_{\text{Hg}} - 1) \quad (\text{X1.21})$$

$$\Delta h_2 = \Delta h_1 + \left((-\Delta H_{\text{g,tail}}) \cdot \left(\frac{a_{\text{out}}}{a_{\text{in}}} + 1 \right) \cdot (G_{\text{Hg}} - 1) \right) \quad (\text{X1.22})$$

where for Δh_1 and Δh_2 :

- $\Delta H_{\text{Hg,1}}$ = the initial difference in height of mercury in the tailwater and headwater tubes of the manometer at the start (t_1) of a given permeation trial, m;
- $\Delta H_{\text{Hg,c}}$ = the positive difference in height of mercury in the tailwater and headwater tubes of the manometer with equal pressure applied to both tubes, m, (This height differential is caused by the difference in capillary pressure of the two tubes making up the mercury U-tube manometer); and,

$-\Delta H_{g_{\text{tail}}}$ = the negative change in height of the mercury level in the tailwater tube of the manometer during a given permeation trial, m.

X2. RELATIONSHIP BETWEEN CHANGE IN AXIAL STRAIN AND HYDRAULIC CONDUCTIVITY OF TEST SPECIMEN

X2.1 Introduction—It is important to understand how hydraulic conductivity (k) changes with changes in void ratio, dry unit weight, or volume change of a given test specimen. With this understanding, one can thereby know the accuracy to which volume changes need to be controlled while testing. For instance, if k is very sensitive to volume changes then the effective consolidation stress needs to be accurately controlled; also, any volume changes during the back-pressuring process would have to be minimized.

X2.1.1 The relationships presented in this appendix are for relatively plastic clays, CL or CH, with a plasticity index greater than about 10, but less than about 50.

X2.2 Change in Void Ratio versus Hydraulic Conductivity—A graphical representation of how hydraulic conductivity (k) varies with void ratio (e) is presented in Fig. X2.1. This figure graphically shows that the k decreases as e decreases. In addition, for a given change in void ratio (Δe) the rate of change in k is much more dramatic in the overconsolidated range than in the normally consolidated range. The definition for the terms/notation presented in this figure are:

- k = given hydraulic conductivity, m/s,
- k_L = lower bound k for given percent change in k , m/s,
- k_U = upper bound k for given percent change in k , m/s,
- Δk_L = lower bound change in k for given percent change (decimal form) in k , m/s/
- % Change = the percent change (decimal form) in k ,
- Δe_L = the compressive change in void ratio (decreasing change)

e_U = the swelling change in void ratio (increasing change), and
 m = the ratio of Δe to $\Delta \log k = \Delta e / \log(k/k_L)$, 1/(m/s).

X2.3 Mathematical Relationship Between e and k —It can be shown that:

$$k_L = k + (-\Delta k_L) = k - \% \text{Change} \times k = (1 - \% \text{Change}) \times k \quad (\text{X2.1})$$

or

$$k_U = (1 + \% \text{Change}) \times k$$

$$-\Delta k_L = \log(1 - \% \text{Change}) \quad (\text{X2.2})$$

or

$$\Delta k_U = \log(1 + \% \text{Change})$$

$$e_L = e + (-\Delta e_L) \quad (\text{X2.3})$$

or

$$e_U = e + \Delta e_U$$

$$-\Delta e_L = m \times -\Delta k_L = m \times \log(1 - \% \text{Change}), \quad (\text{X2.4})$$

or

$$\Delta e_U = m \times \Delta k_U = m \times \log(1 + \% \text{Change})$$

X2.4 Based upon the theory of elasticity, the following relationships between axial strain (ϵ_a) and volumetric strain (ϵ_v), and volumetric strain and change in void ratio (Δe) are:

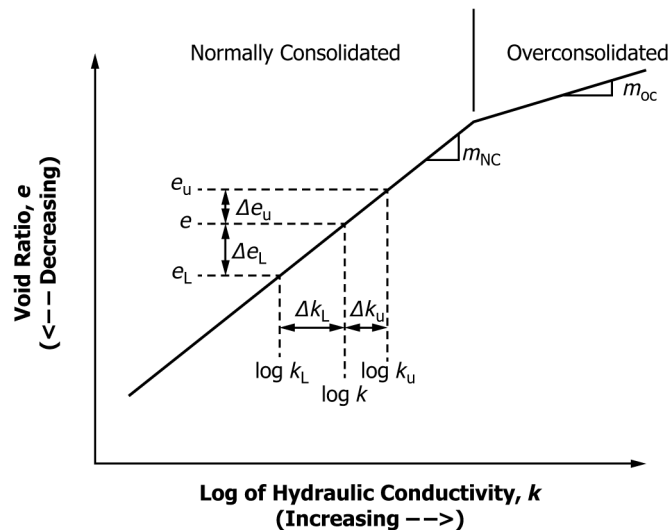


FIG. X2.1 Typical Relationship Between Void Ratio and Logarithm of Hydraulic Conductivity

$$\Delta \varepsilon_a = \frac{1}{3} \cdot E_c \cdot \Delta \varepsilon_v = \left(\frac{1}{3} \cdot E_c \cdot -\Delta \varepsilon_L \right) / (1 - e_o) \quad (\text{X2.5})$$

- $\Delta \varepsilon_a$ = the change in compressive axial strain ($\Delta L/L_o$), m/m.
 E_c = a constant to correct for the non-elastic response of hydraulic-conductivity test specimens. For test specimens having a height to diameter ratio of about one, this value is about 0.8 for normally consolidated (NC) specimens and 0.6 for overconsolidated (OC) specimens,
 $\Delta \varepsilon_v$ = the change in compressive volumetric strain ($\Delta V/V_o$), m³/m³.
 $-\Delta e$ = the compressive change in void ratio; that is, decreasing change,
 e_o = the initial void ratio.

X2.5 Combining the above equations and rearranging:

$$\Delta \varepsilon_{a,NC} = \frac{-1}{3 \cdot E_c} \cdot \frac{m_{NC}}{1 + e_o} \cdot \log(1 - \% \text{Change}) \quad (\text{X2.6})$$

or

$$\Delta \varepsilon_{a,OC} = \frac{-1}{3 \cdot E_c} \cdot \frac{m_{OC}}{1 + e_o} \cdot \log(1 - \% \text{Change})$$

also

$$-\% \text{Change} = 10^{-\left(3 \cdot E_c \cdot \left(\frac{1 + e_o}{m}\right) \cdot \Delta \varepsilon_{a,L}\right)} \quad (\text{X2.7})$$

$$+\% \text{Change} = 10^{-\left(3 \cdot E_c \cdot \left(\frac{1 + e_o}{m}\right) \cdot -\Delta \varepsilon_{a,U}\right)}$$

where:

- m_{NC} = m value in normally consolidated region, 1/[m/s], and
 m_{OC} = m value in the overconsolidated region, 1/[m/s].

X2.5.1 Tavenas, et al. (12) indicates that m is about $(1/3 \text{ to } 1/2) \times e_o$ for normally consolidated clays. For overconsolidated clay, it is assumed that m in the overconsolidated range is reduced by the same ratio that the compression index (C) is when going from the normally consolidated region (C_{NC}) to the overconsolidated region (C_{OC}). Therefore,

$$m_{OC}/m_{NC} = C_{OC}/C_{NC} \sim 0.185 + 0.002 \times PI$$

where PI = plasticity index. This assumption is based on limited data, and in some cases, there was not any significant difference in m between the normally consolidated and overconsolidated regions. Based on the above, an initial void ratio of 0.8 and a plasticity index (PI) of 30, m_{NC} is about 0.33; while m_{OC} is about 0.082.

X2.5.2 Using the above m values and equations, and a % Change equal to 12.5 %, one could assume excess axial strains caused by a poor testing protocol should be less than the following values:

For normally consolidated soils:

$$\Delta \varepsilon_{a,NC} = \frac{-1}{3 \cdot 0.8} \cdot \frac{0.33}{1.8} \cdot \log 0.875 = 0.0044 = 0.4 \%$$

For overconsolidated consolidated soils:

$$\Delta \varepsilon_{a,OC} = \frac{-1}{3 \cdot 0.6} \cdot \frac{0.082}{1.8} \cdot \log 0.875 = 0.00147 = 0.1 \%$$

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SUMMARY OF CHANGES

In accordance with Committee D18 policy, this section identifies the location of changes to this standard since the last edition (2016a) that may impact the use of this standard. (June 15, 2024)

(1) Revised **9.3.3**.

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Designation: D698 – 12 (Reapproved 2021)

Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft³ (600 kN-m/m³))¹

This standard is issued under the fixed designation D698; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ϵ) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the U.S. Department of Defense.

1. Scope

1.1 These test methods cover laboratory compaction methods used to determine the relationship between molding water content and dry unit weight of soils (compaction curve) compacted in a 4 or 6-in. (101.6 or 152.4-mm) diameter mold with a 5.50-lbf (24.5-N) rammer dropped from a height of 12.0 in. (305 mm) producing a compactive effort of 12 400 ft-lbf/ft³ (600 kN-m/m³).

NOTE 1—The equipment and procedures are similar as those proposed by R. R. Proctor (*Engineering News Record*—September 7, 1933) with this one major exception: his rammer blows were applied as “12 inch firm strokes” instead of free fall, producing variable compactive effort depending on the operator, but probably in the range 15 000 to 25 000 ft-lbf/ft³ (700 to 1200 kN-m/m³). The standard effort test (see 3.1.4) is sometimes referred to as the Proctor Test.

1.1.1 Soils and soil-aggregate mixtures are to be regarded as natural occurring fine- or coarse-grained soils, or composites or mixtures of natural soils, or mixtures of natural and processed soils or aggregates such as gravel or crushed rock. Hereafter referred to as either soil or material.

1.2 These test methods apply only to soils (materials) that have 30 % or less by mass of particles retained on the $\frac{3}{4}$ -in. (19.0-mm) sieve and have not been previously compacted in the laboratory; that is, do not reuse compacted soil.

1.2.1 For relationships between unit weights and molding water contents of soils with 30 % or less by mass of material retained on the $\frac{3}{4}$ -in. (19.0-mm) sieve to unit weights and molding water contents of the fraction passing $\frac{3}{4}$ -in. (19.0-mm) sieve, see Practice D4718/D4718M.

1.3 Three alternative methods are provided. The method used shall be as indicated in the specification for the material being tested. If no method is specified, the choice should be based on the material gradation.

1.3.1 Method A:

1.3.1.1 *Mold*—4-in. (101.6-mm) diameter.

1.3.1.2 *Material*—Passing No. 4 (4.75-mm) sieve.

1.3.1.3 *Layers*—Three.

1.3.1.4 *Blows per Layer*—25.

1.3.1.5 *Usage*—May be used if 25 % or less (see 1.4) by mass of the material is retained on the No. 4 (4.75-mm) sieve.

1.3.1.6 *Other Usage*—If this gradation requirement cannot be met, then Method C may be used.

1.3.2 Method B:

1.3.2.1 *Mold*—4-in. (101.6-mm) diameter.

1.3.2.2 *Material*—Passing $\frac{3}{8}$ -in. (9.5-mm) sieve.

1.3.2.3 *Layers*—Three.

1.3.2.4 *Blows per Layer*—25.

1.3.2.5 *Usage*—May be used if 25 % or less (see 1.4) by mass of the material is retained on the $\frac{3}{8}$ -in. (9.5-mm) sieve.

1.3.2.6 *Other Usage*—If this gradation requirement cannot be met, then Method C may be used.

1.3.3 Method C:

1.3.3.1 *Mold*—6-in. (152.4-mm) diameter.

1.3.3.2 *Material*—Passing $\frac{3}{4}$ -in. (19.0-mm) sieve.

1.3.3.3 *Layers*—Three.

1.3.3.4 *Blows per Layer*—56.

1.3.3.5 *Usage*—May be used if 30 % or less (see 1.4) by mass of the material is retained on the $\frac{3}{4}$ -in. (19.0-mm) sieve.

1.3.4 The 6-in. (152.4-mm) diameter mold shall not be used with Method A or B.

NOTE 2—Results have been found to vary slightly when a material is tested at the same compactive effort in different size molds, with the smaller mold size typically yielding larger values of density/unit weight (1, pp. 21+).²

1.4 If the test specimen contains more than 5 % by mass of oversize fraction (coarse fraction) and the material will not be included in the test, corrections must be made to the unit mass and molding water content of the specimen or to the appropriate field-in-place density test specimen using Practice D4718/D4718M.

1.5 This test method will generally produce a well-defined maximum dry unit weight for non-free draining soils. If this

¹ These Test Methods are under the jurisdiction of ASTM Committee D18 on Soil and Rock and are the direct responsibility of Subcommittee D18.03 on Texture, Plasticity and Density Characteristics of Soils.

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² The boldface numbers in parentheses refer to the list of references at the end of this standard.

*A Summary of Changes section appears at the end of this standard

test method is used for free-draining soils the maximum unit weight may not be well defined, and can be less than obtained using Test Methods [D4253](#).

1.6 All observed and calculated values shall conform to the guidelines for significant digits and rounding established in Practice [D6026](#), unless superseded by this standard.

1.6.1 For purposes of comparing measured or calculated value(s) with specified limits, the measured or calculated value(s) shall be rounded to the nearest decimal or significant digits in the specified limits.

1.6.2 The procedures used to specify how data are collected/recorded or calculated, in this standard are regarded as the industry standard. In addition, they are representative of the significant digits that generally should be retained. The procedures used do not consider material variation, purpose for obtaining the data, special purpose studies, or any considerations for the user's objectives; and it is common practice to increase or reduce significant digits of reported data to be commensurate with these considerations. It is beyond the scope of this standard to consider significant digits used in analytical methods for engineering design.

1.7 The values in inch-pound units are to be regarded as the standard. The values stated in SI units are provided for information only, except for units of mass. The units for mass are given in SI units only, g or kg.

1.7.1 It is common practice in the engineering profession to concurrently use pounds to represent both a unit of mass (lbm) and a force (lbf). This implicitly combines two separate systems of units; that is, the absolute system and the gravitational system. It is scientifically undesirable to combine the use of two separate sets of inch-pound units within a single standard. This standard has been written using the gravitational system of units when dealing with the inch-pound system. In this system, the pound (lbf) represents a unit of force (weight). However, the use of balances or scales recording pounds of mass (lbm) or the recording of density in lbm/ft³ shall not be regarded as a nonconformance with this standard.

1.8 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.9 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:³

[C127](#) Test Method for Relative Density (Specific Gravity) and Absorption of Coarse Aggregate

[C136/C136M](#) Test Method for Sieve Analysis of Fine and Coarse Aggregates

[D653](#) Terminology Relating to Soil, Rock, and Contained Fluids

[D854](#) Test Methods for Specific Gravity of Soil Solids by Water Pycnometer

[D2168](#) Practices for Calibration of Laboratory Mechanical-Rammer Soil Compactors

[D2216](#) Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass

[D2487](#) Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)

[D2488](#) Practice for Description and Identification of Soils (Visual-Manual Procedures)

[D3740](#) Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

[D4253](#) Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table

[D4718/D4718M](#) Practice for Correction of Unit Weight and Water Content for Soils Containing Oversize Particles

[D4753](#) Guide for Evaluating, Selecting, and Specifying Balances and Standard Masses for Use in Soil, Rock, and Construction Materials Testing

[D4914/D4914M](#) Test Methods for Density of Soil and Rock in Place by the Sand Replacement Method in a Test Pit

[D5030/D5030M](#) Test Methods for Density of In-Place Soil and Rock Materials by the Water Replacement Method in a Test Pit

[D6026](#) Practice for Using Significant Digits and Data Records in Geotechnical Data

[D6913/D6913M](#) Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis

[E11](#) Specification for Woven Wire Test Sieve Cloth and Test Sieves

[E177](#) Practice for Use of the Terms Precision and Bias in ASTM Test Methods

[E691](#) Practice for Conducting an Interlaboratory Study to Determine the Precision of a Test Method

IEEE/ASTM SI 10 Standard for Use of the International System of Units (SI): the Modern Metric System

3. Terminology

3.1 Definitions:

3.1.1 See Terminology [D653](#) for general definitions.

3.1.2 *molding water content, n* —the adjusted water content of a soil (material) that will be compacted/reconstituted.

3.1.3 *standard effort*—in compaction testing, the term for the 12 400 ft-lbf/ft³ (600 kN-m/m³) compactive effort applied by the equipment and methods of this test.

3.1.4 *standard maximum dry unit weight, $\gamma_{d,max}$ in lb/ft³ (kN/m³)*—in compaction testing, the maximum value defined by the compaction curve for a compaction test using standard effort.

3.1.5 *standard optimum water content, w_{opt} in %*—in compaction testing, the molding water content at which a soil can be compacted to the maximum dry unit weight using standard compactive effort.

³ For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.



3.2 Definitions of Terms Specific to This Standard:

3.2.1 *oversize fraction (coarse fraction), P_C in %*—the portion of total specimen not used in performing the compaction test; it may be the portion of total specimen retained on the No. 4 (4.75-mm) sieve in Method A, $\frac{3}{8}$ -in. (9.5-mm) sieve in Method B, or $\frac{3}{4}$ -in. (19.0-mm) sieve in Method C.

3.2.2 *test fraction (finer fraction), P_F in %*—the portion of the total specimen used in performing the compaction test; it is the fraction passing the No. 4 (4.75-mm) sieve in Method A, passing the $\frac{3}{8}$ -in. (9.5-mm) sieve in Method B, or passing the $\frac{3}{4}$ -in. (19.0-mm) sieve in Method C.

4. Summary of Test Method

4.1 A soil at a selected molding water content is placed in three layers into a mold of given dimensions, with each layer compacted by 25 or 56 blows of a 5.50-lbf (24.47-N) rammer dropped from a distance of 12.00 in. (304.8 mm), subjecting the soil to a total compactive effort of about 12 400 ft-lbf/ft³ (600 kN-m/m³). The resulting dry unit weight is determined. The procedure is repeated for a sufficient number of molding water contents to establish a relationship between the dry unit weight and the molding water content for the soil. This data, when plotted, represents a curvilinear relationship known as the compaction curve. The values of optimum water content and standard maximum dry unit weight are determined from the compaction curve.

5. Significance and Use

5.1 Soil placed as engineering fill (embankments, foundation pads, road bases) is compacted to a dense state to obtain satisfactory engineering properties such as, shear strength, compressibility, or permeability. In addition, foundation soils are often compacted to improve their engineering properties. Laboratory compaction tests provide the basis for determining the percent compaction and molding water content needed to achieve the required engineering properties, and for controlling construction to assure that the required compaction and water contents are achieved.

5.2 During design of an engineered fill, shear, consolidation, permeability, or other tests require preparation of test specimens by compacting at some molding water content to some unit weight. It is common practice to first determine the optimum water content (w_{opt}) and maximum dry unit weight ($\gamma_{d,max}$) by means of a compaction test. Test specimens are compacted at a selected molding water content (w), either wet or dry of optimum (w_{opt}) or at optimum (w_{opt}), and at a selected dry unit weight related to a percentage of maximum dry unit weight ($\gamma_{d,max}$). The selection of molding water content (w), either wet or dry of optimum (w_{opt}) or at optimum (w_{opt}) and the dry unit weight ($\gamma_{d,max}$) may be based on past experience, or a range of values may be investigated to determine the necessary percent of compaction.

5.3 Experience indicates that the methods outlined in 5.2 or the construction control aspects discussed in 5.1 are extremely difficult to implement or yield erroneous results when dealing with certain soils. 5.3.1 – 5.3.3 describe typical problem soils, the problems encountered when dealing with such soils and possible solutions for these problems.

5.3.1 *Oversize Fraction*—Soils containing more than 30 % oversize fraction (material retained on the $\frac{3}{4}$ -in. (19-mm) sieve) are a problem. For such soils, there is no ASTM test method to control their compaction and very few laboratories are equipped to determine the laboratory maximum unit weight (density) of such soils (USDI Bureau of Reclamation, Denver, CO and U.S. Army Corps of Engineers, Vicksburg, MS). Although Test Methods D4914/D4914M and D5030/D5030M determine the “field” dry unit weight of such soils, they are difficult and expensive to perform.

5.3.1.1 One method to design and control the compaction of such soils is to use a test fill to determine the required degree of compaction and the method to obtain that compaction, followed by use of a method specification to control the compaction. Components of a method specification typically contain the type and size of compaction equipment to be used, the lift thickness, acceptable range in molding water content, and the number of passes.

NOTE 3—Success in executing the compaction control of an earthwork project, especially when a method specification is used, is highly dependent upon the quality and experience of the contractor and inspector.

5.3.1.2 Another method is to apply the use of density correction factors developed by the USDI Bureau of Reclamation (2, 3) and U.S. Corps of Engineers (4). These correction factors may be applied for soils containing up to about 50 to 70 % oversize fraction. Each agency uses a different term for these density correction factors. The USDI Bureau of Reclamation uses D ratio (or D -VALUE), while the U.S. Corps of Engineers uses Density Interference Coefficient (I_c).

5.3.1.3 The use of the replacement technique (Test Method D698–78, Method D), in which the oversize fraction is replaced with a finer fraction, is inappropriate to determine the maximum dry unit weight, $\gamma_{d,max}$, of soils containing oversize fractions (4).

5.3.2 *Degradation*—Soils containing particles that degrade during compaction are a problem, especially when more degradation occurs during laboratory compaction than field compaction, as is typical. Degradation typically occurs during the compaction of a granular-residual soil or aggregate. When degradation occurs, the maximum dry-unit weight increases (1, p. 73) so that the laboratory maximum value is not representative of field conditions. Often, in these cases, the maximum dry unit weight is impossible to achieve in the field.

5.3.2.1 Again, for soils subject to degradation, the use of test fills and method specifications may help. Use of replacement techniques is not correct.

5.3.3 *Gap Graded*—Gap-graded soils (soils containing many large particles with limited small particles) are a problem because the compacted soil will have larger voids than usual. To handle these large voids, standard test methods (laboratory or field) typically have to be modified using engineering judgement.

NOTE 4—The quality of the result produced by this standard is dependent on the competence of the personnel performing it, and the suitability of the equipment and facilities used. Agencies that meet the criteria of Practice D3740 are generally considered capable of competent and objective testing/sampling/inspection, and the like. Users of this standard are cautioned that compliance with Practice D3740 does not in itself assure reliable results. Reliable results depend on many factors;

Practice D3740 provides a means of evaluating some of those factors.

6. Apparatus

6.1 Mold Assembly—The molds shall be cylindrical in shape, made of rigid metal and be within the capacity and dimensions indicated in 6.1.1 or 6.1.2 and Figs. 1 and 2. See also Table 1. The walls of the mold may be solid, split, or tapered. The “split” type may consist of two half-round sections, or a section of pipe split along one element, which can be securely locked together to form a cylinder meeting the requirements of this section. The “tapered” type shall have an internal diameter taper that is uniform and not more than 0.200 in./ft (16.7 mm/m) of mold height. Each mold shall have a base plate and an extension collar assembly, both made of rigid metal and constructed so they can be securely attached and easily detached from the mold. The extension collar assembly shall have a height extending above the top of the mold of at least 2.0 in. (51 mm) which may include an upper section that flares out to form a funnel, provided there is at least a 0.75 in. (19 mm) straight cylindrical section beneath it. The extension collar shall align with the inside of the mold. The bottom of the base plate and bottom of the centrally recessed area that accepts the cylindrical mold shall be planar within ± 0.005 in. (± 0.1 mm).

6.1.1 Mold, 4 in.—A mold having a 4.000 ± 0.016 -in. (101.6 ± 0.4 -mm) average inside diameter, a height of 4.584 ± 0.018 in. (116.4 ± 0.5 mm) and a volume of 0.0333 ± 0.0005 ft³ (943.0 ± 14 cm³). A mold assembly having the minimum required features is shown in Fig. 1.

6.1.2 Mold, 6 in.—A mold having a 6.000 ± 0.026 -in. (152.4 ± 0.7 -mm) average inside diameter, a height of 4.584 ± 0.018 in. (116.4 ± 0.5 mm), and a volume of 0.0750 ± 0.0009 ft³ (2124 ± 25 cm³). A mold assembly having the minimum required features is shown in Fig. 2.

6.2 Rammer—A rammer, either manually operated as described further in 6.2.1 or mechanically operated as described in 6.2.2. The rammer shall fall freely through a distance of 12.00 ± 0.05 in. (304.8 ± 1 mm) from the surface of the specimen. The weight of the rammer shall be 5.50 ± 0.02 lbf (24.47 ± 0.09 N, or mass of 2.495 ± 0.009 kg), except that the weight of the mechanical rammers may be adjusted as described in Practices D2168; see Note 5. The striking face of the rammer shall be planar and circular, except as noted in 6.2.2.1,

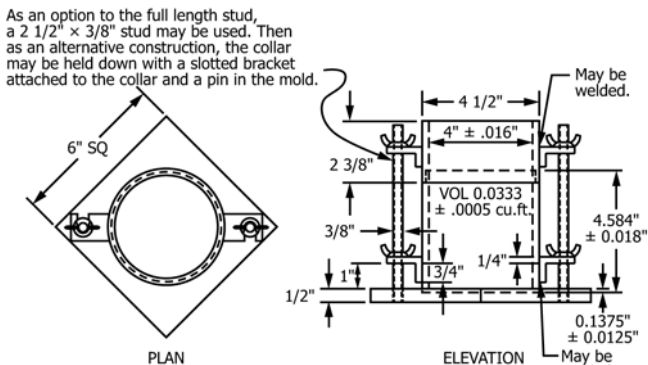


FIG. 1 4.0-in. Cylindrical Mold

As an option to the full length stud, a $2\frac{1}{2}$ " $\times$ $\frac{3}{8}$ " stud may be used. Then as an alternative construction, the collar may be held down with a slotted bracket attached to the collar and a pin in the mold.

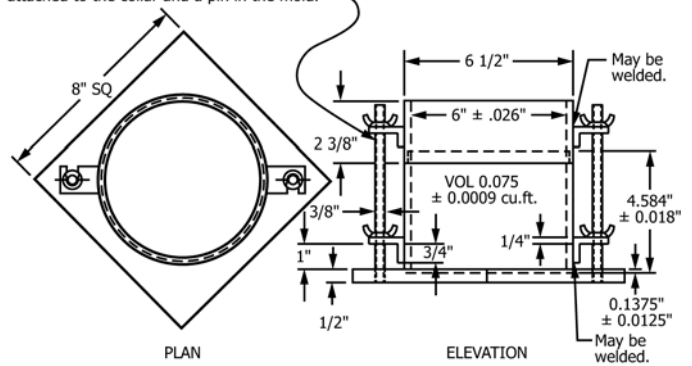


FIG. 2 6.0-in. Cylindrical Mold

TABLE 1 Metric Equivalents for Figs. 1 and 2

in.	mm
0.016	0.41
0.026	0.66
0.032	0.81
0.028	0.71
$\frac{1}{2}$	12.70
$2\frac{1}{2}$	63.50
$2\frac{5}{8}$	66.70
4	101.60
$4\frac{1}{2}$	114.30
4.584	116.43
$4\frac{3}{4}$	120.60
6	152.40
$6\frac{1}{2}$	165.10
$6\frac{5}{8}$	168.30
$6\frac{3}{4}$	171.40
$8\frac{1}{4}$	209.60
ft ³	cm ³
$\frac{1}{30}$ (0.0333)	943
0.0005	14
(0.0750)	2,124
0.0011	31

with a diameter when new of 2.000 ± 0.005 in. (50.80 ± 0.13 mm). The rammer shall be replaced if the striking face becomes worn or bellied to the extent that the diameter exceeds 2.000 ± 0.01 in. (50.80 ± 0.25 mm).

NOTE 5—It is a common and acceptable practice to determine the weight of the rammer using either a kilogram or pound balance and assume 1 lbf is equivalent to 0.4536 kg, 1 lbf is equivalent to 1 lbm, or 1 N is equivalent to 0.2248 lbf or 0.1020 kg.

6.2.1 Manual Rammer—The rammer shall be equipped with a guide sleeve that has sufficient clearance that the free fall of the rammer shaft and head is not restricted. The guide sleeve shall have at least four vent holes at each end (eight holes total) located with centers $\frac{3}{4} \pm \frac{1}{16}$ in. (19 ± 2 mm) from each end and spaced 90 degrees apart. The minimum diameter of the vent holes shall be $\frac{3}{8}$ in. (9.5 mm). Additional holes or slots may be incorporated in the guide sleeve.

6.2.2 Mechanical Rammer-Circular Face—The rammer shall operate mechanically in such a manner as to provide uniform and complete coverage of the specimen surface. There shall be 0.10 ± 0.03 -in. (2.5 ± 0.8 -mm) clearance between the rammer and the inside surface of the mold at its smallest

diameter. The mechanical rammer shall meet the standardization/calibration requirements of Practices D2168. The mechanical rammer shall be equipped with a positive mechanical means to support the rammer when not in operation.

6.2.2.1 Mechanical Rammer-Sector Face—The sector face can be used with the 6-in. (152.4-mm) mold, as an alternative to the circular face mechanical rammer described in 6.2.2. The striking face shall have the shape of a sector of a circle of radius equal to 2.90 ± 0.02 in. (73.7 ± 0.5 mm) and an area about the same as the circular face, see 6.2. The rammer shall operate in such a manner that the vertex of the sector is positioned at the center of the specimen and follow the compaction pattern given in Fig. 3b.

6.3 Sample Extruder (optional)—A jack, with frame or other device adapted for the purpose of extruding compacted specimens from the mold.

6.4 Balance—A Class GP5 balance meeting the requirements of Guide D4753 for a balance of 1-g readability. If the water content of the compacted specimens is determined using a representative portion of the specimen, rather than the whole specimen, and if the representative portion is less than 1000 g, a Class GP2 balance having a 0.1-g readability is needed in order to comply with Test Methods D2216 requirements for determining water content to 0.1 %.

NOTE 6—Use of a balance having an equivalent capacity and a readability of 0.002 lbm as an alternative to a class GP5 balance should not be regarded as nonconformance to this standard.

6.5 Drying Oven—Thermostatically controlled oven, capable of maintaining a uniform temperature of $230 \pm 9^\circ\text{F}$ ($110 \pm 5^\circ\text{C}$) throughout the drying chamber. These requirements typically require the use of a forced-draft type oven. Preferably the oven should be vented outside the building.

6.6 Straightedge—A stiff metal straightedge of any convenient length but not less than 10 in. (250 mm). The total length of the straightedge shall be machined straight to a tolerance of ± 0.005 in. (± 0.1 mm). The scraping edge shall be beveled if it is thicker than $\frac{1}{8}$ in. (3 mm).

6.7 Sieves— $\frac{3}{4}$ in. (19.0 mm), $\frac{3}{8}$ in. (9.5 mm), and No. 4 (4.75 mm), conforming to the requirements of Specification E11.

6.8 Mixing Tools—Miscellaneous tools such as mixing pan, spoon, trowel, spatula, spraying device (to add water evenly),

and (preferably, but optional) suitable mechanical device for thoroughly mixing the subspecimen of soil with increments of water.

7. Standardization/Calibration

7.1 Perform standardizations before initial use, after repairs or other occurrences that might affect the test results, at intervals not exceeding 1,000 test specimens, or annually, whichever occurs first, for the following apparatus:

7.1.1 **Balance**—Evaluate in accordance with Guide D4753.

7.1.2 **Molds**—Determine the volume as described in Annex A1.

7.1.3 **Manual Rammer**—Verify the free fall distance, rammer weight, and rammer face are in accordance with 6.2. Verify the guide sleeve requirements are in accordance with 6.2.1.

7.1.4 **Mechanical Rammer**—Verify and adjust if necessary that the mechanical rammer is in accordance with Practices D2168. In addition, the clearance between the rammer and the inside surface of the mold shall be verified in accordance with 6.2.2.

8. Test Specimen

8.1 The minimum specimen (test fraction) mass for Methods A and B is about 16 kg, and for Method C is about 29 kg of dry soil. Therefore, the field sample should have a moist mass of at least 23 kg and 45 kg, respectively. Greater masses would be required if the oversize fraction is large (see 10.2 or 10.3) or an additional molding water content is taken during compaction of each point (see 10.4.2.1).

8.2 If gradation data is not available, estimate the percentage of material (by mass) retained on the No. 4 (4.75-mm), $\frac{3}{8}$ -in. (9.5-mm), or $\frac{3}{4}$ -in. (19.0-mm) sieve as appropriate for selecting Method A, B, or C, respectively. If it appears the percentage retained of interest is close to the allowable value for a given Method (A, B, or C), then either:

8.2.1 Select a Method that allows a higher percentage retained (B or C).

8.2.2 Using the Method of interest, process the specimen in accordance with 10.2 or 10.3, this determines the percentage retained for that method. If acceptable, proceed, if not go to the next Method (B or C).

8.2.3 Determine percentage retained values by using a representative portion from the total sample, and performing a simplified or complete gradation analysis using the sieve(s) of

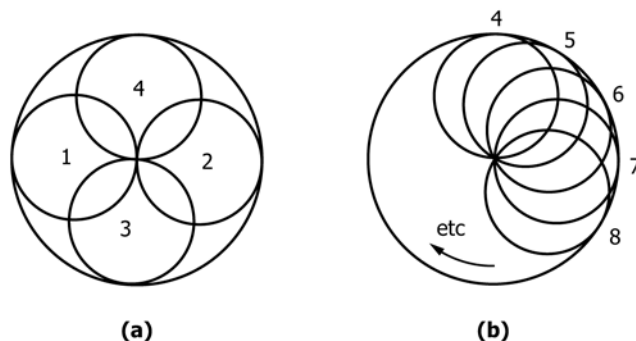


FIG. 3 Rammer Pattern for Compaction in 4 in. (101.6 mm) Mold

interest and Test Methods **D6913/D6913M** or **C136/C136M**. It is only necessary to calculate the retained percentage(s) for the sieve or sieves for which information is desired.

9. Preparation of Apparatus

9.1 Select the proper compaction mold(s), collar, and base plate in accordance with the Method (A, B, or C) being used. Check that its volume is known and determined with or without base plate, free of nicks or dents, and will fit together properly.

NOTE 7—Mass requirements are given in **10.4**.

9.2 Check that the manual or mechanical rammer assembly is in good working condition and that parts are not loose or worn. Make any necessary adjustments or repairs. If adjustments or repairs are made, the rammer must be re-standardized.

10. Procedure

10.1 Soils:

10.1.1 Do not reuse soil that has been previously compacted in the laboratory. The reuse of previously compacted soil yields a significantly greater maximum dry unit weight (**1**, p. 31).

10.1.2 When using this test method for soils containing hydrated halloysite, or in which past experience indicates that results will be altered by air-drying, use the moist preparation method (see **10.2**). In referee testing, each laboratory has to use the same method of preparation, either moist (preferred) or air-dried.

10.1.3 Prepare the soil specimens for testing in accordance with **10.2** (preferred) or with **10.3**.

10.2 *Moist Preparation Method (preferred)*—Without previously drying the sample/specimen, process it over a No. 4 (4.75-mm), $\frac{3}{8}$ -in. (9.5-mm), or $\frac{3}{4}$ -in. (19.0-mm) sieve, depending on the Method (A, B, or C) being used or required as covered in **8.2**. For additional processing details, see Test Methods **D6913/D6913M**. Determine and record the mass of both the retained and passing portions (oversize fraction and test fraction, respectively) to the nearest g. Oven dry the oversize fraction and determine and record its dry mass to the nearest g. If it appears more than 0.5 % of the total dry mass of the specimen is adhering to the oversize fraction, wash that fraction. Then determine and record its oven dry mass to the nearest g. Determine and record the water content of the processed soil (test fraction). Using that water content, determine and record the oven dry mass of the test fraction to the nearest g. Based on these oven dry masses, the percent oversize fraction, P_C , and test fraction, P_F , shall be determined and recorded, unless a gradation analysis has already been performed, see Section **11** on Calculations.

10.2.1 From the test fraction, select and prepare at least four (preferably five) subspecimens having molding water contents such that they bracket the estimated optimum water content. A subspecimen having a molding water content close to optimum should be prepared first by trial additions or removals of water and mixing (see **Note 8**). Select molding water contents for the rest of the subspecimens to provide at least two subspecimens wet and two subspecimens dry of optimum, and molding water contents varying by about 2 %. At least two molding water contents are necessary on the wet and dry side of optimum to define the dry-unit-weight compaction curve (see **10.5**). Some

soils with very high optimum water content or a relatively flat compaction curve may require larger molding water content increments to obtain a well-defined maximum dry unit weight. Molding water content increments should not exceed about 4 %.

NOTE 8—With practice it is usually possible to visually judge a point near optimum water content. Typically, cohesive soils at the optimum water content can be squeezed into a lump that sticks together when hand pressure is released, but will break cleanly into two sections when “bent.” They tend to crumble at molding water contents dry of optimum; while, they tend to stick together in a sticky cohesive mass wet of optimum. The optimum water content is typically slightly less than the plastic limit. While for cohesionless soils, the optimum water content is typically close to zero or at the point where bleeding occurs.

10.2.2 Thoroughly mix the test fraction, then using a scoop select representative soil for each subspecimen (compaction point). Select about 2.3 kg when using Method A or B, or about 5.9 kg for Method C. Test Methods **D6913/D6913M** section on Specimen and Annex A2 gives additional details on obtaining representative soil using this procedure and why it is the preferred method. To obtain the subspecimen’s molding water contents selected in **10.2.1**, add or remove the required amounts of water as follows. To add water, spray it into the soil during mixing; to remove water, allow the soil to dry in air at ambient temperature or in a drying apparatus such that the temperature of the sample does not exceed 140°F (60°C). Mix the soil frequently during drying to facilitate an even water content distribution. Thoroughly mix each subspecimen to facilitate even distribution of water throughout and then place in a separate covered container to stand (cure) in accordance with **Table 2** prior to compaction. For selecting a standing time, the soil may be classified using Practice **D2487**, Practice **D2488**, or data on other samples from the same material source. For referee testing, classification shall be by Practice **D2487**.

10.3 *Dry Preparation Method*—If the sample/specimen is too damp to be friable, reduce the water content by air drying until the material is friable. Drying may be in air or by the use of drying apparatus such that the temperature of the sample does not exceed 140°F (60°C). Thoroughly break up the aggregations in such a manner as to avoid breaking individual particles. Process the material over the appropriate sieve: No. 4 (4.75-mm), $\frac{3}{8}$ -in. (9.5-mm), or $\frac{3}{4}$ -in. (19.0-mm). When preparing the material by passing over the $\frac{3}{4}$ -in. sieve for compaction in the 6-in. mold, break up aggregations sufficiently to at least pass the $\frac{3}{8}$ -in. sieve in order to facilitate the distribution of water throughout the soil in later mixing. Determine and record the water content of the test fraction and all masses covered in **10.2**, as applicable to determine the percent oversize fraction, P_C , and test fraction, P_F .

10.3.1 From the test fraction, select and prepare at least four (preferably five) subspecimens in accordance with **10.2.1** and

TABLE 2 Required Standing Times of Moisturized Specimens

Classification	Minimum Standing Time, h
GW, GP, SW, SP	No Requirement
GM, SM	3
All other soils	16

10.2.2, except for the following: Use either a mechanical splitting or quartering process to obtain the subspecimens. As stated in Test Methods **D6913/D6913M**, both of these processes will yield non-uniform subspecimens compared to the moist procedure. Typically, only the addition of water to each subspecimen will be required.

10.4 Compaction—After standing (curing), if required, each subspecimen (compaction point) shall be compacted as follows:

10.4.1 Determine and record the mass of the mold or mold and base plate, see **10.4.7**.

10.4.2 Assemble and secure the mold and collar to the base plate. Check the alignment of the inner wall of the mold and mold extension collar. Adjust if necessary. The mold shall rest, without wobbling/rocking on a uniform rigid foundation, such as provided by a cylinder or cube of concrete with a weight or mass of not less than 200-lbf or 91-kg, respectively. Secure the base plate to the rigid foundation. The method of attachment to the rigid foundation shall allow easy removal of the assembled mold, collar and base plate after compaction is completed.

10.4.2.1 During compaction, it is advantageous but not required to determine the water content of each subspecimen. This provides a check on the molding water content determined for each compaction point and the magnitude of bleeding, see **10.4.9**. However, more soil will have to be selected for each subspecimen than stated in **10.2.2**.

10.4.3 Compact the soil in three layers. After compaction, each layer should be approximately equal in thickness and extend into the collar. Prior to compaction, place the loose soil into the mold and spread into a layer of uniform thickness. Lightly tamp the soil prior to compaction until it is not in a fluffy or loose state, using either the manual rammer or a $2 \pm$ -in. ($50 \pm$ -mm) diameter cylinder. Following compaction of each of the first two layers, any soil that has not been

compacted; such as adjacent to the mold walls or extends above the compacted surface (up the mold walls) shall be trimmed. The trimmed soil shall be discarded. A knife or other suitable device may be used. The total amount of soil used shall be such that the third compacted layer slightly extends into the collar, but does not extend more than approximately $\frac{1}{4}$ -in. (6-mm) above the top of the mold. If the third layer does extend above this limit, then the compaction point shall be discarded. In addition, the compaction point shall be discarded when the last blow on the rammer for the third layer results in the bottom of the rammer extending below the top of the compaction mold; unless the soil is pliable enough, that this surface can easily be forced above the top of the compaction mold during trimming (see **Note 9**).

10.4.4 Compact each layer with 25 blows for the 4-in. (101.6-mm) mold or with 56 blows for the 6-in. (152.4-mm) mold. The manual rammer shall be used for referee testing.

10.4.5 In operating the manual rammer, take care to avoid lifting the guide sleeve during the rammer upstroke. Hold the guide sleeve steady and within 5° of vertical. Apply the blows at a uniform rate of about 25 blows/min and in such a manner as to provide complete, uniform coverage of the specimen surface. When using a 4-in. (101.6-mm) mold and manual rammer, follow the blow pattern given in **Fig. 3a** and **Fig. 3b**; while for a mechanical rammer, follow the pattern in **Fig. 3b**. When using a 6-in. (152.4-mm) mold and manual rammer, follow the blow pattern given in **Fig. 4** up to the 9th blow, then systematically around the mold (**Fig. 3b**) and in the middle. When using a 6-in. (152.4-mm) mold and a mechanical rammer equipped with a sector face, the mechanical rammer shall be designed to follow the compaction pattern given in **Fig. 3b**. When using a 6-in. (152.4-mm) mold and a mechanical rammer equipped with a circular face, the mechanical rammer shall be designed to distribute the blows uniformly over the

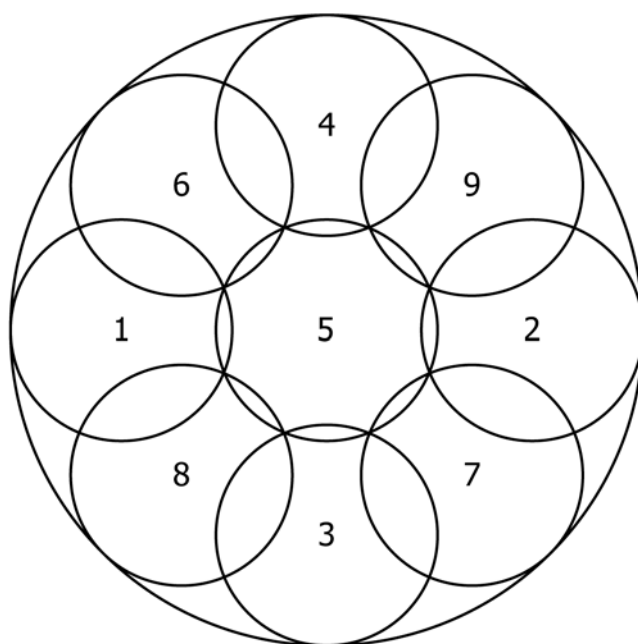


FIG. 4 Rammer Pattern for Compaction in 6 in. (152.4 mm) Mold

surface of the specimen. If the surface of the compacted soil becomes highly uneven (see [Note 9](#)), then adjust the pattern to follow the logic given in [Fig. 3a](#) or [Fig. 4](#). This will most likely void the use of a mechanical rammer for such compaction points.

NOTE 9—When compacting specimens wetter than optimum water content, uneven compacted surfaces can occur and operator judgement is required as to the average height of the specimen and rammer pattern during compaction.

10.4.6 Following compaction of the last layer, remove the collar and base plate (except as noted in [10.4.7](#)) from the mold. A knife may be used to trim the soil adjacent to the collar to loosen the soil from the collar before removal to avoid disrupting the soil below the top of the mold. In addition, to prevent/reduce soil sticking to the collar or base plate, rotate them before removal.

10.4.7 Carefully trim the compacted specimen even with the top of the mold by means of the straightedge scraped across the top of the mold to form a plane surface even with the top of the mold. Initial trimming of the specimen above the top of the mold with a knife may prevent the soil from tearing below the top of the mold. Fill any holes in the top surface with unused or trimmed soil from the specimen, press in with the fingers, and again scrape the straightedge across the top of the mold. If gravel size particles are encountered, trim around them or remove them, whichever is the easiest and reduces the disturbance of the compacted soil. The estimated volume of particles above the surface of the compacted soil and holes in that surface shall be equal, fill in remaining holes as mentioned above. Repeat the appropriate preceding operations on the bottom of the specimen when the mold volume was determined without the base plate. For very wet or dry soils, soil or water may be lost if the base plate is removed. For these situations, leave the base plate attached to the mold. When the base plate is left attached, the volume of the mold must be calibrated with the base plate attached to the mold rather than a plastic or glass plate as noted in [Annex A1, A1.4](#).

10.4.8 Determine and record the mass of the specimen and mold to the nearest g. When the base plate is left attached, determine and record the mass of the specimen, mold and base plate to the nearest g.

10.4.9 Remove the material from the mold. Obtain a specimen for molding water content by using either the whole specimen (preferred method) or a representative portion. When the entire specimen is used, break it up to facilitate drying. Otherwise, obtain a representative portion of the three layers, removing enough material from the specimen to report the water content to 0.1 %. The mass of the representative portion of soil shall conform to the requirements of Table 1, Method B, of Test Methods [D2216](#). Determine the molding water content in accordance with Test Methods [D2216](#).

10.5 Following compaction of the last specimen, compare the wet unit weights to ensure that a desired pattern of obtaining data on each side of the optimum water content will be attained for the dry-unit-weight compaction curve. Plotting the wet unit weight and molding water content of each compacted specimen can be an aid in making the above evaluation. If the desired pattern is not obtained, additional

compacted specimens will be required. Generally, for experienced plotters of compaction curves, one compaction point wet of the optimum water content is adequate to define the maximum wet unit weight, see [11.2](#).

11. Calculations and Plotting (Compaction Curve)

11.1 Fraction Percentages—If gradation data from Test Methods [D6913/D6913M](#) is not available, calculate the dry mass of the test fraction, percentage of oversize fraction and test fraction as covered below and using the data from [10.2](#) or [10.3](#):

11.1.1 Test Fraction—Determine the dry mass of the test fraction as follows:

$$M_{d,tf} = \frac{M_{m,tf}}{1 + \frac{w_{tf}}{100}} \quad (1)$$

where:

- $M_{d,tf}$ = dry mass of test fraction, nearest g or 0.001 kg,
- $M_{m,tf}$ = moist mass of test fraction, nearest g or 0.001 kg, and
- w_{tf} = water content of test fraction, nearest 0.1 %.

11.1.2 Oversize Fraction Percentage—Determine the oversize (coarse) fraction percentage as follows:

$$P_C = \frac{M_{d,of}}{M_{d,of} + M_{d,tf}} \quad (2)$$

where:

- P_C = percentage of oversize (coarse) fraction, nearest %, and
- $M_{d,of}$ = dry mass of oversize fraction, nearest g or 0.001 kg,

11.1.3 Test Fraction Percentage—Determine the test (finer) fraction percentage as follows:

$$P_F = 100 - P_C \quad (3)$$

where:

- P_F = percentage of test (finer) fraction, nearest %.

11.2 Density and Unit Weight—Calculate the molding water content, moist density, dry density, and dry unit weight of each compacted specimen as explained below.

11.2.1 Molding Water Content, w —Calculate in accordance with Test Methods [D2216](#) to nearest 0.1 %.

11.2.2 Density and Unit Weights—Calculate the moist (total) density ([Eq 4](#)), the dry density ([Eq 5](#)), and then the dry unit weight ([Eq 6](#)) as follows:

11.2.2.1 Moist Density:

$$\rho_m = K \times \frac{(M_t - M_{md})}{V} \quad (4)$$

where:

- ρ_m = moist density of compacted subspecimen (compaction point), four significant digits, g/cm³ or kg/m³,
- M_t = mass of moist soil in mold and mold, nearest g,
- M_{md} = mass of compaction mold, nearest g,
- V = volume of compaction mold, cm³ or m³ (see [Annex A1](#)), and

K = conversion constant, depending on density units and volume units.
 Use 1 for g/cm^3 and volume in cm^3 .
 Use 1000 for g/cm^3 and volume in m^3 .
 Use 0.001 for kg/cm^3 and volume in m^3 .
 Use 1000 for kg/m^3 and volume in cm^3 .

11.2.2.2 Dry Density:

$$\rho_d = \frac{\rho_m}{1 + \frac{w}{100}} \quad (5)$$

where:

ρ_d = dry density of compaction point, four significant digits, g/cm^3 or kg/m^3 , and
 w = molding water content of compaction point, nearest 0.1 %.

11.2.2.3 Dry Unit Weight:

$$\gamma_d = K_1 \times \rho_d \text{ in lbf/ft}^3 \quad (6)$$

or

$$\gamma_d = K_2 \times \rho_d \text{ in kN/m}^3 \quad (7)$$

where:

γ_d = dry unit weight of compacted specimen, four significant digits, in lbf/ft^3 or kN/m^3 ,
 K_1 = conversion constant, depending on density units,
 Use 62.428 for density in g/cm^3 , or
 Use 0.062428 for density in kg/m^3 ,
 K_2 = conversion constant, depending on density units,
 Use 9.8066 for density in g/cm^3 , or
 Use 0.0098066 for density in kg/m^3 .

11.3 Compaction Curve—Plot the dry unit weight and molding water content values, the saturation curve (see 11.3.2), and draw the compaction curve as a smooth curve through the points (see example, Fig. 5). For each point on the compaction curve, calculate, record, and plot dry unit weight to the nearest 0.1 lbf/ft^3 (0.02 kN/m^3) and molding water content to the nearest 0.1 %. From the compaction curve, determine the compaction results: optimum water content, to nearest 0.1 % and maximum dry unit weight, to the nearest 0.1 lbf/ft^3 (0.02 kN/m^3). If more than 5 % by mass of oversize material was removed from the sample/specimen, calculate the corrected optimum water content and maximum dry unit weight of the total material using Practice D4718/D4718M. This correction may be made to the appropriate field in-place density test specimen rather than to the laboratory compaction results.

11.3.1 In these plots, the scale sensitivities should remain the same, that is the change in molding water content or dry unit weight per division is constant between plots. Typically, the change in dry unit weight per division is twice that of molding water content's (2 lbf/ft^3 to 1 % w per major division). Therefore, any change in the shape of the compaction curve is a result of testing different material, not the plotting scale. However, a one to one ratio should be used for soils that have a relatively flat compaction curve (see 10.2.1), such as highly plastic soils or relatively free draining ones up to the point of bleeding.

COMPACTION TEST

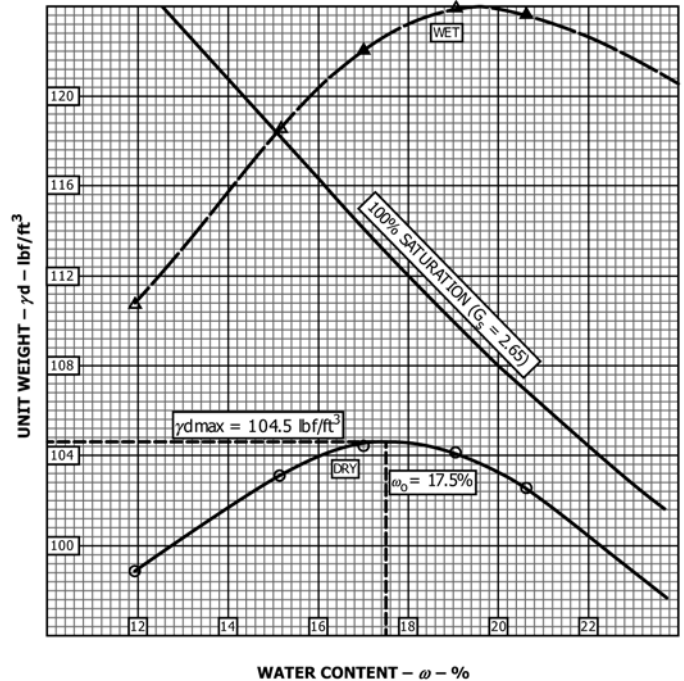


FIG. 5 Example Compaction Curve Plotting

11.3.1.1 The shape of the compaction curve on the wet side on optimum should typically follow that of the saturation curve. The shape of the compaction curve on the dry side of optimum may be relatively flat or up and down when testing some soils, such as relatively free draining ones or plastic soils prepared using the moist procedure and having molding water contents close to or less than the shrinkage limit.

11.3.2 Plot the 100 % saturation curve, based on either an estimated or a measured specific gravity. Values of water content for the condition of 100 % saturation can be calculated as explained in 11.4 (see example, Fig. 5).

NOTE 10—The 100 % saturation curve is an aid in drawing the compaction curve. For soils containing more than about 10 % fines and molding water contents well above optimum, the two curves generally become roughly parallel with the wet side of the compaction curve between 92 to 95 % saturation. Theoretically, the compaction curve cannot plot to the right of the 100 % saturation curve. If it does, there is an error in specific gravity, in measurements, in calculations, in testing, or in plotting. The 100 % saturation curve is sometimes referred to as the zero air voids curve or the complete saturation curve.

11.4 Saturation Points—To calculate points for plotting the 100 % saturation curve or zero air voids curve, select values of dry unit weight, calculate corresponding values of water content corresponding to the condition of 100 % saturation as follows:

$$w_{sat} = \frac{(\gamma_w)(G_s) - \gamma_d}{(\gamma_d)(G_s)} \times 100 \quad (8)$$

where:

w_{sat} = water content for complete saturation, nearest 0.1 %,
 γ_w = unit weight of water, 62.32 lbf/ft^3 (9.789 kN/m^3) at 20°C,

γ_d = dry unit weight of soil, lbf/ft³ (kN/m³), three significant digits, and
 G_s = specific gravity of soil (estimated or measured), to nearest 0.01 value, see 11.4.1.

11.4.1 Specific gravity may be estimated for the test fraction based on test data from other soils having the same soil classification and source or experience. Otherwise, a specific gravity test (Test Methods C127 or D854, or both) is necessary.

12. Report: Data Sheet(s)/Form(s)

12.1 The methodology used to specify how data are recorded on the test data sheet(s)/form(s), as described below, is covered in 1.6.

12.2 The data sheet(s)/form(s) shall contain as a minimum the following information:

- 12.2.1 Method used (A, B, or C).
- 12.2.2 Preparation method used (moist or dry).
- 12.2.3 As received water content if determined, nearest 1 %.
- 12.2.4 Standard optimum water content, Std- w_{opt} to nearest 0.1 %.
- 12.2.5 Standard maximum dry unit weight, Std- $\gamma_{d,max}$ nearest 0.1 lbf/ft³ or 0.02 kN/m³.
- 12.2.6 Type of rammer (manual or mechanical).
- 12.2.7 Soil sieve data when applicable for selection of Method (A, B, or C) used.
- 12.2.8 Description of sample used in test (as a minimum, color and group name and symbol), by Practice D2488, or classification by Practice D2487.
- 12.2.9 Specific gravity and method of determination, nearest 0.01 value.
- 12.2.10 Identification of sample used in test; for example, project number/name, location, depth, and the like.
- 12.2.11 Compaction curve plot showing compaction points used to establish compaction curve, and 100 % saturation curve, value or point of maximum dry unit weight and optimum water content.
- 12.2.12 Percentages for the fractions retained (P_C) and passing (P_F) the sieve used in Method A, B, or C, nearest 1 %. In addition, if compaction data (Std- w_{opt} and Std- $\gamma_{d,max}$) are corrected for the oversize fraction, include that data.

13. Precision and Bias

13.1 *Precision*—Criteria for judging the acceptability of test results obtained by these test methods on a range of soil types are given in Tables 3 and 4. These estimates of precision are based on the results of the interlaboratory program conducted by the ASTM Reference Soils and Testing Program.⁴ In this program, Method A and the Dry Preparation Method were used. In addition, some laboratories performed three replicate tests per soil type (triplicate test laboratory), while other laboratories performed a single test per soil type (single test laboratory). A description of the soils tested is given in 13.1.4. The precision estimates vary with soil type, and may vary with methods used (Method A, B, or C, or wet/dry preparation

TABLE 3 Summary of Test Results from Triplicate Test Laboratories (Standard Effort Compaction)

(1) Number of Triplicate Test Labs			(2) Test Value ^A (Units)	(3) Average Value ^B	(4) Standard Deviation ^C	(5) Acceptable Range of Two Results ^{D,E}
			Soil Type:			
			CH CL ML	CH CL ML	CH CL ML	CH CL ML
Single-Operator Results (Within-Laboratory Repeatability):						
11	12	11	$\gamma_{d,max}$ (pcf)	97.2 109.2 106.3	0.5 0.4 0.5	1.3 1.2 1.3
11	12	11	w_{opt} (%)	22.8 16.6 17.1	0.2 0.3 0.3	0.7 0.9 0.9
Multilaboratory Results (Between-Laboratory Reproducibility):						
11	12	11	$\gamma_{d,max}$ (pcf)	97.2 109.2 106.3	1.4 0.8 0.6	3.9 2.3 1.6
11	12	11	w_{opt} (%)	22.8 16.6 17.1	0.7 0.5 0.5	1.8 1.5 1.3

^A $\gamma_{d,max}$ (pcf) = standard maximum dry unit weight in lbf/ft³ and w_{opt} (%) = standard optimum water in percent.

^B The number of significant digits and decimal places presented are representative of the input data. In accordance with Practice D6026, the standard deviation and acceptable range of results can not have more decimal places than the input data.

^C Standard deviation is calculated in accordance with Practice E691 and is referred to as the 1 s limit.

^D Acceptable range of two results is referred to as the d2s limit. It is calculated as $1.960 \sqrt{2} \cdot 1s$, as defined by Practice E177. The difference between two properly conducted tests should not exceed this limit. The number of significant digits/decimal places presented is equal to that prescribed by this standard or Practice D6026. In addition, the value presented can have the same number of decimal places as the standard deviation, even if that result has more significant digits than the standard deviation.

^E Both values of $\gamma_{d,max}$ and w_{opt} have to fall within values given for the selected soil type.

TABLE 4 Summary of Single Test Results from Each Laboratories (Standard Effort Compaction)^A

(1) Number of Test Laboratories			(2) Test Value (Units)	(3) Average Value	(4) Standard Deviation	(5) Acceptable Range of Two Results
			Soil Type:			
			CH CL ML	CH CL ML	CH CL ML	CH CL ML
Multilaboratory Results (Between-Laboratory Reproducibility):						
26	26	25	$\gamma_{d,max}$ (pcf)	97.3 109.2 106.2	1.6 1.1 1.0	4.5 3.0 2.9
			w_{opt} (%)	22.6 16.4 16.7	0.9 0.7 1.0	2.4 1.8 2.9

^A See footnotes in Table 3.

method). Judgement is required when applying these estimates to another soil, method, or preparation method.

13.1.1 The data in Table 3 are based on three replicate tests performed by each triplicate test laboratory on each soil type. The single operator and multilaboratory standard deviation show in Table 3, Column 4 were obtained in accordance with Practice E691, which recommends each testing laboratory perform a minimum of three replicate tests. Results of two properly conducted tests performed by the same operator on the same material, using the same equipment, and in the shortest practical period of time should not differ by more than the single-operator d2s shown in Table 3, Column 5. For definition of d2s, see footnote D in Table 1. Results of two properly conducted tests performed by different operators and on different days should not differ by more than the multilaboratory d2s limits shown in Table 3, Column 5.

13.1.2 In the ASTM Reference Soils and Testing Program, many of the laboratories performed only a single test on each

⁴ Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D18-1008. Contact ASTM Customer Service at service@astm.org.

soil type. This is common practice in the design and construction industry. The data for each soil type in **Table 4** are based upon the first test result from the triplicate test laboratories and the single test results from the other laboratories. Results of two properly conducted tests performed by two different laboratories with different operators using different equipment and on different days should not vary by more than the d_{2s} limits shown in **Table 4**, Column 5. The results in **Tables 3 and 4** are dissimilar because the data sets are different.

13.1.3 **Table 3** presents a rigorous interpretation of triplicate test data in accordance with Practice **E691** from pre-qualified laboratories. **Table 4** is derived from test data that represents common practice.

13.1.4 *Soil Types*—Based on the multilaboratory test results the soils used in the program are described below in accordance with Practice **D2487**. In addition, the local names of the soils are given.

CH	Fat clay, CH, 99 % fines, LL=60, PI=39, grayish brown, soil had been air dried and pulverized. Local name—Vicksburg Buckshot Clay
CL	Lean clay, CL, 89 % fines, LL=33, PI=13, gray, soil had been air dried and pulverized. Local name—Annapolis Clay
ML	Silt, ML, 99 % fines, LL=27, PI=4, light brown, soil had been air dried and pulverized. Local name—Vicksburg Silt

13.2 *Bias*—There is no accepted reference values for this test method, therefore, bias cannot be determined.

14. Keywords

14.1 compaction characteristics; density; impact compaction; laboratory tests ; moisture-density curves; proctor test; soil; soil compaction; standard effort

ANNEX

(Mandatory Information)

A1. VOLUME OF COMPACTION MOLD

A1.1 Scope

A1.1.1 This annex describes the procedure for determining the volume of a compaction mold.

A1.1.2 The volume is determined by two methods, a water-filled and linear-measurement method.

A1.1.3 The water filling method for the 4-in. (106.5-mm) mold, when using a balance readable to nearest g, does not yield four significant figures for its volume, just three. Based on Practice **D6026**, this limits the density/unit weight determinations previously presented from four to three significant figures. To prevent this limitation, the water filling method has been adjusted from that presented in early versions of this test method.

A1.2 Apparatus

A1.2.1 In addition to the apparatus listed in Section 6 the following items are required:

A1.2.1.1 *Vernier or Dial Caliper*, having a measuring range of at least 0 to 6 in. (0 to 150 mm) and readable to at least 0.001 in. (0.02 mm).

A1.2.1.2 *Inside Micrometer (optional)*, having a measuring range of at least 2 to 12 in. (50 to 300 mm) and readable to at least 0.001 in. (0.02 mm).

A1.2.1.3 *Depth Micrometer (optional)*, having a measuring range of at least 0 to 6 in. (0 to 150 mm) and readable to at least 0.001 in. (0.02 mm).

A1.2.1.4 *Plastic or Glass Plates*—Two plastic or glass plates about 8 in. square by ¼ in. thick (200 by 200 by 6 mm).

A1.2.1.5 *Thermometer or Other Thermometric Device*, having graduation increments of 0.1°C.

A1.2.1.6 *Stopcock Grease*, or similar sealant.

A1.2.1.7 *Miscellaneous Equipment*—Bulb syringe, towels, etc.

A1.3 Precautions

A1.3.1 Perform this method in an area isolated from drafts or extreme temperature fluctuations.

A1.4 Procedure

A1.4.1 *Water-Filling Method*:

A1.4.1.1 Lightly grease the bottom of the compaction mold and place it on one of the plastic or glass plates. Lightly grease the top of the mold. Be careful not to get grease on the inside of the mold. If it is necessary to use the base plate, as noted in **10.4.7**, place the greased mold onto the base plate and secure with the locking studs.

A1.4.1.2 Determine the mass of the greased mold and both plastic or glass plates to the nearest 1 g and record, M_{mp} . When the base plate is being used in lieu of the bottom plastic or glass plate, determine the mass of the mold, base plate and a single plastic or glass plate to be used on top of the mold to the nearest 1 g and record.

A1.4.1.3 Place the mold and the bottom plastic or glass plate on a firm, level surface and fill the mold with water to slightly above its rim.

A1.4.1.4 Slide the second plate over the top surface of the mold so that the mold remains completely filled with water and air bubbles are not entrapped. Add or remove water as necessary with a bulb syringe.

A1.4.1.5 Completely dry any excess water from the outside of the mold and plates.

A1.4.1.6 Determine the mass of the mold, plates and water and record to the nearest 1 g, $M_{mp,w}$.

A1.4.1.7 Determine the temperature of the water in the mold to the nearest 0.1°C and record. Determine and record the density of water from the table given in Test Methods **D854** or as follows:

$$\rho_{w,c} = 1.00034038 - (7.77 \times 10^{-6}) \times T - (4.95 \times 10^{-6}) \times T^2 \quad (\text{A1.1})$$

where:

$\rho_{w,c}$ = density of water, nearest 0.00001 g/cm³, and

T = calibration test temperature, nearest 0.1°C.

A1.4.1.8 Calculate the mass of water in the mold by subtracting the mass determined in **A1.4.1.2** from the mass determined in **A1.4.1.6**.

A1.4.1.9 Calculate the volume of water by dividing the mass of water by the density of water. Record this volume to the nearest 0.1 cm³ for the 4-in. (101.6-mm) mold or nearest 1 cm³ for the 6-in. (152.4-mm) mold. To determine the volume of the mold in m³, multiply the volume in cm³ by 1×10^{-6} . Record this volume, as prescribed.

A1.4.1.10 If the filling method is being used to determine the mold's volume and checked by linear measurement method, repeat this volume determination (**A1.4.1.3** – **A1.4.1.9**) and determine and record the average value, V_w as prescribed.

A1.4.2 Linear Measurement Method:

A1.4.2.1 Using either the vernier caliper or the inside micrometer (preferable), measure the inside diameter (ID) of the mold 6 times at the top of the mold and 6 times at the bottom of the mold, spacing each of the six top and bottom measurements equally around the ID of the mold. Record the values to the nearest 0.001-in. (0.02-mm). Determine and record the average ID to the nearest 0.001-in. (0.02-mm), d_{avg} . Verify that this ID is within specified tolerances, 4.000 ± 0.016 in. (101.6 ± 0.4 mm), if not discard the mold.

A1.4.2.2 Using the vernier caliper or depth micrometer (preferably), measure the inside height of the mold attached to the base plate. In these measurements, make three or more measurements equally spaced around the ID of the mold, and preferably one in the center of the mold, but not required (used the straightedge to facilitate the later measurement and correct measurement for thickness of straightedge). Record these values to the nearest 0.001-in. (0.02-mm). Determine and

record the average of these height measurements to the nearest 0.001 in. (0.02 mm), h_{avg} . Verify that this height is within specified tolerances, 4.584 ± 0.018 in. (116.4 ± 0.5 mm), if not discard the mold.

A1.4.2.3 Calculate the volume of the mold to four significant digits in cm³ as follows:

$$V_{lm} = K_3 \frac{\pi \times h_{avg} \times (d_{avg})^2}{4} \quad (\text{A1.2})$$

where:

V_{lm} = volume of mold by linear measurements, to four significant digits, cm³,

K_3 = constant to convert measurements made in inch (in.) or mm,

Use 16.387 for measurements in inches.

Use 10^{-3} for measurements in mm.

π = 3.14159,

h_{avg} = average height, in. (mm), and

d_{avg} = average of the top and bottom diameters, in. (mm).

A1.4.2.4 If the volume in m³ is required, then multiply the above value by 10^{-6} .

A1.5 Comparison of Results and Standardized Volume of Mold

A1.5.1 The volume obtained by either method should be within the volume tolerance requirements of **6.1.1** and **6.1.2**, using either or cm³ to ft³. To convert cm³ to ft³, divide cm³ by 28 317, record to the nearest 0.0001 ft³.

A1.5.2 The difference between the two methods should not exceed 0.5 % of the nominal volume of the mold, cm³ to ft³.

A1.5.3 Repeat the determination of volume, which is most suspect or both if these criteria are not met.

A1.5.4 Failure to obtain satisfactory agreement, between these methods, even after several trials is an indication the mold is badly deformed and should be replaced.

A1.5.5 Use the volume of the mold determined using the water-filling or linear method, or average of both methods as the standardized volume for calculating the moist density (see **11.4**). This value (V) in cm³ or m³ shall have four significant digits. The use of a volume in ft³, along with masses in lbm shall not be regarded as a nonconformance with this standard.

REFERENCES

- (1) Johnson, A. W., and Sallberg, J. R., *Factors Influencing Compaction Test Results*, Highway Research Board, Bulletin 318, Publication 967, National Academy of Sciences-National Research Council, Washington, DC, 1962.
- (2) *Earth Manual*, United States Bureau of Reclamation, Part 1, Third Edition, 1998, pp. 255–260.
- (3) *Earth Manual*, United States Bureau of Reclamation, Part 2, Third Edition, 1990, USBR 5515.
- (4) Torrey, V. H., and Donaghe, R. T., “Compaction Control of Earth-Rock Mixtures: A New Approach,” *Geotechnical Testing Journal*, GTJODJ, Vol 17, No. 3, September 1994, pp. 371–386.

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Designation: D1557 – 12 (Reapproved 2021)

Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft³ (2,700 kN-m/m³))¹

This standard is issued under the fixed designation D1557; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ϵ) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the U.S. Department of Defense.

1. Scope

1.1 These test methods cover laboratory compaction methods used to determine the relationship between molding water content and dry unit weight of soils (compaction curve) compacted in a 4- or 6-in. (101.6- or 152.4-mm) diameter mold with a 10.00-lbf. (44.48-N) rammer dropped from a height of 18.00 in. (457.2 mm) producing a compactive effort of 56 000 ft-lbf/ft³ (2700 kN-m/m³).

NOTE 1—The equipment and procedures are the same as proposed by the U.S. Corps of Engineers in 1945. The modified effort test (see 3.1.3) is sometimes referred to as the Modified Proctor Compaction Test.

1.1.1 Soils and soil-aggregate mixtures are to be regarded as natural occurring fine- or coarse-grained soils, or composites or mixtures of natural soils, or mixtures of natural and processed soils or aggregates such as gravel or crushed rock. Hereafter referred to as either soil or material.

1.2 These test methods apply only to soils (materials) that have 30 % or less by mass of their particles retained on the 3/4-in. (19.0-mm) sieve and have not been previously compacted in the laboratory; that is, do not reuse compacted soil.

1.2.1 For relationships between unit weights and molding water contents of soils with 30 % or less by weight of material retained on the 3/4-in. (19.0-mm) sieve to unit weights and molding water contents of the fraction passing the 3/4-in. (19.0-mm) sieve, see Practice D4718/D4718M.

1.3 Three alternative methods are provided. The method used shall be as indicated in the specification for the material being tested. If no method is specified, the choice should be based on the material gradation.

1.3.1 Method A:

1.3.1.1 Mold—4-in. (101.6-mm) diameter.

1.3.1.2 Material—Passing No. 4 (4.75-mm) sieve.

1.3.1.3 Layers—Five.

1.3.1.4 Blows per layer—25.

1.3.1.5 Usage—May be used if 25 % or less by mass of the material is retained on the No. 4 (4.75-mm) sieve. However, if 5 to 25 % by mass of the material is retained on the No. 4 (4.75-mm) sieve, Method A can be used but oversize corrections will be required (See 1.4) and there are no advantages to using Method A in this case.

1.3.1.6 Other Use—If this gradation requirement cannot be met, then Methods B or C may be used.

1.3.2 Method B:

1.3.2.1 Mold—4-in. (101.6-mm) diameter.

1.3.2.2 Material—Passing 3/8-in. (9.5-mm) sieve.

1.3.2.3 Layers—Five.

1.3.2.4 Blows per layer—25.

1.3.2.5 Usage—May be used if 25 % or less by mass of the material is retained on the 3/8-in. (9.5-mm) sieve. However, if 5 to 25 % of the material is retained on the 3/8-in. (9.5-mm) sieve, Method B can be used but oversize corrections will be required (See 1.4). In this case, the only advantages to using Method B rather than Method C are that a smaller amount of sample is needed and the smaller mold is easier to use.

1.3.2.6 Other Usage—If this gradation requirement cannot be met, then Method C may be used.

1.3.3 Method C:

1.3.3.1 Mold—6-in. (152.4-mm) diameter.

1.3.3.2 Material—Passing 3/4-in. (19.0-mm) sieve.

1.3.3.3 Layers—Five.

1.3.3.4 Blows per layer—56.

1.3.3.5 Usage—May be used if 30 % or less (see 1.4) by mass of the material is retained on the 3/4-in. (19.0-mm) sieve.

1.3.4 The 6-in. (152.4-mm) diameter mold shall not be used with Method A or B.

NOTE 2—Results have been found to vary slightly when a material is tested at the same compactive effort in different size molds, with the smaller mold size typically yielding larger values of unit weight and density (1).²

1.4 If the test specimen contains more than 5 % by mass of oversize fraction (coarse fraction) and the material will not be

¹ These test methods are under the jurisdiction of ASTM Committee D18 on Soil and Rock and are the direct responsibility of Subcommittee D18.03 on Texture, Plasticity and Density Characteristics of Soils.

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² The boldface numbers in parentheses refer to the list of references at the end of this standard.

*A Summary of Changes section appears at the end of this standard

included in the test, corrections must be made to the unit weight and molding water content of the test specimen or to the appropriate field in-place unit weight (or density) test specimen using Practice **D4718/D4718M**.

1.5 This test method will generally produce a well-defined maximum dry unit weight for non-free draining soils. If this test method is used for free-draining soils the maximum unit weight may not be well defined, and can be less than obtained using Test Methods **D4253**.

1.6 All observed and calculated values shall conform to the guidelines for significant digits and rounding established in Practice **D6026**, unless superseded by these test methods.

1.6.1 For purposes of comparing measured or calculated value(s) with specified limits, the measured or calculated value(s) shall be rounded to the nearest decimal or significant digits in the specified limits.

1.6.2 The procedures used to specify how data are collected/recorded or calculated in this standard are regarded as the industry standard. In addition, they are representative of the significant digits that generally should be retained. The procedures used do not consider material variation, purpose for obtaining the data, special purpose studies, or any considerations for the user's objectives; it is common practice to increase or reduce significant digits of reported data to be commensurate with these considerations. It is beyond the scope of these test methods to consider significant digits used in analytical methods for engineering design.

1.7 The values in inch-pound units are to be regarded as the standard. The values stated in SI units are provided for information only, except for units of mass. The units for mass are given in SI units only, g or kg.

1.7.1 It is common practice in the engineering profession to concurrently use pounds to represent both a unit of mass (lbm) and a force (lbf). This implicitly combines two separate systems of units; that is, the absolute system and the gravitational system. It is scientifically undesirable to combine the use of two separate sets of inch-pound units within a single standard. These test methods have been written using the gravitational system of units when dealing with the inch-pound system. In this system, the pound (lbf) represents a unit of force (weight). However, the use of balances or scales recording pounds of mass (lbm) or the recording of density in lbm/ft³ shall not be regarded as a nonconformance with this standard.

1.8 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.9 **Warning**—Mercury has been designated by EPA and many state agencies as a hazardous material that can cause central nervous system, kidney, and liver damage. Mercury, or its vapor, may be hazardous to health and corrosive to materials. Caution should be taken when handling mercury and mercury containing products. See the applicable product Material Safety Data Sheet (MSDS) for details and EPA's website (<http://www.epa.gov/mercury/faq.htm>) for additional informa-

tion. Users should be aware that selling mercury or mercury containing products or both into your state may be prohibited by state law.

1.10 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:³

- C127** Test Method for Relative Density (Specific Gravity) and Absorption of Coarse Aggregate
- C136/C136M** Test Method for Sieve Analysis of Fine and Coarse Aggregates
- C670** Practice for Preparing Precision and Bias Statements for Test Methods for Construction Materials
- D653** Terminology Relating to Soil, Rock, and Contained Fluids
- D698** Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft³ (600 kN-m/m³))
- D854** Test Methods for Specific Gravity of Soil Solids by Water Pycnometer
- D2168** Practices for Calibration of Laboratory Mechanical-Rammer Soil Compactors
- D2216** Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
- D2487** Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
- D2488** Practice for Description and Identification of Soils (Visual-Manual Procedures)
- D3740** Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
- D4220/D4220M** Practices for Preserving and Transporting Soil Samples
- D4253** Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
- D4718/D4718M** Practice for Correction of Unit Weight and Water Content for Soils Containing Oversize Particles
- D4753** Guide for Evaluating, Selecting, and Specifying Balances and Standard Masses for Use in Soil, Rock, and Construction Materials Testing
- D4914/D4914M** Test Methods for Density of Soil and Rock in Place by the Sand Replacement Method in a Test Pit
- D5030/D5030M** Test Methods for Density of In-Place Soil and Rock Materials by the Water Replacement Method in a Test Pit
- D6026** Practice for Using Significant Digits and Data Records in Geotechnical Data
- D6913/D6913M** Test Methods for Particle-Size Distribution

³ For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

(Gradation) of Soils Using Sieve Analysis
E11 Specification for Woven Wire Test Sieve Cloth and Test Sieves
E319 Practice for the Evaluation of Single-Pan Mechanical Balances
 IEEE/ASTM SI 10 Standard for Use of the International System of Units (SI): The Modern Metric System

3. Terminology

3.1 Definitions:

3.1.1 See Terminology **D653** for general definitions.

3.1.2 *molding water content, n* —the water content of the soil (material) specimen in the mold after it has been reconstituted and compacted.

3.1.3 *modified effort*—in compaction testing, the term for the 56 000 ft-lbf/ft³ (2700 kN-m/m³) compactive effort applied by the equipment and methods of this test.

3.1.4 *modified maximum dry unit weight, $\gamma_{d,max}$ (lb/ft³ (kN/m³))*—in compaction testing, the maximum value defined by the compaction curve for a compaction test using modified effort.

3.1.5 *modified optimum water content, w_{opt} (%)*—in compaction testing, the water content at which the soil can be compacted to the maximum dry unit weight using modified compactive effort.

3.2 Definitions of Terms Specific to This Standard:

3.2.1 *oversize fraction (coarse fraction), P_C (%)*—the portion of total specimen not used in performing the compaction test; it may be the portion of total specimen retained on the No. 4 (4.75-mm) sieve in Method A, $\frac{3}{8}$ -in. (9.5-mm) sieve in Method B, or $\frac{3}{4}$ -in. (19.0-mm) sieve in Method C.

3.2.2 *test fraction (finer fraction), P_F (%)*—the portion of the total specimen used in performing the compaction test; it may be fraction passing the No. 4 (4.75-mm) sieve in Method A, passing the $\frac{3}{8}$ -in. (9.5-mm) sieve in Method B, or passing the $\frac{3}{4}$ -in. (19.0-mm) sieve in Method C.

4. Summary of Test Method

4.1 A soil at a selected molding water content is placed in five layers into a mold of given dimensions, with each layer compacted by 25 or 56 blows of a 10.00-lbf (44.48-N) rammer dropped from a distance of 18.00 in. (457.2 mm), subjecting the soil to a total compactive effort of about 56 000 ft-lbf/ft³ (2700 kN-m/m³). The resulting dry unit weight is determined. The procedure is repeated for a sufficient number of molding water contents to establish a relationship between the dry unit weight and the molding water content for the soil. This data, when plotted, represent a curvilinear relationship known as the compaction curve. The values of optimum water content and modified maximum dry unit weight are determined from the compaction curve.

5. Significance and Use

5.1 Soil placed as engineering fill (embankments, foundation pads, road bases) is compacted to a dense state to obtain satisfactory engineering properties such as shear strength, compressibility, or permeability. In addition, foundation soils

are often compacted to improve their engineering properties. Laboratory compaction tests provide the basis for determining the percent compaction and molding water content needed to achieve the required engineering properties, and for controlling construction to assure that the required compaction and water contents are achieved.

NOTE 3—The degree of soil compaction required to achieve the desired engineering properties is often specified as a percentage of the modified maximum dry unit weight as determined using this test method. If the required degree of compaction is substantially less than the modified maximum dry unit weight using this test method, it may be practicable for testing to be performed using Test Method and to specify the degree of compaction as a percentage of the standard maximum dry unit weight. Since more energy is applied for compaction using this test method, the soil particles are more closely packed than when **D698** is used. The general overall result is a higher maximum dry unit weight, lower optimum moisture content, greater shear strength, greater stiffness, lower compressibility, lower air voids, and decreased permeability. However, for highly compacted fine-grained soils, absorption of water may result in swelling, with reduced shear strength and increased compressibility, reducing the benefits of the increased effort used for compaction (2). Use of **D698**, on the other hand, allows compaction using less effort and generally at a higher optimum moisture content. The compacted soil may be less brittle, more flexible, more permeable, and less subject to effects of swelling and shrinking. In many applications, building or construction codes may direct which test method, **D698** or this one, should be used when specifying the comparison of laboratory test results to the degree of compaction of the in-place soil in the field.

5.2 During design of an engineered fill, testing performed to determine shear, consolidation, permeability, or other properties requires test specimens to be prepared by compacting the soil at a prescribed molding water content to obtain a predetermined unit weight. It is common practice to first determine the optimum water content (w_{opt}) and maximum dry unit weight ($\gamma_{d,max}$) by means of a compaction test. Test specimens are compacted at a selected molding water content (w), either wet or dry of optimum (w_{opt}) or at optimum (w_{opt}), and at a selected dry unit weight related to a percentage of maximum dry unit weight ($\gamma_{d,max}$). The selection of molding water content (w), either wet or dry of optimum (w_{opt}) or at optimum (w_{opt}) and the dry unit weight ($\gamma_{d,max}$) may be based on past experience, or a range of values may be investigated to determine the necessary percent of compaction.

5.3 Experience indicates that the methods outlined in 5.2 or the construction control aspects discussed in 5.1 are extremely difficult to implement or yield erroneous results when dealing with some soils. The following subsections describe typical problem soils, the problems encountered when dealing with such soils and possible solutions for these problems.

5.3.1 *Oversize Fraction*—Soils containing more than 30 % oversize fraction (material retained on the $\frac{3}{4}$ -in. (19-mm) sieve) are a problem. For such soils, there is no ASTM test method to control their compaction and very few laboratories are equipped to determine the laboratory maximum unit weight (density) of such soils (USDI Bureau of Reclamation, Denver, CO and U.S. Army Corps of Engineers, Vicksburg, MS). Although Test Methods **D4914/D4914M** and **D5030/D5030M** determine the “field” dry unit weight of such soils, they are difficult and expensive to perform.

5.3.1.1 One method to design and control the compaction of such soils is to use a test fill to determine the required degree

of compaction and the method to obtain that compaction. Then use a method specification to control the compaction. Components of a method specification typically contain the type and size of compaction equipment to be used, the lift thickness, acceptable range of molding water content, and number of passes.

NOTE 4—Success in executing the compaction control of an earthwork project, especially when a method specification is used, is highly dependent upon the quality and experience of the contractor and inspector.

5.3.1.2 Another method is to apply the use of density correction factors developed by the USDI Bureau of Reclamation (3,4) and U.S. Corps of Engineers (5). These correction factors may be applied for soils containing up to about 50 to 70 % oversize fraction. Both agencies use a different term for these density correction factors. The USDI Bureau of Reclamation uses D ratio (or D – VALUE), while the U.S. Corps of Engineers uses Density Interference Coefficient (I_c).

5.3.1.3 The use of the replacement technique (Test Method D1557–78, Method D), in which the oversize fraction is replaced with a finer fraction, is inappropriate to determine the maximum dry unit weight, γ_{dmax} , of soils containing oversize fractions (5).

5.3.2 *Degradation*—Soils containing particles that degrade during compaction are a problem, especially when more degradation occurs during laboratory compaction than field compaction, the typical case. Degradation typically occurs during the compaction of a granular-residual soil or aggregate. When degradation occurs, the maximum dry-unit weight increases (1) so that the resulting laboratory maximum value is not representative of field conditions. Often, in these cases, the maximum dry unit weight is impossible to achieve in the field.

5.3.2.1 Again for soils subject to degradation, the use of test fills and method specifications may help. Use of replacement techniques is not correct.

5.3.3 *Gap Graded*—Gap-graded soils (soils containing many large particles with limited small particles) are a problem because the compacted soil will have larger voids than usual. To handle these large voids, standard test methods (laboratory or field) typically have to be modified using engineering judgement.

NOTE 5—The quality of the result produced by this standard is dependent on the competence of the personnel performing it, and the

suitability of the equipment and facilities used. Agencies that meet the criteria of Practice D3740 are generally considered capable of competent and objective testing/sampling/inspection/etc. Users of this standard are cautioned that compliance with Practice D3740 does not in itself assure reliable results. Reliable results depend on many factors; Practice D3740 provides a means of evaluating some of those factors.

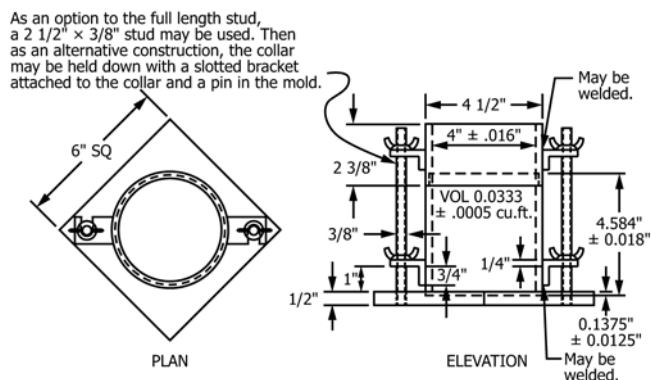
6. Apparatus

6.1 *Mold Assembly*—The molds shall be cylindrical in shape, made of rigid metal and be within the capacity and dimensions indicated in 6.1.1 or 6.1.2 and Fig. 1 and Fig. 2. See also Table 1. The walls of the mold may be solid, split, or tapered. The “split” type may consist of two half-round sections, or a section of pipe split along one element, which can be securely locked together to form a cylinder meeting the requirements of this section. The “tapered” type shall have an internal diameter taper that is uniform and not more than 0.200 in./ft (16.7 mm/m) of mold height. Each mold shall have a base plate and an extension collar assembly, both made of rigid metal and constructed so they can be securely attached and easily detached from the mold. The extension collar assembly shall have a height extending above the top of the mold of at least 2.0 in. (51 mm) which may include an upper section that flares out to form a funnel, provided there is at least a 0.75-in. (19-mm) straight cylindrical section beneath it. The extension collar shall align with the inside of the mold. The bottom of the base plate and bottom of the centrally recessed area that accepts the cylindrical mold shall be planar within ± 0.005 in. (± 0.1 mm).

6.1.1 *Mold, 4 in.*—A mold having a 4.000 ± 0.016 -in. (101.6 ± 0.4 -mm) average inside diameter, a height of 4.584 ± 0.018 in. (116.4 ± 0.5 mm) and a volume of 0.0333 ± 0.0005 ft³ (943.0 ± 14.0 cm³). A mold assembly having the minimum required features is shown in Fig. 1.

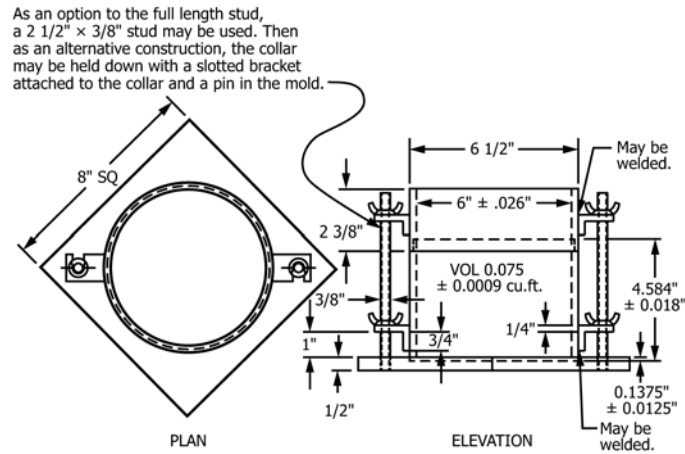
6.1.2 *Mold, 6 in.*—A mold having a 6.000 ± 0.026 -in. (152.4 ± 0.7 -mm) average inside diameter, a height of 4.584 ± 0.018 in. (116.4 ± 0.5 mm), and a volume of 0.0750 ± 0.0009 ft³ (2124 ± 25 cm³). A mold assembly having the minimum required features is shown in Fig. 2.

6.2 *Rammer*—A rammer, either manually operated as described further in 6.2.1 or mechanically operated as described in 6.2.2. The rammer shall fall freely through a distance of 18.00 ± 0.05 in. (457.2 ± 1.3 mm) from the surface of the



NOTE 1—See Table 1 for SI equivalents.

FIG. 1 Cylindrical Mold, 4.0-in.



NOTE 1—See Table 1 for SI equivalents.

FIG. 2 Cylindrical Mold, 6.0-in.

TABLE 1 SI Equivalents for Figs. 1 and 2

in.	mm
0.016	0.41
0.026	0.66
0.032	0.81
0.028	0.71
1/2	12.70
2 3/8	60.33
2 1/2	63.50
2 5/8	66.70
4	101.60
4 1/2	114.30
4.584	116.43
4 3/4	120.60
6	152.40
6 1/2	165.10
6 5/8	168.30
6 3/4	171.40
8 1/4	208.60
ft ³	cm ³
1/30 (0.0333)	943
0.0005	14
1/13.333 (0.0750)	2,124
0.0011	31

specimen. The weight of the rammer shall be 10.00 ± 0.02 lbf (44.48 ± 0.09 N, or mass of 4.5364 ± 0.009 kg), except that the weight of the mechanical rammers may be adjusted as described in Practices D2168 (see Note 6). The striking face of the rammer shall be planar and circular, except as noted in 6.2.2.1, with a diameter when new of 2.000 ± 0.005 in. (50.80 ± 0.13 mm). The rammer shall be replaced if the striking face becomes worn or bellied to the extent that the diameter exceeds 2.000 ± 0.01 in. (50.80 ± 0.25 mm).

NOTE 6—It is a common and acceptable practice to determine the weight of the rammer using either a kilogram or pound balance and assume 1 lbf is equivalent to 0.4536 kg, 1 lbf is equivalent to 1 lbm, or 1 N is equivalent to 0.2248 lbf or 0.1020 kg.

6.2.1 *Manual Rammer*—The rammer shall be equipped with a guide sleeve that has sufficient clearance that the free fall of the rammer shaft and head is not restricted. The guide sleeve shall have at least four vent holes at each end (eight holes total) located with centers $3/4 \pm 1/16$ in. (19 ± 2 mm) from each end and spaced 90° apart. The minimum diameter of the vent holes

shall be $3/8$ in. (9.5 mm). Additional holes or slots may be incorporated in the guide sleeve.

6.2.2 *Mechanical Rammer-Circular Face*—The rammer shall operate mechanically in such a manner as to provide uniform and complete coverage of the specimen surface. There shall be 0.10 ± 0.03 -in. (2.5 ± 0.8 -mm) clearance between the rammer and the inside surface of the mold at its smallest diameter. The mechanical rammer shall meet the standardization/calibration requirements of Practices D2168. The mechanical rammer shall be equipped with a positive mechanical means to support the rammer when not in operation.

6.2.2.1 *Mechanical Rammer-Sector Face*—The sector face can be used with the 6.0-in. (152.4-mm) mold, as an alternative to the circular face mechanical rammer described in 6.2.2. The striking face shall have the shape of a sector of a circle of radius equal to 2.90 ± 0.02 in. (73.7 ± 0.5 mm) and an area about the same as the circular face (see 6.2). The rammer shall operate in such a manner that the vertex of the sector is positioned at the center of the specimen and follow the compaction pattern given in Fig. 3(b).

6.3 *Sample Extruder (optional)*—A jack, with frame or other device adapted for the purpose of extruding compacted specimens from the mold.

6.4 *Balance*—A Class GP5 balance meeting the requirements of Specification D4753 for a balance of 1-g readability. If the water content of the compacted specimens is determined using a representative portion of the specimen, rather than the whole specimen, and if the representative portion is less than 1000 g, a Class GP2 balance having a 0.1-g readability is needed in order to comply with Test Methods D2216 requirements for determining water content to 0.1 %.

NOTE 7—Use of a balance having an equivalent capacity and a readability of 0.002 lbm as an alternative to a class GP5 balance should not be regarded as nonconformance to this standard.

6.5 *Drying Oven*—Thermostatically controlled oven, capable of maintaining a uniform temperature of $230 \pm 9^\circ\text{F}$ ($110 \pm 5^\circ\text{C}$) throughout the drying chamber. These requirements typically require the use of a forced-draft type oven. Preferably the oven should be vented outside the building.

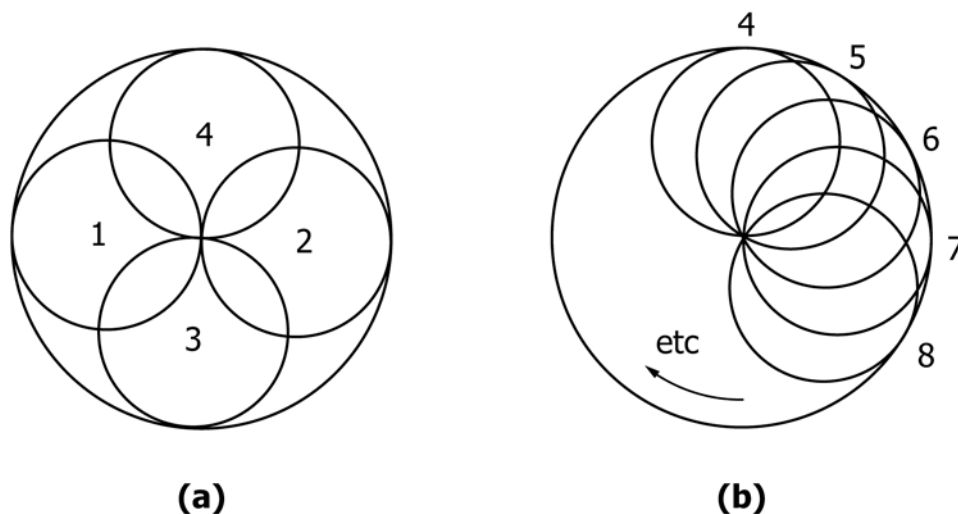


FIG. 3 Rammer Pattern for Compaction in 4-in. (101.6-mm) Mold

6.6 *Straightedge*—A stiff metal straightedge of any convenient length but not less than 10 in. (250 mm). The total length of the straightedge shall be machined straight to a tolerance of ± 0.005 in. (± 0.1 mm). The scraping edge shall be beveled if it is thicker than $\frac{1}{8}$ in. (3 mm).

6.7 *Sieves*— $\frac{3}{4}$ in. (19.0 mm), $\frac{3}{8}$ in. (9.5 mm), and No. 4 (4.75 mm), conforming to the requirements of Specification E11.

6.8 *Mixing Tools*—Miscellaneous tools such as mixing pan, spoon, trowel, spatula, spray device (to add water evenly), and (preferably, but optional) a suitable mechanical device for thoroughly mixing the subspecimen of soil with increments of water.

7. Standardization/Calibration

7.1 Perform standardizations before initial use, after repairs or other occurrences that might affect the test results, at intervals not exceeding 1000 test specimens, or annually, whichever occurs first, for the following apparatus:

7.1.1 *Balance*—Evaluate in accordance with Specification D4753 or Practice E319.

7.1.2 *Molds*—Determine the volume as described in Annex A1.

7.1.3 *Manual Rammer*—Verify the free fall distance, rammer weight, and rammer face are in accordance with 6.2. Verify the guide sleeve requirements in accordance with 6.2.1.

7.1.4 *Mechanical Rammer*—Verify and adjust if necessary that the mechanical rammer in accordance with Practices D2168. In addition, the clearance between the rammer and the inside surface of the mold shall be verified in accordance with 6.2.2.

8. Test Specimen

8.1 The minimum test specimen (test fraction) mass for Methods A and B is about 16 kg, and for Method C is about 29 kg of dry soil. Therefore, the field sample (see Practices D4220/D4220M for practices of preserving and transporting soil samples) should have a moist mass of at least 23 kg and 45

kg, respectively. Greater masses would be required if the oversize fraction is large (see 10.2 or 10.3) or an additional molding water content is taken during compaction of each point (see 10.4.1).

8.2 If gradation data is not available, estimate the percentage of material (by mass) retained on the No. 4 (4.75-mm), $\frac{3}{8}$ -in. (9.5-mm), or $\frac{3}{4}$ -in. (19.0-mm) sieve as appropriate for selecting Method A, B, or C, respectively. If it appears the percentage retained of interest is close to the allowable value for a given Method (A, B, or C), then either:

8.2.1 Select a Method that allows a higher percentage retained (B or C).

8.2.2 Using the sieve size designated for the Method of interest, process the specimen in accordance with 10.2 or 10.3 herein. This determines the percentage of material retained for that method. If the percentage retained is acceptable, proceed. If the percentage retained is not acceptable, go to Method B or C using the next larger sieve size.

8.2.3 Determine percentage retained values using a representative portion from the total sample, and performing a simplified or complete gradation analysis using the sieve(s) of interest and Test Method D6913/D6913M or C136/C136M. It is only necessary to calculate the retained percentage(s) for the sieve or sieves for which information is desired.

9. Preparation of Apparatus

9.1 Select the proper compaction mold(s), collar, and base plate in accordance with the Method (A, B, or C) being used. Check that the volume of the mold is known and whether the volume was determined with or without the base plate. Also, check that the mold is free of nicks or dents, and that the mold will fit together properly with the collar and base plate.

9.2 Check that the manual or mechanical rammer assembly is in good working condition and that parts are not loose or worn. Make any necessary adjustments or repairs. If adjustments or repairs are made, the rammer must be restandardized.

10. Procedure

10.1 Soils:

10.1.1 Do not reuse soil that has been previously compacted in the laboratory. The reuse of previously compacted soil yields a significantly greater maximum dry unit weight (1).

10.1.2 When using this test method for soils containing hydrated halloysite, or in which past experience indicates that results will be altered by air-drying, use the moist preparation method (see Section 10.2). In referee testing, each laboratory has to use the same method of preparation, either moist (preferred) or air-dried.

10.1.3 Prepare the soil specimens for testing in accordance with 10.2 (preferred) or with 10.3.

10.2 *Moist Preparation Method (preferred)*—Without previously drying the sample/specimen, process it over a No. 4 (4.75-mm), $\frac{3}{8}$ -in. (9.5-mm), or $\frac{3}{4}$ -in. (19.0-mm) sieve, depending on the Method (A, B, or C) being used or required as covered in 8.2. For additional processing details, see Test Method D6913/D6913M. Determine and record the mass of both the retained and passing portions (oversize fraction and test fraction, respectively) to the nearest g. Oven dry the oversize fraction and determine and record its dry mass to the nearest g. If it appears more than 0.5 % of the total dry mass of the specimen is adhering to the oversize fraction, wash that fraction. Then determine and record its oven dry mass to the nearest g. Determine and record the water content of the processed soil (test fraction). Using that water content, determine and record the oven dry mass of the test fraction to the nearest g. Based on these oven dry masses, the percent oversize fraction, P_C , and test fraction, P_F , shall be determined and recorded, unless a gradation analysis has already been performed. See Section 11 on Calculations.

10.2.1 From the test fraction, select and prepare at least four (preferably five) subspecimens having molding water contents such that they bracket the estimated optimum water content. A subspecimen having a molding water content close to optimum should be prepared first by trial additions or removals of water and mixing (see Note 8). Select molding water contents for the rest of the subspecimens to provide at least two subspecimens wet and two subspecimens dry of optimum, and molding water contents varying by about 2 %. At least two molding water contents are necessary on the wet and dry side of optimum to define the dry-unit-weight compaction curve (see 10.5). Some soils with very high optimum water content or a relatively flat compaction curve may require larger molding water content increments to obtain a well-defined maximum dry unit weight. Molding water content increments should not exceed about 4 %.

NOTE 8—With practice it is usually possible to visually judge a point near optimum water content. Typically, cohesive soils at the optimum water content can be squeezed into a lump that barely sticks together when hand pressure is released, but will break cleanly into two sections when “bent.” They tend to crumble at molding water contents dry of optimum; they tend to stick together in a sticky cohesive mass wet of optimum. For cohesive soils, the optimum water content is typically slightly less than the plastic limit. For cohesionless soils, the optimum water content is typically close to zero or at the point where bleeding occurs.

10.2.2 Thoroughly mix the test fraction, then using a scoop select representative soil for each subspecimen (compaction

point). Select about 2.3 kg when using Method A or B, or about 5.9 kg for Method C. Test Method D6913/D6913M section on Specimen and Annex A2 give additional details on obtaining representative soil using this procedure and the reason it is the preferred method. To obtain the subspecimen’s molding water contents selected in 10.2.1, add or remove the required amounts of water as follows: To add water, spray it into the soil during mixing; to remove water, allow the soil to dry in air at ambient temperature or in a drying apparatus such that the temperature of the sample does not exceed 140°F (60°C). Mix the soil frequently during drying to facilitate an even water content distribution. Thoroughly mix each subspecimen to facilitate even distribution of water throughout and then place in a separate covered container to stand (cure) in accordance with Table 2 prior to compaction. For selecting a standing time, the soil may be classified using Practice D2487, Practice D2488 or data on other samples from the same material source. For referee testing, classification shall be by Practice D2487.

10.3 *Dry Preparation Method*—If the sample/specimen is too damp to be friable, reduce the water content by air drying until the material is friable. Drying may be in air or by the use of drying apparatus such that the temperature of the sample does not exceed 140°F (60°C). Thoroughly break up the aggregations in such a manner as to avoid breaking individual particles. Process the material over the appropriate sieve: No. 4 (4.75 mm), $\frac{3}{8}$ in. (9.5 mm), or $\frac{3}{4}$ in. (19.0 mm). When preparing the material by passing over the $\frac{3}{4}$ -in. sieve for compaction in the 6-in. mold, break up aggregations sufficiently to at least pass the $\frac{3}{8}$ in. sieve in order to facilitate the distribution of water throughout the soil in later mixing. Determine and record the water content of the test fraction and all masses covered in 10.2, as applicable to determine the percent oversize fraction, P_C , and test fraction, P_F .

10.3.1 From the test fraction, select and prepare at least four (preferably five) subspecimens in accordance with 10.2.1 and 10.2.2, except for the following: Use either a mechanical splitting or quartering process to obtain the subspecimens. As stated in Test Method D6913/D6913M, both of these processes will yield non-uniform subspecimens compared to the moist procedure. Typically, only the addition of water to each subspecimen will be required.

10.4 *Compaction*—After standing (curing), if required, each subspecimen (compaction point) shall be compacted as follows:

10.4.1 Determine and record the mass of the mold or mold and base plate, see 10.4.7.

10.4.2 Assemble and secure the mold and collar to the base plate. Check the alignment of the inner wall of the mold and mold extension collar. Adjust if necessary. The mold shall rest without wobbling/rocking on a uniform rigid foundation, such as provided by a cylinder or cube of concrete with a weight or

TABLE 2 Required Standing Times of Moisturized Specimens

Classification	Minimum Standing Time, h
GW, GP, SW, SP	no requirement
GM, SM	3
All other soils	16

mass of not less than 200 lb or 91 kg, respectively. Secure the base plate to the rigid foundation. The method of attachment to the rigid foundation shall allow easy removal of the assembled mold, collar and base plate after compaction is completed.

10.4.2.1 During the compaction procedure, it is advantageous but not required to determine the water content of each subspecimen immediately prior to compaction. This provides a check on the molding water content determined for each compaction point and the magnitude of bleeding. See 10.4.9. However, more soil will have to be selected for each subspecimen than stated in 10.2.2.

10.4.3 Compact the soil in five layers. After compaction, each layer should be approximately equal in thickness and the final layer shall extend slightly into the collar. Prior to compaction, place the loose soil into the mold and spread into a layer of uniform thickness. Lightly tamp the soil prior to compaction until it is not in a fluffy or loose state, using either the manual rammer or a cylinder approximately 2 in. (50 mm) in diameter. Following compaction of each of the first four layers, any soil that has not been compacted, such as adjacent to the mold walls, or extends above the compacted surface (up the mold walls) shall be trimmed. The trimmed soil shall be discarded. A knife or other suitable device may be used. The total amount of soil used shall be such that the fifth compacted layer slightly extends into the collar, but does not extend more than approximately $\frac{1}{4}$ in. (6 mm) above the top of the mold. If the fifth layer does extend above this limit, then the compaction point shall be discarded. In addition, the compaction point shall be discarded when the last blow on the rammer for the fifth layer results in the bottom of the rammer extending below the top of the compaction mold, unless the soil is pliable enough that this surface can easily be forced above the top of the compaction mold during trimming. See Note 9.

10.4.4 Compact each layer with 25 blows for the 4-in. (101.6-mm) mold or with 56 blows for the 6-in. (152.4-mm) mold. The manual rammer shall be used for referee testing.

10.4.5 In operating the manual rammer, take care to avoid lifting the guide sleeve during the rammer upstroke. Hold the guide sleeve steady and within 5° of vertical. Apply the blows at a uniform rate of about 25 blows/min and in such a manner as to provide complete, uniform coverage of the specimen surface. When using a 4-in. (101.6-mm) mold and manual rammer, follow the blow pattern given in Fig. 3(a) and Fig. 3(b) while for a mechanical rammer, follow the pattern in Fig. 3(b). When using a 6-in. (152.4-mm) mold and manual rammer, follow the blow pattern given in Fig. 4 up to the 9th blow, then systematically around the mold (Fig. 3(b)) and in the middle. When using a 6-in. (152.4-mm) mold and a mechanical rammer equipped with a sector face, the mechanical rammer shall be designed to follow the compaction pattern given in Fig. 3(b). When using a 6-in. (152.4-mm) mold and a mechanical rammer equipped with a circular face, the mechanical rammer shall be designed to distribute the blows uniformly over the surface of the specimen. If the surface of the compacted soil becomes highly uneven (see Note 9) then adjust the pattern to follow the logic given in Fig. 3(a) or Fig. 4. This will most likely void the use of a mechanical rammer for such compaction points.

NOTE 9—When compacting specimens wetter than optimum water content, uneven compacted surfaces can occur and operator judgment is required as to the average height of the specimen and rammer pattern during compaction.

10.4.6 Following compaction of the last layer, remove the collar and base plate (except as noted in 10.4.7) from the mold. A knife may be used to trim the soil adjacent to the collar to loosen the soil from the collar before removal to avoid disrupting the soil below the top of the mold. In addition, to prevent/reduce soil sticking to the collar or base plate, rotate them before removal.

10.4.7 Carefully trim the compacted specimen even with the top of the mold by means of the straightedge scraped across the top of the mold to form a plane surface even with the top of the

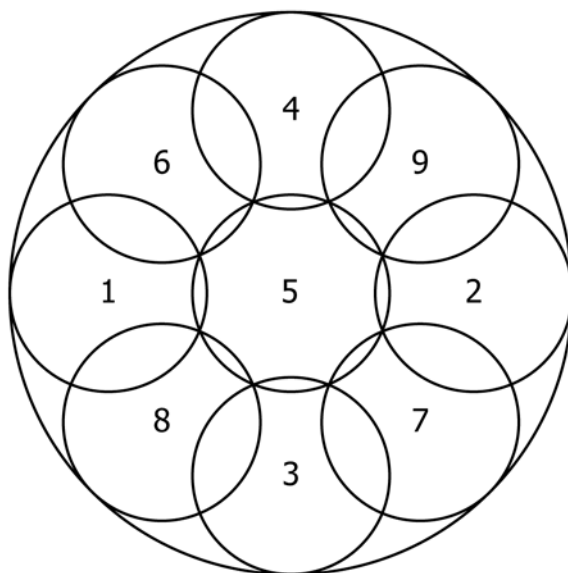


FIG. 4 Rammer Pattern for Compaction in 6-in. (152.4-mm) Mold

mold. Initial trimming of the specimen above the top of the mold with a knife may prevent tearing out soil below the top of the mold. Fill any holes in either surface with unused or trimmed soil from the specimen, press in with the fingers, and again scrape the straightedge across the top of the mold. If gravel size particles are encountered, trim around them or remove them, whichever is the easiest and reduces the disturbance of the compacted soil. The estimated volume of particles above the surface of the compacted soil and holes in that surface shall be equal. Fill in remaining holes as mentioned above. Repeat the appropriate preceding operations on the bottom of the specimen when the mold volume was determined without the base plate. For very wet or dry soils, soil or water may be lost if the base plate is removed. For these situations, leave the base plate attached to the mold. When the base plate is left attached, the volume of the mold must be calibrated with the base plate attached to the mold rather than a plastic or glass plate as noted in **Annex A1 (A1.4.1)**.

10.4.8 Determine and record the mass of the specimen and mold to the nearest g. When the base plate is left attached, determine and record the mass of the specimen, mold and base plate to the nearest g.

10.4.9 Remove the material from the mold. Obtain a specimen for molding water content by using either the whole specimen (preferred method) or a representative portion. When the entire specimen is used, break it up to facilitate drying. Otherwise, obtain a representative portion of the five layers, removing enough material from the specimen to report the water content to 0.1 %. The mass of the representative portion of soil shall conform to the requirements of Table 1, Method B, of Test Methods **D2216**. Determine the molding water content in accordance with Test Method **D2216**.

10.5 Following compaction of the last specimen, compare the wet unit weights to ensure that a desired pattern of obtaining data on each side of the optimum water content will be attained for the dry-unit-weight compaction curve. Plotting the wet unit weight and molding water content of each compacted specimen can be an aid in making the above evaluation. If the desired pattern is not obtained, additional compacted specimens will be required. Generally, for experienced plotters of compaction curves, one compaction point wet of the optimum water content is adequate to define the maximum wet unit weight. See **11.2**.

11. Calculation and Plotting (Compaction Curve)

11.1 *Fraction Percentages*—If gradation data from Test Method **D6913/D6913M** is not available, calculate the dry mass of the test fraction, percentage of oversize fraction, and test fraction as covered below and using the data from **10.2** or **10.3**:

11.1.1 *Test Fraction*—Determine the dry mass of the test fraction as follows:

$$M_{d,tf} = \frac{M_{m,tf}}{1 + \frac{w_{tf}}{100}} \quad (1)$$

where:

$M_{d,tf}$ = dry mass of test fraction, nearest g or 0.001 kg,

$M_{m,tf}$ = moist mass of test fraction, nearest g or 0.001 kg, and

w_{tf} = water content of test fraction, nearest 0.1 %.

11.1.2 *Oversize Fraction Percentage*—Determine the oversize (coarse) fraction percentage as follows:

$$P_c = \frac{M_{d,of}}{M_{d,of} + M_{d,tf}} \quad (2)$$

where:

P_c = percentage of oversize (coarse) fraction, nearest %, $M_{d,of}$ = dry mass of oversize fraction, nearest g or 0.001 kg.

11.1.3 *Test Fraction Percentage*—Determine the test (finer) fraction percentage as follows:

$$P_f = 100 - P_c \quad (3)$$

where:

P_f = percentage of test (finer) fraction, nearest %

11.2 *Density and Unit Weight*—Calculate the molding water content, moist density, dry density, and dry unit weight of each compacted specimen as explained below.

11.2.1 *Molding Water Content, w*—Calculate in accordance with Test Method **D2216** to nearest 0.1 %.

11.2.2 *Density and Unit Weights*—Calculate the moist (total) density (**Eq 4**), the dry density (**Eq 5**), and then the dry unit weight (**Eq 6**) as follows:

11.2.2.1 *Moist Density*:

$$\rho_m = K \times \frac{(M_t - M_{md})}{V} \quad (4)$$

where:

ρ_m = moist density of compacted subspecimen (compaction point), four significant digits, g/cm³ or kg/m³,

M_t = mass of moist soil in mold and mold, nearest g,

M_{md} = mass of compaction mold, nearest g,

V = volume of compaction mold, cm³ or m³ (see **Annex A1**), and

K = conversion constant, depending on density units and volume units. Use 1 for g/cm³ and volume in cm³. Use 1000 for g/cm³ and volume in m³. Use 0.001 for kg/cm³ and volume in m³. Use 1000 for kg/m³ and volume in cm³.

11.2.2.2 *Dry Density*:

$$\rho_d = \frac{\rho_m}{1 + \frac{w}{100}} \quad (5)$$

where:

ρ_d = dry density of compaction point, four significant digits, g/cm³ or kg/m³, and

w = molding water content of compaction point, nearest 0.1 %.

11.2.2.3 *Dry Unit Weight*:

$$\gamma_d = K_1 \times \rho_d \quad (6)$$

in lbf/ft³, or,

$$\gamma_d = K_2 \times \rho_d \quad (7)$$

in kN/m³,

where:

- γ_d = dry unit weight of compacted specimen, four significant digits, in lb/ft^3 or kN/m^3 ,
 K_1 = conversion constant, depending on density units. Use 62.428 for density in g/cm^3 , or use 0.062428 for density in kg/m^3 ,
 K_2 = conversion constant, depending on density units. Use 9.8066 for density in g/cm^3 , or use 0.0098066 for density in kg/m^3 .

11.3 Compaction Curve—Plot the dry unit weight and molding water content values, the saturation curve (see 11.3.2), and draw the compaction curve as a smooth curve through the points (see example, Fig. 5). For each point on the compaction curve, calculate, record, and plot dry unit weight to the nearest 0.1 lb/ft^3 (0.02 kN/m^3) and molding water content to the nearest 0.1 %. From the compaction curve, determine the compaction results: optimum water content, to nearest 0.1 % and maximum dry unit weight, to the nearest 0.1 lb/ft^3 (0.02 kN/m^3). If more than 5 % by mass of oversize material was removed from the sample/specimen, calculate the corrected optimum water content and maximum dry unit weight of the total material using Practice D4718/D4718M. This correction may be made to the appropriate field in-place density test specimen rather than to the laboratory compaction results.

11.3.1 In these plots, the scale sensitivities should remain the same, that is, the change in molding water content or dry unit weight per division is constant between plots. Typically, the change in dry unit weight per division is twice that of

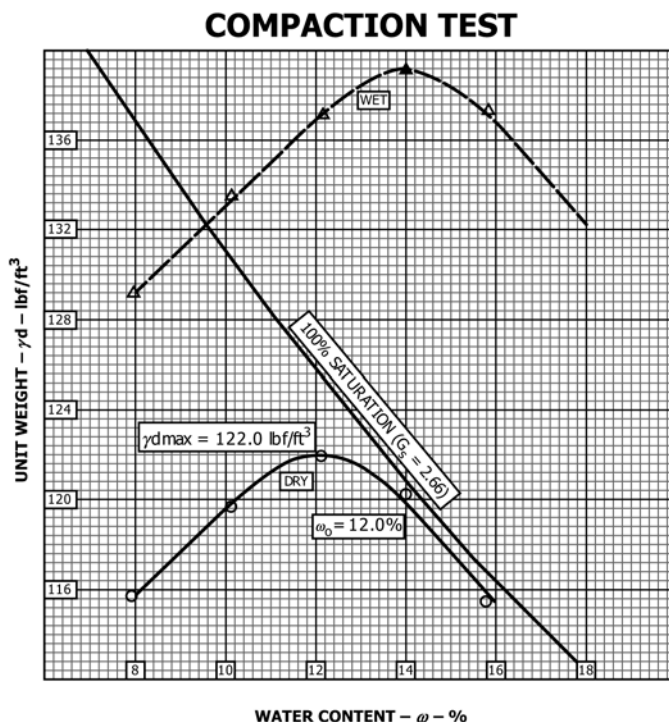
molding water content's (2 lb/ft^3 to 1 % w per major division). Therefore, any change in the shape of the compaction curve is a result of testing different material, not the plotting scale. However, a one to one ratio should be used for soils have a relatively flat compaction curve (see 10.2.1), such as highly plastic soils or relatively free draining ones up to the point of bleeding.

11.3.1.1 The shape of the compaction curve on the wet side on optimum should typically follow that of the saturation curve. The shape of the compaction curve on the dry side of optimum may be relatively flat or up and down when testing some soils, such as relatively free draining ones or plastic soils prepared using the moist procedure and having molding water contents close to or less than the shrinkage limit.

11.3.2 Plot the 100 % saturation curve, based on either an estimated or a measured specific gravity. Values of water content for the condition of 100 % saturation can be calculated as explained in 11.4 (see example, Fig. 5).

NOTE 10—The 100 % saturation curve is an aid in drawing the compaction curve. For soils containing more than about 10 % fines and molding water contents well above optimum, the two curves generally become roughly parallel with the wet side of the compaction curve between 92 % to 95 % saturation. Theoretically, the compaction curve cannot plot to the right of the 100 % saturation curve. If it does, there is an error in specific gravity, in measurements, in calculations, in testing, or in plotting. The 100 % saturation curve is sometimes referred to as the zero air voids curve or the complete saturation curve.

11.4 Saturation Points—To calculate points for plotting the 100 % saturation curve or zero air voids curve, select values of



NOTE 1—Wet Unit Weights are usually not plotted. They are plotted here for informational purposes only. Also notice that the compaction points may not all lie exactly on the compaction curve.

FIG. 5 Example Compaction Curve Plotting

dry unit weight, calculate corresponding values of water content corresponding to the condition of 100 % saturation as follows:

$$w_{sat} = \frac{(\gamma_w)(G_s) - \gamma_d}{(\gamma_d)(G_s)} \times 100 \quad (8)$$

where:

- w_{sat} = water content for complete saturation, nearest 0.1 %,
- γ_w = unit weight of water, 62.32 lbf/ft³ (9.789 kN/m³) at 20°C,
- γ_d = dry unit weight of soil, lbf/ft³ (kN/m³), three significant digits, and
- G_s = specific gravity of soil (estimated or measured), to nearest 0.01 value, see 11.4.1.

11.4.1 Specific gravity may be estimated for the test fraction based on test data from other soils having the same soil classification and source or experience. Otherwise, a specific gravity test (Test Method C127, Test Method D854, or both) is necessary.

12. Report: Data Sheet(s)/Form(s)

12.1 The methodology used to specify how data are recorded on the test data sheet(s)/form(s), as described below, is covered in Section 1.6.

12.2 The data sheet(s)/form(s) shall contain as a minimum the following information:

- 12.2.1 Method used (A, B, or C).
- 12.2.2 Preparation method used (moist or dry).
- 12.2.3 As-received water content, if determined, nearest 1 %.
- 12.2.4 Modified optimum water content, Mod- w_{opt} to nearest 0.1 %.
- 12.2.5 Modified maximum (optimum) dry unit weight, Mod- $\gamma_{d,max}$ nearest 0.1 lbf/ft³ or 0.02 kN/m³.
- 12.2.6 Type of rammer (manual or mechanical).
- 12.2.7 Soil sieve data when applicable for selection of Method (A, B, or C) used.
- 12.2.8 Description of sample used in test (as a minimum, color and group name and symbol), by Practice D2488, or classification by Test Method D2487.
- 12.2.9 Specific gravity and method of determination, nearest 0.01 value.
- 12.2.10 Identification of sample used in test, for example, project number/name, location, depth, and the like.
- 12.2.11 Compaction curve plot showing compaction points used to establish compaction curve, and 100 % saturation curve, value or point of maximum dry unit weight and optimum water content.
- 12.2.12 Percentages for the fractions retained (P_C) and passing (P_F) the sieve used in Method A, B, or C, nearest 1 %. In addition, if compaction data (Mod- w_{opt} and Mod- $\gamma_{d,max}$) are corrected for the oversize fraction, include that data.

NOTE 11—The Data Sheet(s)/Form requirements in Section 12 are not intended as requirements for reporting final test results to the requesting agency. The requirements apply to testing records for measurements, for intermediate calculations and for compaction points used to plot the

compaction curve. It has been attempted in this test method to determine all measurements and calculations to four significant figures. The purpose is to ensure that precision is not lost due to rounding prior to plotting the compaction curve and that data sheets and forms retained by the laboratory contain that same degree of precision.

13. Precision and Bias⁴

13.1 *Precision*—Criteria for judging the acceptability of maximum unit weight and optimum water content results obtained by this method are given in Table 3.

13.1.1 *Single-Operator Precision (Repeatability)*—The figures in Column 2 of Table 3 are the standard deviations that have been found to be appropriate for the conditions of test described in Column 1. Two results obtained in the same laboratory, by the same operator using the same equipment, in the shortest practical period of time, should not be considered suspect unless the difference in the two results exceeds the values given in Table 3, Column 3.

13.1.2 *Multilaboratory Precision (Reproducibility)*—The figures in Column 2 of Table 3 are the standard deviations that have been found to be appropriate for the conditions of test described in Column 1. Two results submitted by two different operators testing the same material in different laboratories shall not be considered suspect unless the difference in the two results exceeds the values given in Table 3, Column 3.

13.2 *Bias*—It is not possible to present information on bias because there is no other method of determining the values of modified maximum unit weight and modified optimum water content.

14. Keywords

14.1 compaction characteristics; density; impact compaction using modified effort; laboratory tests; modified proctor test; moisture-density curves; soil compaction

⁴ Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D18-1017.

TABLE 3 Precision Estimates

Conditions of Test and Test Property	Standard Deviation (1s) ⁴	Acceptable Range of Two Test Results (d2s) ⁴
Single Operator Precision:		
Maximum Unit Weight (lbf/ft ³)	0.6	1.8
Optimum Water Content (percent)	0.4	1.0
Multilaboratory Precision:		
Maximum Unit Weight (lbf/ft ³)	1.6	4.4
Optimum Water Content (percent)	0.7	2.1

⁴These numbers represent, respectively, the (1s) and (d2s) limits as described in Practice C670 for Preparing Precision and Bias Statements for Test Methods for Construction Materials.

NOTE 1—The precision estimates given in Table 3 are based on the analysis of test results from three pairs of AMRL proficiency samples. The data analyzed consisted of results from 144 to 253 laboratories for each of the three pairs of samples. The analysis included two classifications of fine grained soil (group symbol CL): lean clay with sand and sandy lean clay. Average maximum unit weights ranged from 125.8 lbf/ft³ to 132.6 lbf/ft³. Average optimum water contents ranged from 8.0 percent to 10.4 percent.

ANNEX

(Mandatory Information)

A1. VOLUME OF COMPACTION MOLD

A1.1 Scope

A1.1.1 This annex describes the method for determining the volume of a compaction mold.

A1.1.2 The volume is determined by two methods, a water-filled method and a linear-measurement method.

A1.1.3 The water filling method for the 4-in. (106.5-mm) mold, when using a balance readable to nearest g, does not yield four significant figures for its volume, just three. Based on Practice **D6026**, this limits the density/unit weight determinations previously presented from four to three significant figures. To prevent this limitation, the water filling method has been adjusted from that presented in early versions of this test method.

A1.2 Apparatus

A1.2.1 In addition to the apparatus listed in Section 6, the following items are required:

A1.2.1.1 *Vernier or Dial Caliper*, having a measuring range of at least 0 to 6 in. (0 to 150 mm) and readable to at least 0.001 in. (0.02 mm).

A1.2.1.2 *Inside Micrometer (optional)*, having a measuring range of at least 2 to 12 in. (50 to 300 mm) and readable to at least 0.001 in. (0.02 mm).

A1.2.1.3 *Depth Micrometer (optional)*, having a measuring range of at least 0 to 6 in. (0 to 150 mm) and readable to at least 0.001 in. (0.02 mm).

A1.2.1.4 *Plastic or Glass Plates*—Two plastic or glass plates about 8 in. by 8 in. by ¼ in. thick (200 mm by 200 mm by 6 mm).

A1.2.1.5 *Thermometer or other Thermometric Device*, having a readability of 0.1°C and a maximum permissible error of 0.5°C.

A1.2.1.6 *Stopcock Grease* or similar sealant.

A1.2.1.7 *Distilled Water or De-ionized Water*—Either type of water may be used to fill the mold when determining the mold volume using the water-filling method. Distilled water or de-ionized water may be purchased and is available in most grocery stores. In the procedure for the water-filling method, distilled water, or de-ionized water, is referred to as water.

A1.2.1.8 *Miscellaneous equipment*—Bulb syringe, towels, etc.

A1.3 Precautions

A1.3.1 Perform this method in an area isolated from drafts or extreme temperature fluctuations.

A1.4 Procedure

A1.4.1 Water-Filling Method:

A1.4.1.1 Lightly grease the bottom of the compaction mold and place it on one of the plastic or glass plates. Lightly grease the top of the mold. Be careful not to get grease on the inside

of the mold. If it is necessary to use the base plate, as noted in **10.4.7**, place the greased mold onto the base plate and secure with the locking studs.

A1.4.1.2 Determine the mass of the greased mold and both plastic or glass plates to the nearest 1 g and record, M_{mp} . When the base plate is being used in lieu of the bottom plastic or glass plate determine the mass of the mold, base plate and a single plastic or glass plate to be used on top of the mold to the nearest 1 g and record.

A1.4.1.3 Place the mold and the bottom plate on a firm, level surface and fill the mold with water to slightly above its rim.

A1.4.1.4 Slide the second plate over the top surface of the mold so that the mold remains completely filled with water and air bubbles are not entrapped. Add or remove water as necessary with a bulb syringe.

A1.4.1.5 Completely dry any excess water from the outside of the mold and plates.

A1.4.1.6 Determine the mass of the mold, plates and water and record to the nearest 1 g, $M_{mp,w}$.

A1.4.1.7 Determine the temperature of the water in the mold to the nearest 0.1°C and record. Determine and record the density of water from the table given in **D854** or as follows:

$$\rho_{w,c} = 1.00034038 - (7.77 \times 10^{-6}) \times T - (4.95 \times 10^{-6}) \times T^2 \quad (\text{A1.1})$$

where:

$\rho_{w,c}$ = density of water, nearest 0.00001 g/cm³, and
 T = calibration test temperature, nearest 0.1°C.

A1.4.1.8 Calculate the mass of water in the mold by subtracting the mass determined in **A1.4.1.2** from the mass determined in **A1.4.1.6**.

A1.4.1.9 Calculate the volume of water by dividing the mass of water by the density of water. Record this volume to the nearest 0.1 cm³ for the 4-in. (101.6-mm) mold or nearest 1 cm³ for the 6-in. (152.4-mm) mold. To determine the volume of the mold in m³, multiply the volume in cm³ by 1×10^{-6} . Record this volume, as prescribed.

A1.4.1.10 If the water-filling method is being used to determine the mold's volume and checked by linear measurement method, repeat this volume determination (**A1.4.1.3 – A1.4.1.9**) and determine and record the average value, V_w , as prescribed.

A1.4.2 Linear Measurement Method:

A1.4.2.1 Using either the vernier caliper or the inside micrometer (preferable), measure the inside diameter (ID) of the mold six times at the top of the mold and six times at the bottom of the mold spacing each of the six top and bottom measurements equally around the inside circumference of the mold. Record the values to the nearest 0.001 in. (0.02 mm). Determine and record the average ID to the nearest 0.001 in.

(0.02 mm), d_{avg} . Verify that this ID is within specified tolerances, 4.000 ± 0.016 in. (101.6 ± 0.4 mm); if not, discard the mold.

A1.4.2.2 Using the vernier caliper or depth micrometer (preferably), measure the inside height of the mold to the base plate. In these measurements, make three or more measurements equally spaced around the inside circumference of the mold, and preferably one in the center of the mold, but not required (use the straightedge to facilitate the latter measurement and correct the measurement for the thickness of the straightedge). Record these values to the nearest 0.001 in. (0.02 mm). Determine and record the average of these height measurements to the nearest 0.001 in. (0.02 mm), h_{avg} . Verify that this height is within specified tolerances, 4.584 ± 0.018 in. (116.4 ± 0.5 mm); if not, discard the mold.

A1.4.2.3 Calculate the volume of the mold to four significant digits in cm^3 as follows:

$$V_{lm} = K_3 \frac{\pi \times h_{avg} \times (d_{avg})^2}{4} \quad (\text{A1.2})$$

where:

- V_{lm} = volume of mold by linear measurements, to four significant digits, cm^3 ,
- K_3 = constant to convert measurements made in inch (in.) or mm. Use 16.387 for measurements in inches. Use 10^{-3} for measurements in mm.
- π = 3.14159,

- h_{avg} = average height, in. (mm), and
- d_{avg} = average of the top and bottom diameters, in. (mm).

If the volume in m^3 is required, then multiply the above value by 10^{-6} .

A1.5 Comparison of Results and Standardized Volume of Mold

A1.5.1 The volume obtained by either method should be within the volume tolerance requirements of 6.1.1 and 6.1.2, using either or cm^3 or ft^3 . To convert cm^3 to ft^3 , divide cm^3 by 28 317, record to the nearest 0.0001 ft^3 .

A1.5.2 The difference between the two methods should not exceed 0.5 % of the nominal volume of the mold, cm^3 or ft^3 .

A1.5.3 Repeat the determination of volume which is most suspect, or both, if these criteria are not met.

A1.5.4 Failure to obtain satisfactory agreement between the two methods, even after several trials, is an indication that the mold is badly deformed and should be replaced.

A1.5.5 Use the volume of the mold determined using the water-filling method or linear method, or average of both methods as the standardized volume for calculating the moist density (see 11.2.2.1). This value (V) in cm^3 or m^3 shall have four significant digits. The use of a volume in ft^3 , along with masses in lbm shall not be regarded as a nonconformance with this standard.

REFERENCES

- (1) Johnson, A.W., and Sallberg, J.R., *Factors Influencing Compaction Test Results*, Highway Research Board, Bulletin 318, Publication 967, National Academy of Sciences-National Research Council, Washington, DC, 1962, p. 73.
- (2) Head, K. H., *Manual of Soil Laboratory Testing: Volume 1 Soil Classification and Compaction Tests*, Pentech Press, London, Second Edition, 1992, p. 313.
- (3) *Earth Manual*, United States Bureau of Reclamation, Part 1, Third Edition, 1998, pp. 255-260.
- (4) *Earth Manual*, United States Bureau of Reclamation, Part 2, Third Edition, 1990.
- (5) Torrey, V.H., and Donaghe, R.T., "Compaction Control of Earth-Rock Mixtures: A New Approach," *Geotechnical Testing Journal*, GTJODJ, Vol 17, No 3, September 1994, pp. 371-386.

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Designation: D2434 – 22

Standard Test Methods for Measurement of Hydraulic Conductivity of Coarse-Grained Soils¹

This standard is issued under the fixed designation D2434; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ϵ) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the U.S. Department of Defense.

1. Scope*

1.1 These test methods cover laboratory measurement of the hydraulic conductivity (also referred to as *coefficient of permeability*) of water-saturated coarse-grained soils (for example, sands and gravels) with $k > 10^{-7}$ m/s. The test methods utilize low hydraulic gradient conditions.

1.2 This standard describes two methods (A and B) for determining hydraulic conductivity of coarse-grained soils. Method A incorporates use of a rigid wall permeameter and Method B incorporates the use of a flexible wall permeameter. A single- or dual-ring rigid wall permeameter may be used in Method A. A dual-ring permeameter may be preferred over a single-ring permeameter when adverse effects from short-circuiting of permeant water along the sidewalls of the permeameter (that is, prevent sidewall leakage) are suspected by the user of this standard.

1.3 The test methods are used under constant head conditions.

1.4 The test methods are used under saturated soil conditions.

1.5 Water is used to permeate the test specimen with these test methods.

1.6 *Units*—The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

NOTE 1—Hydraulic conductivity has traditionally been reported in cm/s in the US, even though the official SI unit for hydraulic conductivity is m/s.

1.7 The observed and calculated values shall conform to the guidelines for significant digits and rounding established in Practice D6026.

1.8 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the*

responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

1.9 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 *ASTM Standards*:²

D653 Terminology Relating to Soil, Rock, and Contained Fluids

D2487 Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)

D3740 Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

D4753 Guide for Evaluating, Selecting, and Specifying Balances and Standard Masses for Use in Soil, Rock, and Construction Materials Testing

D5101 Test Method for Measuring the Filtration Compatibility of Soil-Geotextile Systems

D5716/D5716M Test Method for Measuring the Rate of Well Discharge by Circular Orifice Weir

D6026 Practice for Using Significant Digits and Data Records in Geotechnical Data

D6913/D6913M Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis

3. Terminology

3.1 *Definitions*:

3.1.1 For definitions of common technical terms in this standard, refer to Terminology D653.

3.1.2 *hydraulic conductivity, k , n* —(also referred to as coefficient of permeability or permeability) the rate of discharge of

¹ This test method is under the jurisdiction of ASTM Committee D18 on Soil and Rock and is the direct responsibility of Subcommittee D18.04 on Hydrologic Properties and Hydraulic Barriers.

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² For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

*A Summary of Changes section appears at the end of this standard

water under laminar flow conditions through a unit cross-sectional area of porous medium under a unit hydraulic gradient and standard temperature conditions (20°C).

3.1.3 *hydraulic gradient, i , n* —the change in total head (head loss, Δh) per unit distance (L) in the direction of fluid flow, in which $i = \Delta h/L$.

3.1.4 *permeameter, n* —the apparatus (cell) containing the test specimen in a hydraulic conductivity test.

3.1.5 *nominal particle size, n* —the particle size corresponding to 75 % passing [that is, D_{75}] per methods in Test Methods **D6913/D6913M**.

4. Summary of Test Method

4.1 The standard includes methods for determining the hydraulic conductivity of free-draining soils (for example, sands and gravels with low fines content) by permeating samples with water under constant head conditions. Multiple methods are presented that include different permeameters (rigid wall, dual-ring rigid wall, and flexible wall). Different options for constant head systems include reservoirs, Mariotte Bottle, and a flow pump.

5. Significance and Use

5.1 These test methods are used to measure one-dimensional vertical flow of water through initially saturated coarse-grained, pervious (that is, free-draining) soils under an applied hydraulic gradient. Hydraulic conductivity of coarse-grained soils is used in various civil engineering applications. These test methods are suitable for determination of hydraulic conductivity for soils with $k > 10^{-7}$ m/s.

NOTE 2—Clean coarse-grained soils that are classified using Practice **D2487-17** as GP, GW, SP, and SW can be tested using these test methods. Depending on fraction and characteristics of fine-grained particles present in soils, these test methods may be suitable for testing other soil types with fines content greater than 5 % (for example, GP-GC, SP-SM).

5.2 Coarse-grained soils are to be tested at a void ratio representative of field conditions. For engineered fills, compaction specification can be used to provide target test conditions, whereas for natural soils, field testing of in-situ density can be used to provide target test conditions.

5.3 Use of a dual-ring permeameter is included in these test methods in addition to a single-ring permeameter for the rigid wall test apparatus. The dual-ring permeameter allows for reducing potential adverse effects of sidewall leakage on measured hydraulic conductivity of the test specimens. The use of a plate at the outflow end of the specimen that contains a ring with a diameter smaller than the diameter of the permeameter and the presence of two outflow ports (one from the inner ring, one from the annular space between the inner ring and the permeameter wall) allows for separating the flow from the central region of the test specimen from the flow near the sidewall of the permeameter.

NOTE 3—Sidewall leakage has been reported to have significant influence on flow conditions for coarse-grained soils due to presence of larger voids at the boundary and higher void ratio in this region of the specimen. Three modifications that have been used to reduce this effect in rigid wall permeameters include: *i*) placing a piping barrier (for example, caulk rings along every approximately 25-mm length of sidewall), *ii*)

spreading a layer of bentonite and petroleum jelly mixture along the entire surface area of the sidewall, and *iii*) using a closed-cell neoprene liner attached to the inside wall of the permeameter.

5.4 Use of a flexible wall permeameter is included in these test methods in addition to the rigid wall permeameters. The flexible wall permeameter reduces potential adverse effects of sidewall leakage on measured hydraulic conductivity of the test specimens and allows for application of hydrostatic confining stress conditions on the specimen during the hydraulic conductivity test. Confining stress allows for representing field conditions (that is, simulating stress states in the subgrade that may affect values of k).

5.5 Darcy's law is assumed to apply to the test conditions, flow is assumed to be laminar (see **Note 4**), and the hydraulic conductivity is assumed to be considered independent of hydraulic gradient. The validity of these assumptions may be evaluated by measuring the hydraulic conductivity of a specimen at three different hydraulic gradients. The discharge velocity ($v = k \times i$) is plotted against the applied hydraulic gradient. If the resulting relationship is linear and the measured hydraulic conductivity values are similar (that is, within 25 %), then these assumptions are considered valid.

NOTE 4—Previous studies suggest that the limit between turbulent flow and laminar flow for soils occurs for Reynolds numbers between 1 and 10 (**1 and 2**)³. A formulation for Reynolds number (and division for laminar and turbulent flow conditions) for flow through packed beds has been reported (**3**). The formulation is presented for uniformly graded, spherical particles in **Eq 1**.

$$Re^* = \frac{Dv\rho_f}{\mu(1 - n)} \quad (1)$$

where:

- Re^* = Reynolds Number for packed bed flow,
- D = granule or particle diameter (m),
- v = superficial fluid velocity (that is, Darcy velocity) through bed (m/s),
- ρ_f = fluid density (kg/m³),
- μ = liquid viscosity (dynamic viscosity) (Pa s), and
- n = porosity of bed (expressed as a ratio).

Provisions are provided in (**3**) for establishing equivalent particle diameter for use in this equation for nonuniform particle size distributions and nonspherical particles.

NOTE 5—Using sufficiently low gradients has been demonstrated to be important for obtaining representative results. Hydraulic gradients less than 0.05 have been reported (**4**). Using a long test specimen (on the order of 1.5 m) has been reported as an effective method for achieving appropriately low hydraulic gradients for materials with $k > 0.01$ m/s.

NOTE 6—The quality of the result produced by this standard is dependent of the competence of the personnel performing it and the suitability of the equipment and facilities used. Agencies that meet the criteria of Practice **D3740** are generally considered capable of competent and objective testing, sampling, inspection, etc. Users of this standard are cautioned that compliance with Practice **D3740** does not in itself result in reliable values. Reliable results depend on many factors; Practice **D3740** provides a means of evaluating some of those factors.

6. Apparatus

6.1 *Constant-Head Hydraulic System*—The hydraulic system is used to apply, maintain, and measure heads and resulting

³ The boldface numbers in parentheses refer to a list of references at the end of this standard.

hydraulic gradients in a test. The hydraulic system mainly consists of reservoirs that hold water and associated piping, tubing, valves, and connections. See [Note 7](#). Pressure application setups also may be used to pressurize influent and effluent liquids. The system shall be capable of maintaining constant hydraulic head to within $\pm 5\%$ or better and shall include means to measure the hydraulic pressures to within the prescribed tolerances. In addition, the head loss must be held constant to $\pm 5\%$ or better and shall be measured with instruments providing the same accuracy and readability or better. Large-scale specimens have high flow capacities that may require specialized systems such as large reservoirs to provide adequate flow rates (see [8.1](#)). Two alternate means of achieving constant head conditions for k testing included in this standard are: a) Mariotte Bottle and b) use of a flow pump.

NOTE 7—Use of reservoirs with an overflow function (such as an outlet pipe or overflow along the perimeter of a level reservoir) that remains at a set elevation for the duration of testing have been demonstrated to work well and will reduce the potential for changes in head during a test.

6.1.1 The head shall be measured with a rigid measuring tape, graduated reservoir, engineer's scale, pressure gage, electronic pressure transducer, or other device that has the resolution and accuracy required for the determination of head to the tolerances provided above. Hydraulic heads shall be measured at points along the length of a test specimen using piezometers and from the inflow and outflow elevations in the case of a rigid-wall permeameter. Hydraulic heads shall be measured from the inflow and outflow elevations in the case of a flexible-wall permeameter. If measurements at inflow and outflow measurement locations are to be used, the procedure specified in [8.1](#) shall be conducted to verify that negligible head loss occurs through the tubing and system residing between the specimen and the inflow and outflow measurement locations. A general configuration for piezometers is provided in Test Method [D5716/D5716M](#). A porous filter is required for piezometers used in hydraulic conductivity tests to avoid soils clogging the piping. A screen placed on the inside of the nipple assembly has been reported for such a filter (U.S. Army Corps of Engineers (1980) Engineering and Design Laboratory Soils Testing, Engineer Manual no. 1110-2-1906).

6.1.2 System De-airing—The hydraulic system shall be designed to facilitate rapid and complete removal of free air bubbles from flow lines. This removal can be accomplished, for example, by using tubing and ball valves that are large enough to prevent entrapment of air bubbles, are large enough not to cause head losses as described in [8.1](#), and using fittings without pipe threads. Placement of valve(s) at points of high elevation within the hydraulic system can facilitate venting of air from the system. If de-aired water is used as permeant water, use a system with sufficient capacity to produce de-aired water for the test duration. Recirculated permeant water shall not be used in the test.

6.2 Flow-Measurement System—The flow-measurement system is used to determine the amount of flow through a specimen during a test. The measurement device shall allow for the measurement of the quantity of flow (inflow, outflow, or both inflow and outflow, if selected for verification of flow conditions) over an interval of time to within $\pm 5\%$. Flow-

measurement system may consist of a graduated accumulator, Mariotte bottle, electromagnetic flow meter, flow pump (if used to apply constant head), or other mass/volume-measuring device that has the resolution and accuracy required to determine flow to the tolerances provided above. In most cases, these devices are common to the hydraulic system.

6.2.1 De-airing and Dimensional Stability of the System—The flow-measurement system shall contain minimal dead space (volumetric space in the system that does not contribute to hydraulic fluid flow) and shall be equipped to allow for complete and rapid de-airing so that the system remains de-aired for the duration of testing. Dimensional stability of the system with respect to changes in pressure shall be accomplished by using a stiff flow-measurement system that includes glass pipe or rigid metallic or thermoplastic tubing.

6.3 Pressure Application System—The system (if used) for applying pressure on the coarse-grained soil specimen in the permeameter shall allow for applying and controlling the pressure to within $\pm 5\%$ of the set value. For a rigid wall permeameter (Method A), a vertical pressure application system is used. The vertical pressure application system may include a dead-weight load application setup; a hydraulic load application system; or other system that allows for application of the desired level of pressure to a specimen via the top of the specimen. The vertical effective stress on the test specimen (which is the difference between the applied vertical pressure and the pore water pressure—provided a system to control pore water pressure is used) shall be maintained to the desired value within $\pm 10\%$ of set value. For a flexible wall permeameter (Method B), a system for pressurizing the permeameter cell shall be capable of applying and controlling the cell pressure to within $\pm 5\%$ of the set value. The effective stress on the test specimen (which is the difference between the cell pressure and the pore water pressure) shall be maintained to the desired value within $\pm 10\%$ of set value. The device for pressurizing the cell may consist of a reservoir connected to the permeameter cell and partially filled with de-aired water, with the upper part of the reservoir connected to a compressed gas supply or other source of pressure. A minimum of 2 to 3 m of water-filled distance within the apparatus between the pressurized gas and the specimen is required (see [Note 8](#)). The gas pressure shall be controlled by a pressure regulator and measured by a pressure gage, electronic pressure transducer, or other device capable of measuring to the prescribed tolerances. A hydraulic system pressurized by dead weight acting on a piston or other pressure device capable of applying and controlling the permeameter cell pressure within the tolerances prescribed in this section may be used.

NOTE 8—De-aired water is commonly used for the cell fluid to reduce the potential for diffusive air transport through the membrane into the specimen. Other fluids that have low gas solubilities such as oils, are also acceptable, provided they do not react with components of the permeameter. Also, use of a sufficiently long tube connecting the pressurized cell liquid to the cell helps to delay the appearance of air in the cell fluid and to reduce the flow of dissolved air into the cell.

6.4 Permeameter—The permeameter shall consist of a cell for containing the test specimen and attached equipment that allow for connecting the cell to the hydraulic system, the flow-measurement system, and the pressure application system

(if applicable), as well as provisions to support a specimen and to permeate the specimen. The permeameter shall consist of a cylindrical cell (rigid for Method A, flexible for Method B), cover plate, base plate, and attachments to hold the components together without leakage during a test. The diameter of the permeameter shall be determined based on the D_{95} (that is, the particle size corresponding to 95 % passing per methods in [D6913/D6913M](#)) of the soil to be tested. A permeameter diameter at least 10 times D_{95} is required. The permeameter length shall be greater than 6 times D_{95} .

6.5 Rigid Wall Permeameter (Method A)—The permeameter shall consist of a rigid-wall cell into which the soil specimen to be tested is placed and in which the test specimen is permeated. The permeameter shall be constructed of a rigid material such as steel, aluminum, brass, or plastic that will not be damaged during placement/compression/compaction of the specimen in the cell. The cross-sectional area along the direction of flow shall not vary by more than $\pm 2\%$ and the height shall not vary by more than $\pm 2\%$. The permeameter shall be designed and operated such that permeant water flows upward or downward through the test specimen. If upward flow operation is used, protect the top of the specimen from upward movement of soil particles using a rigid porous element. Provisions may be included along the sidewall of the permeameter to directly attach the device to the constant-head hydraulic system or the flow-measurement system, or both. Schematic diagrams of a typical permeameter cell are presented in [Figs. 1 and 2](#) for single- and dual-ring rigid wall permeameter cells, respectively.

6.5.1 Top Plate—The top plate shall be constructed of a rigid material that does not react adversely with the test material or permeant water. The top plate may be sealed to the rigid-wall permeameter cell using an O-ring or similar prevent-

ing leakage or the plate may be perforated and not sealed to the permeameter cell based on the design of the test setup. A sealed top plate is used when the hydraulic or flow measurement system, or both, are connected to the top plate (or the permeameter cell) through leak-proof ports or valves, whereas a perforated top plate is used when water is ponded directly above a specimen. The perforated plate shall not impede flow. The procedure specified in [8.1](#) shall be conducted to verify that this criterion is satisfied. The top plate shall be designed to distribute permeant water to the cross section of the specimen such that flow through the test specimen is one-dimensional.

6.5.2 Base Plate—The bottom plate shall be constructed of a rigid material that does not react adversely with the test material or permeant water. The base plate shall be sealed to the rigid-wall permeameter cell using an O-ring or similar preventing leakage. The plate shall be designed to distribute permeant water to the cross section of the specimen such that flow through the test specimen is one-dimensional.

6.5.3 If a dual-ring permeameter is used, the plate corresponding to the outflow end of the specimen (that is, bottom plate for downward flow conditions) shall contain a ring extending outward from the plate in a perpendicular orientation to the plate. The ring shall be constructed of stiff material (for example, steel, aluminum) of as thin as practical configuration to prevent deformation upon placement (for example, compaction) of soil in the permeameter and not disrupt overall flow regime within the permeameter. It is recommended that the diameter of the ring on the plate at the outflow end of the specimen be no larger than 85 % of the diameter of the permeameter cell. The diameter of the ring shall not vary by more than 2 %. The ring shall be concentric to the permeameter. The annular space between the ring and the permeameter

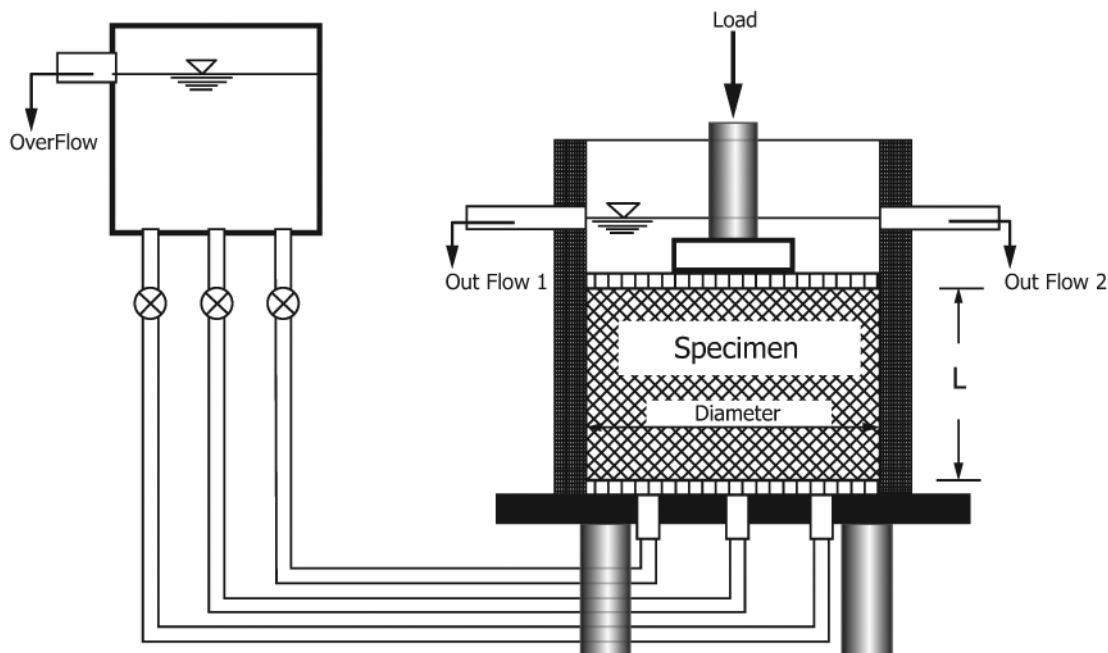


FIG. 1 Example Single-Ring Rigid Wall Permeameter (This schematic depicts a system with multiple inflow and outflow ports to the specimen, which is not necessary to maintain desired head conditions, but can accommodate large flow volumes.)

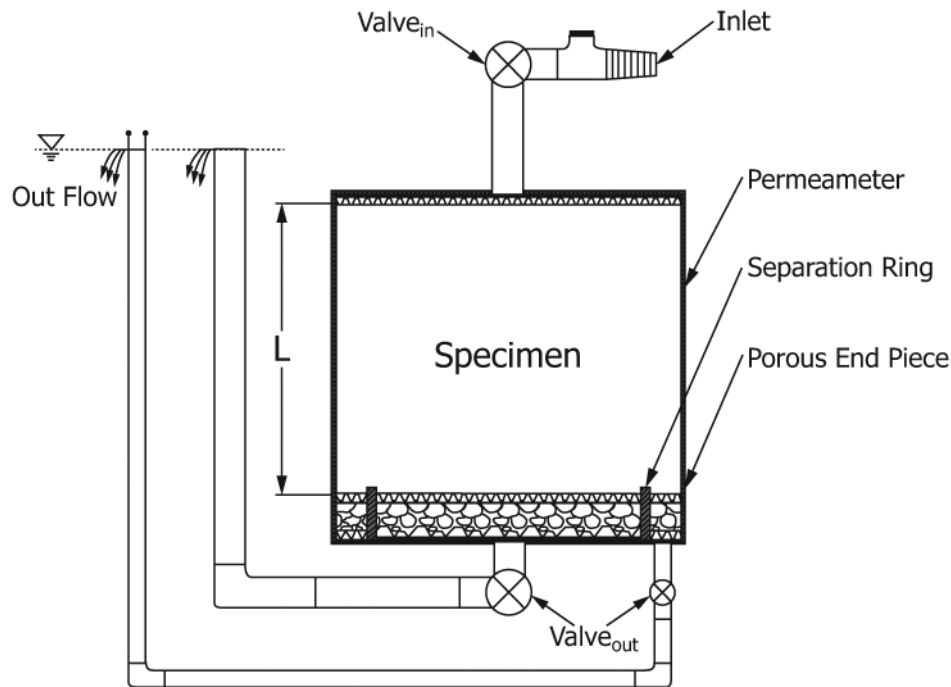


FIG. 2 Example Dual-Ring Rigid Wall Permeameter

wall shall be at least as wide as the nominal particle size. The height of the ring shall be greater than the nominal particle size.

6.5.4 Porous End Pieces—The specimen shall be overlain and underlain by porous end pieces. Porous end pieces shall be used to distribute permeant water uniformly over the surfaces of a test specimen (that is, areas perpendicular to the direction of flow) and shall not limit flow (that is, impedance to flow shall be less through the end pieces than through the system). Porous end pieces shall be constructed of a material that does not react with the specimen or the permeant water. Geosynthetic materials such as geotextiles, geonets, and geotextile-geonet composites (that is, geocomposites) may be used when flow through the system is less than the flow capacity of the geosynthetics. A flow distributor plate manufactured from plastic also has been demonstrated to be effective. Holes in the flow distributor plate shall be large enough as to not inhibit flow through the system. In addition, a metal mesh may be used. The top porous end piece shall have the same diameter ($\pm 5\%$ or better) as the specimen, and shall have sufficient strength (that is, thickness) to prevent breaking. If a dual-ring permeameter is used, two porous end pieces are required at the outflow end of the specimen due to the use of the plate containing the ring. The first porous end piece with a circular shape shall have a diameter within 5 % of the diameter of the inner ring. The second porous end piece with a ring shape shall have a width within 5 % of the width of the annular space between the inner collector ring and the permeameter cell. The end pieces shall be free from clogging. The flow capacity of the porous end pieces shall be at least five times greater than that of the specimen to be tested. The requirements outlined in 8.1 shall be applied to verify that this criterion is satisfied.

6.5.5 Filter Paper—If necessary to prevent intrusion of soil into the pores of the porous end pieces, one or more sheets of filter paper shall be placed between the top and bottom porous end pieces and the specimen. The paper shall have a negligibly small hydraulic impedance. The requirements outlined in 8.1 shall be applied to verify that the impedance is sufficiently small.

6.6 Flexible Wall Permeameter (Method B)—An apparatus shall be provided in which the specimen and porous end pieces, enclosed by a membrane sealed to the cap and base, are subjected to controlled fluid pressures. A schematic diagram of a typical flexible wall permeameter cell is presented in Fig. 3.

6.6.1 In order to facilitate gas removal, and thus saturation of the hydraulic system, four drainage lines leading to the specimen, two each to the base and top caps, are recommended. The drainage lines shall be controlled by no-volume-change valves, such as ball valves, and shall be designed to reduce the potential for presence of dead space (volumetric space in the system that does not contribute to hydraulic fluid flow) in the lines.

6.6.2 Top Cap and Base—An impermeable, rigid top cap and base shall be used to support the specimen and provide for transmission of permeant water to and from the specimen. The diameter or width of the top cap and base shall be equal to the diameter or width of the specimen to $\pm 5\%$ or better. The base shall prevent leakage, lateral motion, or tilting, and the top cap shall be designed to receive the piston or extensometer, if used, such that the piston-to-top cap contact area is concentric with the cap. The surfaces of the base and top cap that contacts the membrane to form a seal shall be smooth and free of scratches.

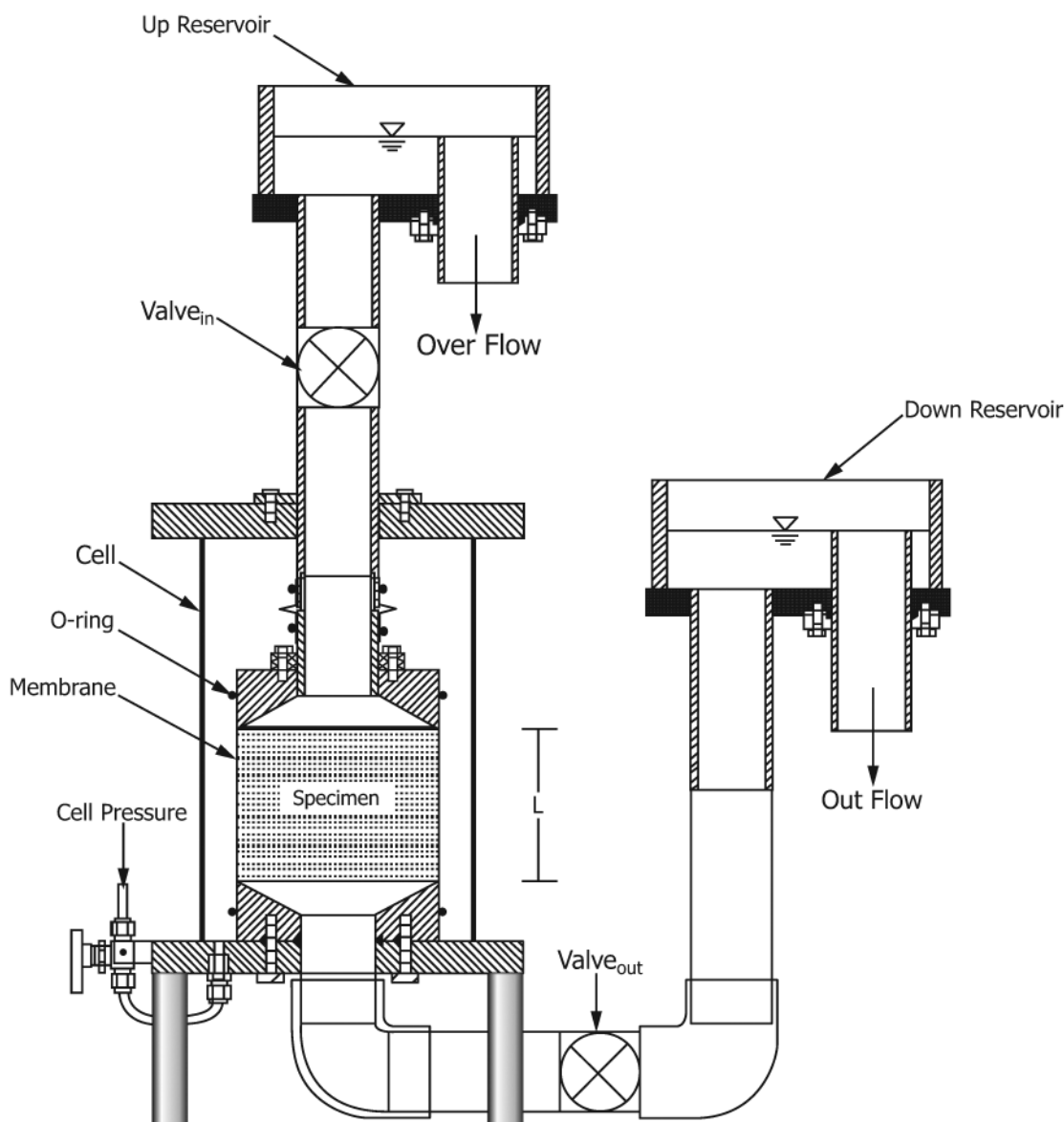


FIG. 3 Example Flexible Wall Permeameter

6.6.3 Flexible Membrane—The flexible membrane used to encase the specimen shall provide reliable protection against leakage. The membrane shall be carefully observed prior to use. If flaws or pinholes are evident, the membrane shall be discarded. To reduce the potential restraint to the specimen, the diameter or width of the non-stretched membrane shall be between 90 and 95 % of that of the specimen. The membrane shall be sealed to the specimen base and cap with rubber O-rings for which the unstressed, inside diameter or width is less than 95 % of the diameter or width of the base and cap, or by other method that will produce an adequate seal.

NOTE 9—Membranes may be tested for flaws by placing them around a form sealed at both ends with rubber O-rings, subjecting them to a small air pressure on the inside, and then dipping them into water and monitoring for presence of air bubbles emitting from the membrane.

6.6.4 Porous End Pieces—The specimen shall be overlain and underlain by porous end pieces. Porous end pieces shall be

used to distribute water uniformly over the surfaces of a test specimen (that is, areas perpendicular to the direction of flow) and shall not limit flow (that is, flow through system shall be less than flow through end pieces). Porous end pieces shall be constructed of a material that does not react with the specimen or the permeant water. Geosynthetic materials such as geotextiles, geonets, and geotextile-geonet composites (that is, geocomposites) may be used when flow through the system is less than the flow capacity of the geosynthetics. A flow distributor plate manufactured from plastic also has been demonstrated to be effective. Holes in the flow distributor plate shall be large enough as to not inhibit flow through the system. In addition, a metal mesh may be used. The top porous end piece shall have the same diameter (± 5 % or better) as the specimen, and shall have sufficient strength (that is, thickness) to prevent breaking. The end pieces shall be free from clogging. The hydraulic conductivity of the porous end pieces

shall be at least 5 times greater than that of the specimen to be tested. The requirements outlined in 8.1 shall be applied to verify that this criterion is satisfied.

6.6.5 Filter Paper—If necessary to prevent intrusion of soil into the pores of the porous end pieces, one or more sheets of filter paper shall be placed between the top and bottom porous end pieces and the specimen. The paper shall have a negligibly small hydraulic impedance. The requirements outlined in 8.1 shall be applied to verify that the impedance is sufficiently small.

6.7 Specimen Length and Deformation Measurement—The permeameter shall be equipped for determination of axial length of a specimen during placement or during a test. The length of a specimen shall be determined to the nearest 1 mm. The length of a specimen may be monitored by direct observation through the cell wall using a cathetometer, camera setup, or other instrument that has the resolution and accuracy required for the determination of length as prescribed above. The deformation (that is, change in length) of a specimen may also be monitored using a deformation gage connected to the top plate above a specimen or a deformation gage attached to a loading piston connected to the top plate above a specimen. The deformation can be determined using a dial gage, LVDT, or other device that has the resolution and accuracy required for the determination of deformation as prescribed above.

6.8 Mariotte Bottle—The bottle assembly, if used, shall be large enough to permit unimpeded water flow during permeation. The bottle assembly also shall be large enough to permit unimpeded air flow through the bubble tube during permeation. The bottle design shall provide acceptable reading accuracy for fluid levels (in relation to 6.1 and 6.2).

6.9 Balances—The balance shall be suitable for determining the mass of a specimen. The balance shall be selected based on the guidelines provided in Specification D4753. The mass of specimens shall be determined to the nearest 0.1 %.

6.10 Time Measurement Devices—Devices to measure the duration of each permeation trial, which shall be readable to 1 s, such as a clock with second hand or stopwatch (or equivalent), or both. For trials that occur over a duration less than 60 s, a clock with both accuracy and readability to 0.1 s shall be used.

6.11 Thermometer—An instrument to determine temperature to an accuracy of $\pm 1^\circ\text{C}$ with a readability of 1°C or less.

6.12 Vacuum Pump—A vacuum pump may be used to assist with de-airing of permeant water or saturation of specimens.

7. Reagents

7.1 Permeant Water:

7.1.1 The permeant water is the liquid used to permeate the test specimen and is also the liquid used in backpressure saturation of the specimen.

7.1.2 The type of permeant water shall be specified by the requestor. If no specification is made, potable tap water shall be used for the permeant water. The type of permeant water utilized shall be indicated in the test data sheet/form.

8. Verification of System Flow

8.1 Determination of System Flow Capacity—Excessive head losses in the tubes, valves, porous end pieces, and filter papers (collectively described in Section 6) may limit flow in the test system and lead to errors in measurements during permeation. System flow capacity shall be determined using the actual permeant water that will be used in the test. The permeameter shall be assembled without a specimen and then water shall be passed through the system. The hydraulic heads that will be used in testing a specimen shall be applied, and the rate of flow shall be measured to within $\pm 5\%$. The rate of flow shall be at least five times greater when no specimen is within the permeameter than the rate of flow that is measured when a specimen is placed inside the permeameter with the same hydraulic heads applied. Determination of system flow shall be conducted annually at a minimum. Retesting of the system flow capacity shall be conducted subsequent to any change to the system configuration (for example, hardware setup, observed clogging during previous testing). If five times flow of the system without a soil specimen (as compared to with a soil specimen) is not obtained, the permeameter can still be used for testing if *a*) flow through the system without a soil specimen is greater than through the system with soil and *b*) for head loss measurements taken at the reservoirs, a flow verification curve is developed for the system and used together with the test results to determine a net change in head on the soil specimen, as presented in Annex A1.

NOTE 10—Further context on interpretation of parasitic head losses during hydraulic conductivity tests is provided in (5).

9. Procedure

9.1 Rigid Wall Permeameter (Method A):

9.1.1 Soak the porous end pieces in a container of permeant water.

9.1.2 Place the porous end piece on the base plate. If a dual-ring permeameter is used, place the two porous end pieces on the base plate. If a multi-component permeameter is used (that is, mold separate from end plates), place the permeameter mold over the base plate and secure the assembly. Place the pre-determined mass of soil specimen in the permeameter, in stages with compaction per layer, so as to obtain the desired dry unit weight or void ratio (consideration for application of pressure as described in 9.1.3 may influence target dry unit weight at this stage of the specimen preparation). See Note 11 for methods of soil placement in the permeameter. Place the top porous end piece and top plate over the specimen. Secure the components in place and fully assemble the permeameter cell.

NOTE 11—Use of a shake table has been demonstrated to be effective for densifying coarse-grained soil specimens in rigid or flexible wall tests. Tamping or application of static load are other methods to reconstitute granular soil to target dry unit weights in rigid or flexible wall tests. To obtain a uniform energy associated during soil placement, establishing a set method of preparation (that is, compaction in equal lifts) that is repeatable is recommended. For example, the mass of soil and number of tamps (or duration of shaking) per lift can be controlled to obtain uniform preparation with depth throughout the specimen. Specifics in this regard are left to the user to obtain representative testing conditions.

9.1.3 Attach the pressure application system to the permeameter, if applicable. If being used for the test, apply

pressure to the specimen at the onset of a test and maintain this pressure during the test. Application of pressure may affect the void ratio and length of the specimen.

9.1.4 Determine the length of the test specimen at three locations around the circumference of the specimen prior to initiation of flow. Determine the diameter of the specimen prior to initiation of flow. These dimensions shall be measured to the nearest 1 mm.

9.1.5 Attach the hydraulic system to the inflow and outflow lines of the permeameter.

9.1.6 Connect the permeameter to the flow measurement system.

NOTE 12—In some test setups, the flow measurement system will not be physically attached to the permeameter, but rather be hydraulically positioned to measure flow (for example, collection vessel placed below outflow tube).

9.2 Flexible Wall Permeameter (Method B):

9.2.1 Soak the porous end pieces in a container of permeant water.

9.2.2 Place the porous end piece on the base plate. Place the flexible membrane over the base plate and attach with O-rings. Place the pre-determined mass of soil specimen in the membrane so as to achieve the desired dry unit weight or void ratio (see [Note 11](#) and [Note 13](#); consideration for application of pressure as described in [9.2.3](#) may influence target dry unit weight at this stage of the specimen preparation). A rigid cylindrical shell (that is, rigid specimen mold or membrane stretcher) may be used to control volume of the flexible membrane during preparation of the specimen. Place the top porous end piece and top plate over the specimen. Attach membrane to top plate with O-rings. Secure the components in place and fully assemble the permeameter cell.

NOTE 13—Care is advised during specimen preparation to not puncture the flexible membrane for flexible wall permeameters.

9.2.3 Attach the pressure application system to the permeameter, if applicable. If being used for the test, apply pressure to the specimen at the onset of a test and maintain this pressure during the test. Application of pressure may affect the void ratio of the specimen.

9.2.4 Determine the length of the test specimen at three locations around the circumference of the specimen prior to initiation of flow. Determine the diameter of the specimen prior to initiation of flow. These dimensions shall be measured to the nearest 1 mm.

9.2.5 Attach the hydraulic system to the inflow and outflow lines of the permeameter.

9.2.6 Connect the permeameter to the flow measurement system. See [Note 12](#).

9.3 Saturation—Saturation step applies to both Methods A and B. Specimen shall be saturated prior to the permeation step (hydraulic conductivity measurement). Saturation shall be attained by initiating flow through the system and bleeding air from the system. During the saturation step, the direction of flow may be opposite of the flow direction during the permeation step. It is easier to drive air out of a specimen if the flow is from bottom to top of the specimen. During the saturation step, a specimen may be inundated with permeant water under

partial vacuum applied to the top of the specimen. Water under atmospheric pressure shall be applied to the specimen via the appropriate ports depending on the saturation technique (that is, if vacuum is applied at the top of the specimen, water under atmospheric pressure shall be applied at the bottom; if positive pressure is applied at the bottom, water under atmospheric pressure shall be applied at the top). Regardless of saturation technique used during the saturation step, the magnitude of the hydraulic gradient across the specimen shall be less than the gradient that will be used during the permeation step. See [Note 14](#) related to importance of obtaining saturated conditions for testing and a method for ensuring that saturation is obtained during a test.

NOTE 14—A method for assessing the degree of saturation within the tested specimen (after tests are conducted) is presented in [\(6\)](#). While most commercial permeameters are constructed to be watertight, the equipment is not always airtight, which may prevent reaching saturation before testing and maintaining saturation during testing. Results for rigid wall hydraulic conductivity testing presented in [\(7\)](#) indicate that significant increases in measured k (400 to 500 %) are possible for soil undergoing modest increases in degree of saturation (80 to 100 %). Use of CO₂ to promote saturation has been reported (for example, [D5101](#)).

9.4 Hydration Period—If specimen contains greater than 5 % fines content (that is, passing 75 μm sieve) by mass, a 24-hour hydration period shall be provided before permeation. This allows time for swelling and reactions to occur in the fine-grained soils. If multiple tests are being conducted with the same soil that contains greater than 5 % fines content, confirmation that less than 25 % difference in measured k value before and after 24-hour hydration period can mitigate the need for providing the hydration period for that set of tests.

9.5 Permeation—Permeation step applies to both Methods A and B. Head determination measurements during permeation may be made by reading difference of inflow and outflow water levels in a test setup. The last digit may be estimated when directly reading liquid levels from graduated reservoirs. The use of estimated digits for such measurements are provided in [Practice D6026](#). When the last digit is estimated in a measurement where the reading is a function of the elevation/location of the eye, a mirror or another device is advised to reduce the reading error caused by parallax.

9.5.1 Adjust/apply the overburden pressure (or test cell pressure for flexible wall permeameters), if a value other than atmospheric pressure is being used for the test, to the value to be used during the test to the specimen in accordance with the testing requirements.

9.5.2 Adjust/set up the constant-head hydraulic system to apply the hydraulic gradient to the value to be used during the test.

9.5.3 Open the influent and effluent valves and allow flow of water through the specimen. Measure and record the heads applied to the specimen at the inflow and outflow ends to the resolution stated in [6.1](#) at the start and end of each permeation trial (as a minimum). Determine the head loss across the specimen using measured heads. The head loss across the permeameter shall be kept constant to within $\pm 5\%$ or less during permeation. The volumes of flow shall be measured and recorded to the resolution stated in [6.2](#) for the permeation duration.

9.5.4 Schematic diagrams of permeation setups and measurement of parameters needed to maintain hydraulic gradient are presented in Figs. 4-6.

9.5.5 Continue permeation with consideration to criteria regarding steadiness of hydraulic conductivity and degree of saturation. The hydraulic conductivity shall be considered steady if four or more consecutive hydraulic conductivity determinations fall within $\pm 25\%$ or less of the mean value and a plot or tabulation of the hydraulic conductivity versus time shows no significant upward or downward trend. The degree of saturation for the specimen shall remain at 100% for the duration of each trial. Further information related to degree of saturation is provided in 9.7 and Note 14.

NOTE 15—The pore volumes of flow may also be measured over the duration of a test. For chemically reactive soils, tests with long durations corresponding to multiple pore volumes of flow through the specimen allow interaction between pore fluid and soil matrix.

9.5.6 Close inflow and outflow lines to end permeation.

9.5.7 The time at the start and end of each permeation trial shall be measured and recorded. The permeation duration shall be measured to the nearest second except for tests that last less than 60 s where permeation duration shall be measured to the nearest 0.1 s.

9.5.8 The test temperature shall be measured and recorded to the nearest 1°C for each permeation trial. The test temperature can be determined by measuring either the temperature of the test specimen or alternatively the temperature of outflow fluid if measured during or immediately following permeation trial.

9.5.9 If permeation trial lasts for extended duration, temperature shall be measured and recorded multiple times to provide representative values for testing conditions. Permeation trials shall be completed with less than 3°C variation in temperature over the duration of the trials.

9.5.10 Measure and record the length of the test specimen to the nearest 1 mm at 3 locations around the circumference of the specimen upon completion of permeation.

9.5.11 Repeat permeation at the next level of hydraulic gradient (see 5.5). At least three different hydraulic gradients shall be tested.

9.6 Migration of fines, including clogging of pores of outflow end of test specimen or clogging of apparatus (for example, filter paper or porous end pieces) can occur in hydraulic conductivity testing of coarse-grained soils, potentially impacting test conditions and resulting determinations of k .

9.6.1 Evidence of migration of fines can be documented as: possibly a trend of reducing k with time; visual observation of particulate accumulation on filter paper/filter assembly after the test is completed; possible trapping of migrating air bubbles; and observation of fines migration if a clear-walled permeameter is used.

9.6.2 If migration of fines is confirmed, test results for affected trials shall be rejected.

NOTE 16—Recommendations for avoiding problems with clogging and migration of fines include: *a*) using upward flow through the permeameter; *b*) using sufficiently low hydraulic gradient (see Note 5), as this directly influences migration of fines; and *c*) measuring head loss using piezometers that are placed within the length of the test specimen, avoiding measurement of head loss associated with the end pieces and filter assemblies.

9.7 The saturation step is required to conduct the test. Due to the various end uses of these test results, interpretation of the test results may require additional information about the conduct of the saturation step, including a tested degree of saturation. See Note 14.

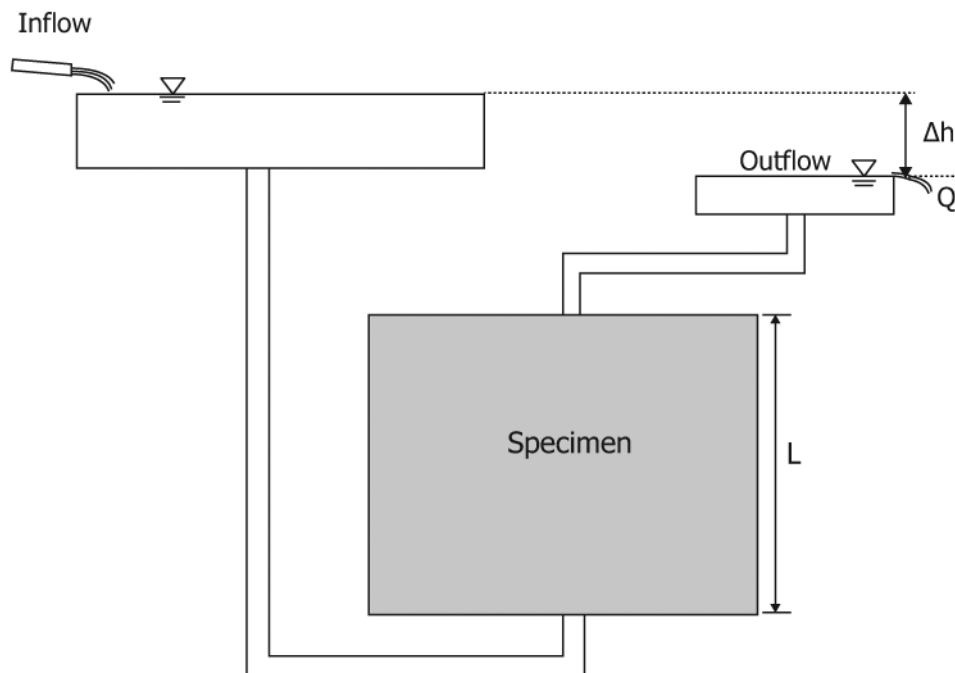


FIG. 4 Schematic of Constant Head Hydraulic Systems and Measurement of Hydraulic Gradients: Reservoir

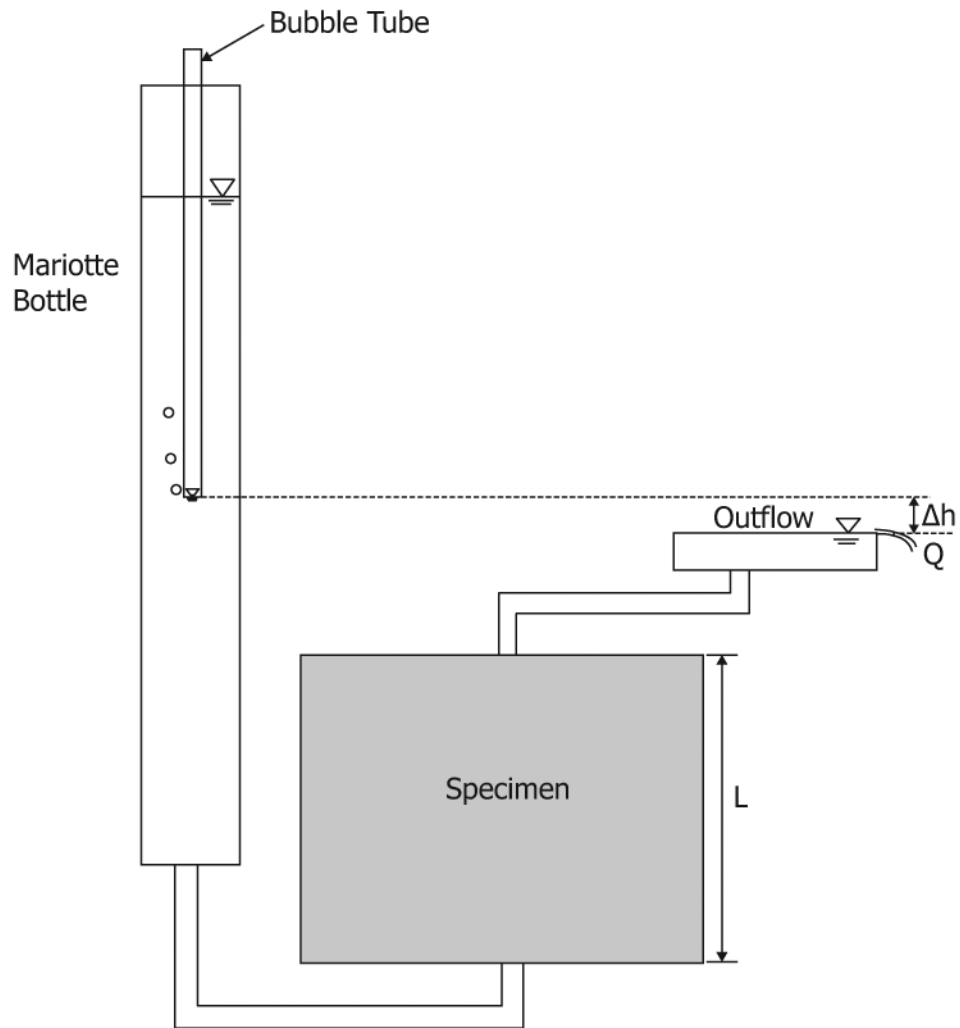


FIG. 5 Schematic of Constant Head Hydraulic Systems and Measurement of Hydraulic Gradients: Mariotte Bottle (example configuration shown; alternative configuration such as specimen located within Mariotte Bottle, typically at bottom of bottle, is also possible)

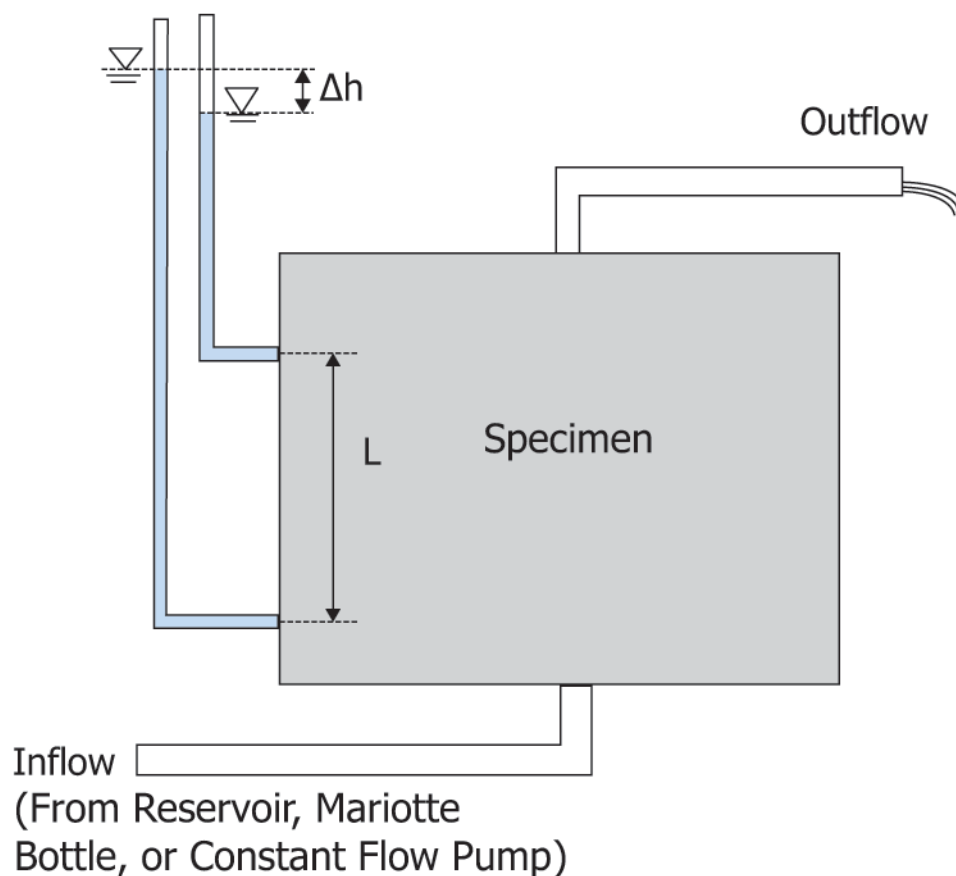


FIG. 6 Schematic of Constant Head Hydraulic Systems and Measurement of Hydraulic Gradients: Piezometers

10. Calculation

10.1 For each permeation trial, calculate the hydraulic conductivity, k , using Eq 2:

$$k = \frac{\Delta V \cdot L}{A \cdot \Delta h \cdot \Delta t} \quad (2)$$

where:

- k = hydraulic conductivity (m/s),
- ΔV = volume of permeant water collected for given time interval, Δt ($t_2 - t_1$), (m^3); if a dual-ring permeameter is used, this is the volume of water that flowed through the central zone of the specimen contained within the ring (m^3),
- L = average length of specimen for period of hydration along which the hydraulic head loss Δh (m) is measured, accounting for possible temporal variability (use length measurements for both pre- and post-permeation conditions),
- A = cross-sectional area of specimen (m^2); if a dual-ring permeameter is used, this is the area confined by the inner ring (m^2),
- Δh = constant head loss across the specimen length (m), and
- Δt = interval of time ($t_2 - t_1$) associated with the volume of permeant water collected (s).

The value of k shall be calculated to two significant digits.

10.2 Correct the hydraulic conductivity to that for 20°C by multiplying k (see 10.1) by R_T , the ratio of the viscosity of water at test temperature to the viscosity of water at 20°C (see Note 17).

NOTE 17—RT may be calculated using Eq 3.

$$R_T = 2.2902(0.9842^T)T^{0.1702} \quad (3)$$

Where T = test temperature (°C) as determined in 9.5.8 and 9.5.9.

10.3 If pore volumes are determined (Note 15), Eq 4 shall be used for the calculation.

$$PVF = \frac{\Sigma V_{outflow}}{V_v} \quad (4)$$

where:

- PVF = pore volumes of flow,
- $\Sigma V_{outflow}$ = cumulative amount of effluent collected during test (m^3), and
- V_v = volume of voids in the specimen (m^3) [can be determined using weight-volume relationships for the test specimen before or after permeation, see (7) for methodology on determining V_v]

11. Report: Test Data Sheet(s)/Form(s)

11.1 Record as a minimum the following general information (data): Sample/specimen identifying information, such as project number, soil source/characteristics, and other relevant information. Record information on permeameter (configuration, pressure control method) and information on permeant water.

11.2 Person(s) conducting test.

11.3 Record as a minimum the following test specimen data:

11.3.1 Specimen size characteristics: length and diameter to two significant digits, and gradation of soil,

11.3.2 Initial void ratio (or dry unit weight).

11.4 Record detailed method of placement/densification of soil in permeameter.

11.5 Record as a minimum the following test conditions and permeation data:

11.5.1 Overburden stress (or confining stress for flexible wall permeameter).

11.5.2 Completion of the saturation step.

11.5.3 Hydraulic head (to nearest mm) and gradient (to two significant digits).

11.6 For each permeation trial, record the following:

11.6.1 The start and end time and elapsed duration (nearest second, unless permeation trial less than 60 s, then to nearest 0.1 s).

11.6.2 The test temperature (nearest 1°C).

11.7 Generate a plot of the discharge velocity ($v = ki$) versus the applied hydraulic gradient ($i = \Delta h/L$). See example plot in [Appendix X1](#).

11.8 Report the calculated hydraulic conductivity, corrected to 20°C, in scientific notation to two significant digits in units of m/s. Note the equation used to correct the calculated hydraulic conductivity.

11.9 If corrected hydraulic conductivity values from individual permeation trials are within 25 % of each other and a noticeable trend (upwards or downwards) in hydraulic conductivity values is not present (based on consecutive trials), report final corrected hydraulic conductivity as the average of the trials that match these criteria. Report k in scientific notation in units of m/s (for example, 8.3×10^{-4} m/s). Also, report in units of cm/s (for example, 8.3×10^{-2} cm/s) or other units requested for a project.

12. Precision and Bias

12.1 *Precision*—Test data on precision is not presented due to the nature of the (soil) materials tested by this test method. It is either not feasible or too costly at this time to have ten or more laboratories participate in a round-robin testing program.

12.1.1 Subcommittee D18.04 is seeking any data from the users of these test methods that might be used to make a limited statement on precision.

12.2 *Bias*—There is no accepted reference value for these test methods, therefore bias cannot be determined.

13. Keywords

13.1 constant head; gravel; hydraulic conductivity; permeameter; sand

ANNEX

(Mandatory Information)

A1. SYSTEM FLOW VERIFICATION CURVE

A1.1 A flow verification curve can be used to determine the net change in head over the soil specimen length. An example plot is presented in [Fig. A1.1](#).

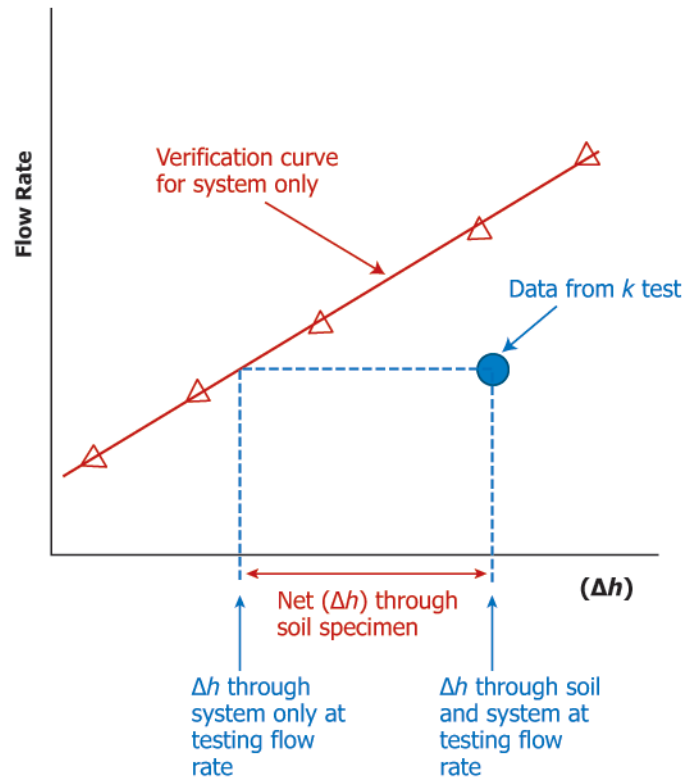


FIG. A1.1 Use of Flow Verification Curve to Determine Net Δh on Specimen

APPENDIX

(Nonmandatory Information)

X1. EXAMPLE FORMAT FOR DATA RECORDING AND PROCESSING

X1.1 See Fig. X1.1.

X1.1.1 For each permeation trial, record the following:

X1.1.1.1 The date of the trial; specimen diameter; starting specimen length (average of three measurements); elapsed duration of the flow measurement period; the test temperature; measured outflow quantity (mass or volume); values of measured head for the inflow and outflow ends; and ending specimen length. Calculate the average specimen length (average of starting and ending); hydraulic gradient; hydraulic

conductivity; corrected hydraulic conductivity (for 20°C); and discharge velocity for each trial.

X1.1.2 Generate a plot of the discharge velocity ($v = ki$) versus the applied hydraulic gradient ($i = \Delta h/L$) to confirm that testing is being conducted at a velocity consistent with laminar range of flow by observing a linear trend on this plot. A sample plot of v versus i and sample data recording and calculations table is in Fig. X1.2 and Fig. X1.3.

Example Data Sheet*

*The type of data collected will vary depending on type of permeameter and type of constant-head hydraulic system used for testing. This example datasheet is set up for simple conditions of rigid wall permeameter and reservoirs for constant-head control.

Project number	
Soil source/characteristics	
Other relevant sample information	
Permeameter configuration	
Constant head system type	
Permeant water	
Person(s) conducting test	
Initial specimen length (mm, to 2 significant digits)	
Specimen diameter (mm, to 2 significant digits)	
Description of gradation of soil	
Specific gravity (indicate whether tested or estimated)	
Initial void ratio (or alternatively, dry unit weight)	
Method of placement/densification of soil in permeameter	
Initial moisture content	
Overburden stress (or confining stress for flexible wall permeameter), if used in test	
Completion of saturation step	
Hydraulic Head (to nearest mm)	
Hydraulic Gradient (to 2 significant digits)	
Final corrected k value (m/s)	
Final corrected k value (cm/s) or other units	

FIG. X1.1 Example Data Sheet

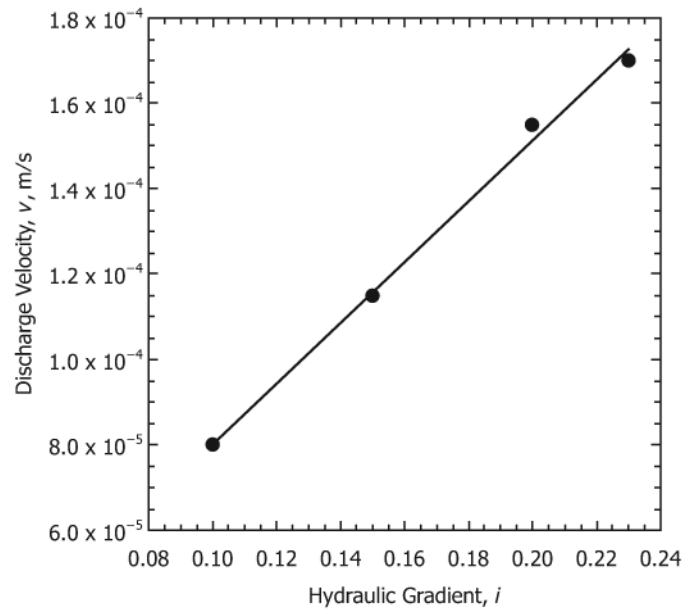


FIG. X1.2 Plot of Discharge Velocity (v) with Hydraulic Gradient (i) to Confirm Laminar Flow Conditions

Trial	Date	Specimen diameter (mm)	Starting specimen length (mm)	Δ time (s)	Temp. (°C)	Outflow mass (g) or volume (mL)	Head measurements (mm)		Δ head (mm)	Ending specimen length (mm)	Average specimen length (mm)	Hydraulic gradient	k (m/s)	k_{20} (m/s)	v (m/s)
							Inflow	Outflow							
1															
2															
3															
4															
5															
6															
7															
8															
9															
10															

FIG. X1.3 Example of Data Recording and Calculations Sheet

REFERENCES

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- (5) Duhaime, F., Chapuis, R. P., and Weber, S. (2015). "Parasitic head losses during laboratory permeability tests," *Geotechnical Testing Journal*, 38(2): 255-260.
- (6) Chapuis, R. P., Baass, K., and Davenne, L. (1989). "Granular soils in rigid-wall permeameters: method for determining the degree of saturation," *Canadian Geotechnical Journal*, 26(1): 71-79.
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SUMMARY OF CHANGES

Committee D18 has identified the location of selected changes to this standard since the last issue (D2434–19) that may impact the use of this standard. (March 15, 2022)

(1) This technical revision incorporates additional permeameter units to be included in the standard (previously just rigid wall permeameter; revised to include rigid wall, dual-ring rigid-wall, and flexible-wall permeameters). Additionally, different hydraulic systems for application of constant head are included (previously just reservoirs; revised to include reservoirs, Mariotte Bottle, and a constant flow pump). Guid-

ance has been added for various aspects of hydraulic conductivity testing of coarse-grained soils including laminar versus turbulent flow conditions, migration of fines, presence of sidewall leakage, obtaining saturated conditions for testing, changes in specimen length during testing, and system flow capacity verification.

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Designation: D2216 – 19

Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass¹

This standard is issued under the fixed designation D2216; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ϵ) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the U.S. Department of Defense.

1. Scope*

1.1 These test methods cover the laboratory determination of the water (moisture) content by mass of soil, rock, and similar materials where the reduction in mass by drying is due to loss of water except as noted in 1.4, 1.5, and 1.8. For simplicity, the word “material” shall refer to soil, rock or aggregate whichever is most applicable.

1.2 Some disciplines, such as soil science, need to determine water content on the basis of volume. Such determinations are beyond the scope of this test method.

1.3 The water content of a material is the ratio of the mass of water contained in the pore spaces of soil or rock material, to the solid mass of particles, expressed as a percentage.

1.4 The term “solid material” as used in geotechnical engineering is typically assumed to mean naturally occurring mineral particles of soil and rock that are not readily soluble in water. Therefore, the water content of materials containing extraneous matter (such as cement etc.) may require special treatment or a qualified definition of water content. In addition, some organic materials may be decomposed by oven drying at the standard drying temperature for this method ($110 \pm 5^\circ\text{C}$). Materials containing gypsum (calcium sulfate dihydrate) or other compounds having significant amounts of hydrated water, may present a special problem as this material slowly dehydrates at the standard drying temperature ($110 \pm 5^\circ\text{C}$) and at very low relative humidity, forming a compound (such as calcium sulfate hemihydrate) that is not normally present in natural materials except in some desert soils. In order to reduce the degree of dehydration of gypsum in those materials containing gypsum or to reduce decomposition in highly/fibrous organic soils, it may be desirable to dry the materials at 60°C or in a desiccator at room temperature. When a drying temperature is used which is different from the standard drying

temperature as defined by this test method, the resulting water content may be different from the standard water content determined at the standard drying temperature of $110 \pm 5^\circ\text{C}$.

NOTE 1—Test Method D2974 provides an alternate procedure for determining water content of peat materials.

1.5 Materials containing water with substantial amounts of soluble solids (such as salt in the case of marine sediments) when tested by this method will give a mass of solids that includes the previously soluble dissolved solids. These materials require special treatment to remove or account for the presence of precipitated solids in the dry mass of the specimen, or a qualified definition of water content must be used. For example, see Test Method D4542 regarding information on marine sediments.

1.6 This test standard requires several hours for proper drying of the water content specimen. Test Methods D4643, D4944 and D4959 provide less time-consuming processes for determining water content. See Gilbert² for details on the background of Test Method D4643.

1.7 Two test methods are provided in this standard. The methods differ in the significant digits reported and the size of the specimen (mass) required. The method to be used may be specified by the requesting authority; otherwise Method A shall be performed.

1.7.1 *Method A*—The water content by mass is recorded to the nearest 1 %. For cases of dispute, Method A is the referee method.

1.7.2 *Method B*—The water content by mass is recorded to the nearest 0.1 %.

1.8 This standard requires the drying of material in an oven. If the material being dried is contaminated with certain chemicals that may react violently or emit hazardous gases when heated, health and safety hazards may exist. Therefore, this standard should not be used in determining the water content of contaminated soils unless adequate health and safety precautions are exercised.

¹ This test method is under the jurisdiction of ASTM Committee D18 on Soil and Rock and is the direct responsibility of Subcommittee D18.03 on Texture, Plasticity and Density Characteristics of Soils.

Current edition approved March 1, 2019. Published March 2019. Originally approved in 1963. Last previous edition approved in 2010 as D2216–10. DOI: 10.1520/D2216-19.

² Gilbert, P.A., “Computer Controlled Microwave Oven System for Rapid Water Content Determination,” Tech. Report GL-88–21, Department of the Army, Waterways Experiment Station, Corps of Engineers, Vicksburg, MS, November 1988.

*A Summary of Changes section appears at the end of this standard

1.9 *Units*—The values stated in SI units shall be regarded as standard except the Alternative Sieve Sizes listed in **Table 1** are used. No other units of measurement are included in this test method.

1.10 All observed and calculated values shall conform to the guidelines for significant digits and rounding established in Practice **D6026**, unless superseded by this test method.

1.10.1 This is especially important if the water content will be used to calculate other relationships such as moist mass to dry mass or vice versa, wet unit weight to dry unit weight or vice versa, and total density to dry density or vice versa. For example, if four significant digits are required in any of the above calculations, then the water content must be recorded to the nearest 0.1 %. This occurs since 1 plus the water content (not in percent) will have four significant digits regardless of what the value of the water content is; that is, 1 plus 0.1/100 = 1.001, a value with four significant digits. While, if three significant digits are acceptable, then the water content can be recorded to the nearest 1 %.

1.10.2 If water content data is to be used to calculate other relationships, such as moist or dry mass, wet or dry unit weight or total or dry density, then the specimen mass up to 200 g must be determined using a balance accurate to 0.01 g.

1.11 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.12 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:³

D653 Terminology Relating to Soil, Rock, and Contained Fluids

D2974 Test Methods for Moisture, Ash, and Organic Matter of Peat and Other Organic Soils

D3740 Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

D4220 Practices for Preserving and Transporting Soil Samples

D4318 Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils

D4542 Test Methods for Pore Water Extraction and Determination of the Soluble Salt Content of Soils by Refractometer

D4643 Test Method for Determination of Water Content of Soil and Rock by Microwave Oven Heating

D4753 Guide for Evaluating, Selecting, and Specifying Balances and Standard Masses for Use in Soil, Rock, and Construction Materials Testing

D4944 Test Method for Field Determination of Water (Moisture) Content of Soil by the Calcium Carbide Gas Pressure Tester

D4959 Test Method for Determination of Water Content of Soil By Direct Heating

D5079 Practices for Preserving and Transporting Rock Core Samples (Withdrawn 2017)⁴

D6026 Practice for Using Significant Digits in Geotechnical Data

D7263 Test Methods for Laboratory Determination of Density (Unit Weight) of Soil Specimens

3. Terminology

3.1 Definitions:

3.1.1 For definitions of common technical terms used in this standard, refer to Terminology **D653**.

3.2 Definitions of Terms Specific to This Standard:

3.2.1 *constant dry mass (of a solid material)*—the state that a water content specimen has attained when further heating results in less than 1 % or 0.1 % additional loss in mass for Method A or B respectively. The time necessary to obtain constant dry mass will vary depending on numerous factors such as the type of material being tested, the size of the specimen and type of oven being used (forced draft or gravity

³ For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

⁴ The last approved version of this historical standard is referenced on www.astm.org.

TABLE 1 Minimum Requirements for Mass of Test Specimens, and Balance Readability^A

Maximum Particle Size (100 % Passing)		Method A Water Content Recorded to ± 1 %		Method B Water Content Recorded to ± 0.1 %	
Sieve Size	Alternative Sieve Size	Minimum Specimen Mass	Balance Readability (g)	Minimum Specimen Mass (g)	Balance Readability (g)
75.0 mm	3 in.	5 kg	10	50 kg	10
37.5 mm	1-1/2 in.	1 kg	10	10 kg	10
19.0 mm	3/4 in.	250 g	0.1	2.5 kg	0.1
9.5 mm	3/8 in.	50 g	0.1	500 g	0.1
4.75 mm	No. 4			100 g	0.01
2.00 mm	No. 10			20 g	0.01

^ASee 1.10.2.

type). The influence of these factors generally can be established by good judgment, and experience with the materials being tested and the apparatus being used.

4. Summary of Test Method

4.1 The mass of a moist test specimen is determined. The specimen is then dried in an oven at a temperature of $110 \pm 5^\circ\text{C}$ until a constant mass is achieved. The loss of mass, due to drying, is considered to be water. The water content is calculated using the mass of water to the mass of the dry specimen expressed in percent.

5. Significance and Use

5.1 For many materials, the water content is one of the most significant properties used in establishing a correlation between soil behavior and its index properties.

5.2 The water content of a material is used in expressing the phase relationships of air, water, and solids in a given volume of material.

5.3 In fine-grained soils, the consistency of a given soil type depends on its water content. The water content of a soil, along with its liquid and plastic limits as determined by Test Method **D4318**, is used to express its relative consistency or liquidity index.

NOTE 2—The quality of the result produced by this standard is dependent on the competence of the personnel performing it, and the suitability of the equipment and facilities used. Agencies that meet the criteria of Practice **D3740** are generally considered capable of competent and objective testing/sampling/inspection/etc. Users of this standard are cautioned that compliance with Practice **D3740** does not in itself assure reliable results. Reliable results depend on many factors; Practice **D3740** provides a means of evaluating some of those factors.

6. Apparatus

6.1 *Drying Oven*—Vented, thermostatically-controlled, preferably of the forced-draft type and capable of maintaining a uniform temperature of $110 \pm 5^\circ\text{C}$ throughout the drying chamber. The oven shall have a means of indicating the oven drying chamber temperature when in operation. This can be accomplished by such means as an electronic display, an analog thermometer, remote temperature recording device or any other means to determine the current drying chamber temperature while in operation.

6.1.1 Ovens in excess of 30 cubic feet shall have the temperature verified for adherence to the temperature requirements in the four quadrants and the center of the oven. Smaller ovens shall have the temperature verified in a single center location. Oven temperature verification shall follow the schedule as outlined in **D3740** or following such things as repairs or questionable operation.

6.2 *Balances*—All balances must meet the requirements of Specification **D4753** and this section. A Class GP1 balance of 0.01 g readability is required for specimens having a mass of up to 200 g (excluding mass of specimen container) and a Class GP2 balance of 0.1 g readability is required for specimens having a mass over 200 g. If desired, a Class GP1 balance may be used for specimens exceeding 200 g providing the specimen

size is within the capacity of the balance. However, the balance used may be controlled by the number of significant digits needed (see **1.10**).

6.3 *Specimen Containers*—Suitable containers made of material resistant to corrosion and change in mass upon repeated heating, cooling, exposure to materials of varying pH, and cleaning. Unless a desiccator is used, containers with close-fitting lids shall be used for testing specimens having a mass of 200 g or less; while for specimens having a mass greater than 200 g, containers without lids may be used (see **Note 3**). One uniquely numbered (identified) container or number-matched container and lid combination as required is needed for each water content determination.

NOTE 3—The purpose of close-fitting lids is to prevent loss of moisture from specimens before initial mass determination, and to prevent absorption of moisture from the atmosphere following drying and before final mass determination.

6.4 *Desiccator (Optional)*—A desiccator cabinet or large desiccator jar of suitable size containing silica gel or anhydrous calcium sulfate. It is preferable to use a desiccant that changes color when it needs to be recharged.

NOTE 4—Anhydrous calcium sulfate is sold under the trade name Drierite.

6.5 *Container Handling Apparatus*, heat resistant gloves, tongs, or suitable holder for moving and handling hot containers after drying.

6.6 *Miscellaneous*, knives, spatulas, scoops, quartering cloth, wire saws, etc., as required.

7. Samples

7.1 Soil samples shall be preserved and transported in accordance with Practice **D4220** Section 8 Groups B, C, or D soils. Rock samples shall be preserved and transported in accordance with Practice **D5079** section 7.5.2.1, Special Care Rock. Keep the samples that are stored prior to testing in non-corrodible airtight containers at a temperature between 3 and 30°C and in an area that is without direct sunlight. Disturbed samples in jars or other containers shall be stored in such a way as to minimize moisture condensation on the insides of the containers.

7.2 The water content determination should be done as soon as practical after sampling, especially if potentially corrodible containers (such as thin-walled steel tubes, paint cans, etc.) or plastic sample bags are used.

8. Test Specimen

8.1 For water contents being determined in conjunction with another ASTM method, the specimen mass requirement stated in that method shall be used if one is provided. If no minimum specimen mass is provided in that method then the values outlined in **Table 1** and **1.10.2** shall apply. See Howard⁵ for background data for the values listed.

⁵ Howard, A. K., "Minimum Test Specimen Mass for Moisture Content Determination," *Geotechnical Testing Journal*, ASTM., Vol. 12, No. 1, March 1989, pp. 39-44.

8.2 The minimum specimen mass of moist material selected to be representative of the total sample is based on visual maximum particle size in the sample and the Method (Method A or B) used to record the data. Minimum specimen mass and balance readability shall be in accordance with [Table 1](#).

8.3 Using a test specimen smaller than the minimum indicated in [Table 1](#) and [8.2](#) requires discretion, though it may be adequate for the purposes of the test. It shall be noted on the test data forms or test data sheets of any specimen used not meeting these requirements.

8.4 When working with a specimen weighing less than 200 g and containing a relatively large gravel particle, it is appropriate not to include the gravel particle in the test specimen. However, any discarded material shall be described and noted on the test data form/sheet.

8.5 For those samples consisting entirely of intact rock or gravel-size aggregate, the minimum specimen mass shall be 500 g. Representative portions of the sample may be broken into smaller particles. The particle size is dictated by the specimen mass, the container volume and the balance being used to determine constant mass, see [10.4](#). Specimen masses as small as 200 g may be tested if water contents of only two significant digits are acceptable.

9. Test Specimen Selection

9.1 When the test specimen is a portion of a larger amount of material, the specimen must be selected to be representative of the water condition of the entire sample. The manner in which the test specimen is selected is dependent on the purpose and application of the test, the type of material being tested, the water condition, and the type of sample (from another test, bag, block, etc.).

9.2 For disturbed samples such as trimmings, bag samples, etc; obtain the test specimen by one of the following methods (listed in order of preference):

9.2.1 If the material is such that it can be manipulated and handled without significant moisture loss and segregation, the material shall be mixed thoroughly. Select a representative portion using a scoop of a size that no more than a few scoops are required to obtain the proper size of specimen defined in [8.2](#). Combine all the portions for the test specimen.

9.2.2 If the material is such that it cannot be thoroughly mixed or mixed and sampled by a scoop, form a stockpile of the material, mixing as much as possible. Take at least five portions of material at random locations using a sampling tube, shovel, scoop, trowel, or similar device appropriate to the maximum particle size present in the material. Combine all the portions for the test specimen.

9.2.3 If the material or conditions are such that a stockpile cannot be formed, take as many portions of the material as practical, using random locations that will best represent the moisture condition. Combine all the portions for the test specimen.

9.3 For intact samples such as block, tube, split barrel, etc, obtain the test specimen by one of the following methods depending on the purpose and potential use of the sample:

9.3.1 Using a knife, wire saw, or other sharp cutting device, trim the outside portion of the sample a sufficient distance to expose the underlying material to determine if it is layered or if there are exposed anomalies. Trimming the sample in this manner also removes material that may be more dry or more wet than the main portion of the sample. If the existence of layering is questionable, slice the sample in half. If the material is layered, see [9.3.3](#).

9.3.2 If the material is not layered, obtain the specimen meeting the mass requirements in [Table 1](#) and [8.2](#) by: (1) taking all or one-half of the interval being tested; (2) trimming a representative slice from the interval being tested; or (3) trimming the exposed surface of one-half or from the interval being tested.

NOTE 5—Migration of moisture in some cohesionless soils may require that the entire sample be tested.

9.3.3 If a layered material (or more than one material type is encountered), select an average specimen, or individual specimens, or both dependent on project goals. Specimens must be properly identified as to location, or what they represent, and appropriate remarks entered on the test data forms or test data sheets.

10. Procedure

10.1 Determine and record the mass of the clean and dry specimen container and its lid, if used, along with its identification number.

10.2 Select representative test specimens in accordance with [Section 9](#).

10.3 Place the moist test specimen in the container and, if used, set the lid securely in position. Determine the mass of the container and moist specimen using a balance (see [Table 1](#) and [8.2](#)) selected on the basis of the specimen mass or required significant digits. Record this value.

NOTE 6—To assist in the oven drying of large test specimens, they should be placed in containers having a large surface area (such as pans) and the material broken up into smaller aggregations.

10.4 Remove the lid (if used) and place the container with the moist specimen in the drying oven. Dry the specimen to a constant mass. Maintain the drying oven at $110 \pm 5^\circ\text{C}$ unless otherwise specified (see [1.4](#)). The time required to obtain constant mass will vary depending on the type of material, size of specimen, oven type and capacity, and other factors. The influence of these factors generally can be established by good judgment and experience with the materials being tested and the apparatus being used.

10.4.1 In most cases, drying a test specimen overnight (about 12 to 16 h) is sufficient, especially when using forced draft ovens. In cases where there is doubt concerning the adequacy of drying to a constant dry mass, see [3.2.1](#) and check for additional loss in mass with additional oven drying over an adequate time period. A minimum time period of two hours shall be used. Increasing the drying time may be necessary with greater specimen mass. A rapid check to see if a relatively large specimen (> than about 100 g of material) is dry; place a small strip of torn paper on top of the material while it is in the oven or just upon removal from the oven. If the paper strip

curls the material is **not** dry and requires additional drying time. Specimens of sand may often be dried to constant mass in a period of about 4 h, when a forced-draft oven is used.

10.4.2 Since some dry materials may absorb moisture from other drying specimens that still retain moisture, dried specimens shall be removed before placing moist specimens in the same oven; unless they are being dried for an extended period of time such as overnight.

10.5 After the specimen has dried to constant mass, remove the container from the oven and replace the lid if used or place the specimen in a desiccator. Allow the specimen and container to cool to room temperature or until the container can be handled comfortably with bare hands and the operation of the balance will not be affected by convection currents or heat transmission. Determine the mass of the container and oven-dried specimen using the same type/capacity balance used in 10.3 and record this value. If the possibility exists that the specimen may absorb moisture from the air prior to the determination of its dry mass, tight fitting lids shall be used even after the specimen has been allowed to cool in a desiccator. Good judgment and experience with testing specific soil types may assist in determining if a lid should be used after cooling in a desiccator. Specimens that are allowed to cool with a lid in place shall be weighed with the lid on.

10.5.1 Cooling in a desiccator is acceptable in place of tight fitting lids since it greatly reduces absorption of moisture from the atmosphere during cooling.

10.6 A copy of a sample data sheet is shown in **Appendix X1**. Any data sheet can be used, provided the form contains all the required data.

11. Calculation

11.1 Calculate the water content of the material as follows:

$$w = [(M_{cms} - M_{cds}) / (M_{cds} - M_c)] \times 100 = (M_w / M_s) \times 100 \quad (1)$$

where:

- w = water content, %,
- M_{cms} = mass of container and moist specimen, g,
- M_{cds} = mass of container and oven dry specimen, g,
- M_c = mass of container, g,
- M_w = mass of water ($M_w = M_{cms} - M_{cds}$), g, and
- M_s = mass of oven dry specimen ($M_s = M_{cds} - M_c$), g.

12. Report: Test Data Sheet(s)/Form(s)

12.1 The methodology used to specify how data are recorded on the test data sheet(s)/form(s), as given below, is covered in Practice **D6026**. These requirements do not consider in situ material variation, use of the data, special purpose studies, or any considerations for the user's objectives. It is common practice to increase or reduce significant digits of reported data commensurate with these considerations. It is beyond the scope of the standard to consider significant digits used in analysis method for engineering design.

12.2 Record as a minimum the following general information (data):

12.2.1 Project information such as project number or identification, the name of the project, the client if applicable.

12.2.2 Person that conducted the test.

12.2.3 Date the test was conducted.

12.2.4 Identification of the sample (material) being tested, such as boring number, sample number, test number, container number etc.

12.2.5 Water content of the specimen to the nearest 1 % for Method A or 0.1 % for Method B, as appropriate based on the minimum mass of the specimen. If this method is used in concert with another method, the water content of the specimen should be reported to the value required by the test method for which the water content is being determined. Refer to Practice **D6026** for guidance concerning significant digits, especially if the value obtained from this test method is to be used to calculate other relationships such as unit weight or density. For instance, if it is desired to express dry unit weight, as determined by **D7263** to the nearest 0.1 lbf/f³ (0.02 kN/m³), it may be necessary to use a balance with a greater readability or use a larger specimen mass to obtain the required significant digits for the mass of water so that the water content can be determined to the required significant digits. Also, the significant digits in Practice **D6026** may need to be increased when calculating phase relationships requiring four significant digits.

12.2.6 Indicate if test specimen had a mass less than the minimum required as indicated in **Table 1** and **8.2**.

12.2.7 Indicate if test specimen contained more than one material type (layered, etc.).

12.2.8 Indicate the drying temperature if different from $110 \pm 5^\circ\text{C}$.

12.2.9 Indicate if any material (size and amount) was excluded from the test specimen.

12.3 When reporting water content in tables, figures, etc., any data not meeting the requirements of this test method shall be noted, such as not meeting the mass, balance, or temperature requirements or a portion of the material is excluded from the test specimen.

13. Precision and Bias

13.1 *Statements on Precision*⁶:

13.1.1 *Precision*—Test data on precision is not presented due to the nature of the soil or rock materials tested by this test method. It is either not feasible or too costly at this time to have ten or more laboratories participate in a round-robin testing program. Any variation observed in the data is just as likely to be due to specimen variation as to operator or laboratory testing variation.

13.1.2 Subcommittee D18.03 is seeking any data from the users of this test method that might be used to make a limited statement on precision.

13.1.3 *Bias*—There is no accepted reference value for this test method, therefore, bias cannot be determined.

14. Keywords

14.1 aggregate; consistency; index property; laboratory; moisture analysis; moisture content; soil; water content

⁶ Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D13-1108.

(Nonmandatory Information)

X1. WATER CONTENT OF SOIL AND ROCK SAMPLE DATA SHEET

Project Name: _____	Project Number: _____	Date: _____
	Test Method: <u> X </u> _____	Method A Method B

Laboratory Number	04-725-S	04-726-S	04-727-S		
Boring Number	B-1	B-2	B-2		
Field Number	SPT-1	SPT-2	SPT-2a		
Container / Lid Number	725	726	727		
Container Mass, g M_c	770.1	731.7	770.6		
Container+Moist Specimen Mass, g M_{cms}	1895.3	2008.4	1827.9		
Initial Container Dry Specimen Mass, g	1721.4	1872.1	1707.6		
Secondary Container Dry Specimen Mass, g	1721.4	1801.2	1660.8		
Final Container Dry Specimen Mass, g, M_{cds}	1721.4	1801.2	1660.8		
Mass of Water, g, $M_w = M_{cms} - M_{cds}$	173.9	207.2	167.1		
Mass of Solids, g, $M_s = M_{cds} - M_c$	951.3	1069.5	890.2		
Water Content, %, $w = (M_w/M_s) \times 100$	18	19	19		
Unified Soil Classification Group Symbol (Visual)	GC	GC	GC		
Approximate Maximum Particle Size (Visual)					
Oven Temperature if Other Than $110 \pm 5^\circ\text{C}$	—	—	—		

Remarks: _____

Tested By: _____

Dry Mass By: _____

Calculated By: _____

Date: _____

Date: _____

Date: _____

Checked By: _____

Spot Checked: _____

Reviewed By: _____

SUMMARY OF CHANGES

Committee D18 has identified the location of selected changes to these test methods since the last issue, D2216–10, that may impact the use of these test methods. (March 1, 2019)

- (1) Added a more detailed description of water content in the Scope.
- (2) Included statement for use of water content when used in conjunction with other soil relationships.
- (3) Eliminated the reference to E145.
- (4) Revised oven requirements in Apparatus Section.
- (5) Revised Report Section to conform to D18 SPM.
- (6) Incorporated grammatical and spelling revisions.

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